Real Analysis

An Undergraduate Notebook

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Contents

Part I

Foundations of Analysis

Chapter 1

Introductory Set Theory

In this chapter, we lay the groundwork for our exposition of real analysis. We will introduce the fundamental notions of **sets**, **relations**, and **maps**, which underlie not only all of analysis but also all of modern mathematics as well.

1.1 What Is a Set?

Since the early 20th century, the practice of mathematics has undergone a radical transformation, which grew out of the idea that the whole of the mathematical canon can and should be rephrased in the language of set theory. Informally, a **set** is a collection of objects, which are called its **elements**. If an object x is an element of a set S, then we write $x \in S$; otherwise, if x is not an element of S, then we write $x \notin S$.

1.1.1 Definitions of Sets

The careful reader may complain that the terms "collection" and "object" are vague and undefined, and they would be right to object; the above characterization of sets is definitely not as rigorous as we might desire, and is in fact internally inconsistent. However, the technical difficulties associated with building a rigorous and internally consistent set theory are outside the scope of this course and in this context, far more trouble than they are worth. In fact, it's entirely possible that such an internally consistent set theory does not exist!

As such, we will leave this ambiguity in favor of moving on to more salient topics. Even though we will not encounter any such issues in this text, the "definition" above does lead to some interesting inconsistencies. We will explore one such contradiction, called Russell's paradox, shortly. First, however, we introduce some shorthand notation which will be useful in specifying sets we wish to discuss.

Convention 1.1.1 Roster notation. It's easy to completely describe a set which contains only a finite number of elements; we can simply write a list of those elements down! Concretely, if X is the set whose elements are precisely the objects $x_1, \ldots, x_n$, then we may write

$$X = \{x_1, \ldots, x_n\}.$$

The above notation for specifying sets is called **roster notation**.

Not all sets are finite, however, and it is obviously impossible to write down a complete roster of all of the elements of an infinite set. We can carefully use ellipses to indicate a set with an infinite roster whose elements follow some discernible pattern. For example, we will see in Section **??** Peano Arithmetic and the Natural Numbers [Skip] that the set $\mathbb{N}$ of natural numbers can be written

$$\mathbb{N} = \{0, 1, 2, 3, \ldots\}.$$

Unfortunately, this extension of roster notation still does not suffice for all sets. For example, we will see in Section **??** Cardinality and Countability [SKIP] that there is no such roster for the set $\mathbb{R}$ of real numbers.

Definition 1.1.2 Definable collection. Let Φ be a **unary Boolean predicate** (a statement whose truth value depends on a single variable x). The collection of objects x which satisfy Φ is said to be **defined** by Φ, and is denoted

$$\{x \mid \Phi(x)\}.$$

Such a collection of objects which satisfy such a Boolean predicate is called a **definable collection**. We will see that the collection of objects that satisfy a given Boolean predicate is not necessarily a set; this is the content of Russell's paradox. ◆

Example 1.1.3 Sets. The following are examples of sets:

- The **empty set**, denoted $\emptyset$ or $\{\}$, is the set with no elements; that is, $x \notin \emptyset$ for all objects x.
- The set $\{1, 2, \text{red}, \text{blue}\}$ is a **non-empty** set with only finitely many elements. Note that this set contains different types of objects. In general, we will not place any explicit restrictions on the types of objects allowed inside a set, although some formalizations of axiomatic set theories rely on this to avoid inconsistencies such as Russell's paradox.
- The set $\mathbb{N} = \{0, 1, 2, \ldots\}$ of **natural numbers** (also sometimes called the **whole numbers** or **counting numbers**) is a non-empty set with infinitely many elements. We will characterize the natural numbers $\mathbb{N}$ in the usual manner of Peano arithmetic later in this chapter.

▲

As we mentioned before, the set theory based on the "definition" above, which is called **naïve set theory**, is in fact inconsistent; taking this "definition" as the basis for an axiomatic system, we can derive a contradiction, which implies that we may in fact derive all possible statements, including those which are false! This inconsistency was first demonstrated by Bertrand Russell at the beginning of the 20th century, and bears his name.

Proposition 1.1.4 Russell's paradox. (Bertrand Russell) *Not all definable collections are sets.*

Proof. By contradiction. Let

$$\mathfrak{R} = \{X \mid X \notin X\}$$

be the definable collection of sets which do not contain themselves, and suppose for a contradiction that $\mathfrak{R}$ is a set. The question of whether or not $\mathfrak{R}$ contains itself is a natural one; however, we will see that both the positive and negative answers lead to contradictions.

This is because, if $\mathfrak{R} \in \mathfrak{R}$, then $\mathfrak{R} \notin \mathfrak{R}$ by the definition of $\mathfrak{R}$. On the other hand, if $\mathfrak{R} \notin \mathfrak{R}$, then $\mathfrak{R} \in \mathfrak{R}$ by the same reasoning. Therefore, $\mathfrak{R} \in \mathfrak{R}$ if and only if $\mathfrak{R} \notin \mathfrak{R}$. This contradiction implies that $\mathfrak{R}$ is not a set, as desired. ■

Russell's paradox tells us that naïve set theory is inconsistent. However, as we stated before, it is no easy task to construct a consistent set theory which is strong enough to describe the sophisticated structures we wish to investigate while also weak enough to be internally consistent. However, since the question of whether modern formal set theories are consistent is beyond the scope of this text. With this in mind, we will continue formulating definitions and results in the language of naïve set theory, keeping in mind that rigorous systems of formal set theory (which do not suffer from Russell's paradox) exist in which they can be expressed and proven.

1.1.2 Set Containment

As we have just seen, not all definable collections of mathematical objects are sets. However, any collection of objects which themselves belong to a set is also a set in its own right; such a collection of elements of a given set is called a **subset**.

Definition 1.1.5 Set containment. A set U is a **subset** of a set X if for all objects x, if $x \in U$, then $x \in X$. In this case, we say that U is **contained** in X, and we write $U \subseteq X$. Less commonly, we may also call X a **superset** of U, say that X contains U, and write $X \supseteq U$.

A set U is a **proper subset** of a set X if $U \subseteq X$ but $X \nsubseteq U$. In this case, we say that U is **strictly contained** in X, and we write $U \subsetneq X$. Less commonly, we may also call X a **proper superset** of U, say that X strictly contains U, and write $X \supsetneq U$. ◆

Convention 1.1.6 Set containment notation. Some authors use the symbols $\subset$ and $\supset$ in place of $\subseteq$ and $\supseteq$ to indicate set containment, and some authors use them in place of $\subsetneq$ and $\supsetneq$ to indicate strict set containment. We will avoid this ambiguity by opting not to write $\subset$ and $\supset$ at all.

Definition 1.1.7 Set equality. Two sets are considered **equal** if they contain exactly the same elements; that is, one set X is equal to another set Y when objects are elements of X if and only if they are elements of Y. This occurs precisely when both $X \subseteq Y$ and $Y \subseteq X$. ◆

Example 1.1.8 Subsets. The following are examples of subsets of sets:

- For any set X, the empty set $\emptyset$ and X itself are subsets of X.
- The set $\{1, 3, 5, \ldots\}$ of odd natural numbers is a proper subset of the set $\mathbb{N}$ of natural numbers. Written more concisely,
$$\{1, 3, 5, \ldots\} \subsetneq \mathbb{N}.$$

▲

An important property of the set containment relationship is its transitivity, which is the content of the following lemma:

Lemma 1.1.9 Transitivity of set containment. *For any sets U, V, and X, if $U \subseteq V$ and $V \subseteq X$, then $U \subseteq X$.*

Proof. Let $u \in U$. Since $u \in U$ and $U \subseteq V$, $u \in V$. Moreover, since $u \in V$ and $V \subseteq X$, $u \in X$. In summary, we have shown that for $u \in X$ for all elements $u \in U$, and so $U \subseteq X$. ■

Set containment is also useful for constructing new notations for specifying sets as definable collections. Crucially, although not all definable collections are sets, any definable subcollection of a set is itself a set.

Convention 1.1.10 Set-builder notation. Given a set X and a unary Boolean predicate Φ, the definable collection

$$\{x \mid x \in X \text{ and } \Phi(x)\}$$

is a set (and in fact a subset of X). This subset will also be denoted in **set-builder notation** by

$$\{x \in X \mid \Phi(x)\} = \{x \mid x \in X \text{ and } \Phi(x)\}.$$

After learning about Russell's paradox, the careful reader may have some trepidation about sets whose elements are themselves sets. However, not all such definable collections lead to contradictions. For example, the collection of all subsets of a given set also form a set.

Definition 1.1.11 Power set. The **power set** of a set X is the set of all subsets of X; this set is denoted $\mathcal{P}(X)$, so that

$$\mathcal{P}(X) = \{U \mid U \subseteq X\}.$$

◆

Lemma 1.1.12 Power sets are non-empty. *The power set $\mathcal{P}(X)$ of any set X is non-empty; that is, $\mathcal{P}(X) \neq \emptyset$.*

Proof. Fix a set X, and note that the statement $x \in X$ for all $x \in \emptyset$ is vacuously true, so that $\emptyset \subseteq X$. Thus $\emptyset \in \mathcal{P}(X)$, and so $\mathcal{P}(X) \neq \emptyset$ is non-empty. ■

Proof. Fix a set X, and note that the statement $x \in X$ for all $x \in X$ is a tautology, so that $X \subseteq X$. Thus $X \in \mathcal{P}(X)$, and so $\mathcal{P}(X) \neq \emptyset$ is non-empty. ■

Both of the above proofs of Lemma **??** Power sets are non-empty are correct, although they are nonequivalent. Both are valuable for different reasons; the first proof shows that the power set of a set is nonempty since the empty set $\emptyset$ is a subset of any set, and the second proof shows the same conclusion since any set is a subset of itself. These proofs are called **constructive**, in that they explicitly exhibited an object satisfied the desired properties (being an element of the power set of a given set). Proofs which are not constructive are called **non-constructive**.

In general, sets are likely to have many subsets, so that power sets of sets are likely to have a great many elements. We will investigate the notion of the number of elements of a set, called its **cardinality**, more precisely and in far greater depth in Section **??** Cardinality and Countability [SKIP] at the end of this chapter.

1.1.3 Set Operations

We now introduce several ways to construct new sets from old ones which loosely correspond to the logical operators of **conjunction** $\wedge$, **disjunction** $\vee$, and **negation** $\neg$, pronounced "and", "or" and "not", respectively.

Definition 1.1.13 Set operations. Let S and T be sets.

- *Intersection.*

  The **intersection** $S \cap T$ of two sets S and T is the set whose elements are *both* elements of S and elements of T; that is,

$$S \cap T = \{x \mid x \in S \wedge x \in T\} = \{x \mid x \in S \text{ and } x \in T\}.$$

- *Union.*

  The **union** $S \cup T$ of two sets S and T is the set whose elements are *either* elements of S and elements of T (or both!); that is,

  $$S \cup T = \{x \mid x \in S \vee x \in T\} = \{x \mid x \in S \text{ or } x \in T\}.$$

- *Complement.*

  The **complement** $S \setminus T$ of T in S is the set whose elements are elements of S but not elements of T; that is,

  $$S \setminus T = \{x \mid x \in S \wedge \neg(x \in T)\} = \{x \mid x \in S \text{ and } x \notin T\}.$$

◆

Informally, the intersection $S \cap T$ of two sets S and T is the largest set which is contained in both S and T; the union of S and T is the smallest set which contains both S and T; and the complement $S \setminus T$ is the largest set which is contained in S but disjoint from T. These informal characterizations are formalized in the following result:

Proposition 1.1.14 Set operations and containment. *Let S, T, and U be sets.*

1. *$U \subseteq S \cap T$ if and only if both $U \subseteq S$ and $U \subseteq T$.*

2. *$S \cup T \subseteq U$ if and only if both $S \subseteq U$ and $T \subseteq U$.*

3. *$U \subseteq S \setminus T$ if and only if both $U \subseteq S$ and $U \cap T = \emptyset$.*

Proof.

1. First suppose that $U \subseteq S \cap T$. Since $S \cap T \subseteq S$, Lemma **??** Transitivity of set containment implies that $U \subseteq S$. Similarly, since $S \cap T \subseteq T$, Lemma **??** Transitivity of set containment implies that $U \subseteq T$.

   Conversely, now suppose that $U \subseteq S$ and $U \subseteq T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Similarly, since $x \in U$ and $U \subseteq T$, $x \in T$. Since $x \in S$ and $x \in T$, we conclude that $x \in S \cap T$. In summary, we have shown that $x \in S \cap T$ for all $x \in U$, so that $U \subseteq S \cap T$.

2. First suppose that $S \cup T \subseteq U$. Since $S \subseteq S \cup T$, Lemma **??** Transitivity of set containment implies that $S \subseteq U$. Similarly, since $T \subseteq S \cup T$, Lemma **??** Transitivity of set containment implies that $T \subseteq U$.

   Conversely, now suppose that $S \subseteq U$ and $T \subseteq U$. Let $x \in S \cup T$, so that $x \in S$ or $x \in T$. If $x \in S$, then $x \in U$, since $S \subseteq U$. Similarly, if $x \in T$, then $x \in U$, since $T \subseteq U$. We conclude that $x \in U$. In summary, we have shown that $x \in U$ for all $x \in S \cup T$, so that $S \cup T \subseteq U$.

3. First suppose that $U \subseteq S \setminus T$, and let $x \in U$. Since $x \in U$ and $U \subseteq S \setminus T$, $x \in S \setminus T$, so that $x \in S$ and $x \notin T$. In summary, we have shown that $x \in S$ for all $x \in U$, so that $U \subseteq S$. Moreover, we have shown that $x \notin T$ for all $x \in U$, so that no element of U is an element of T; that is, $U \cap T = \emptyset$.

   Conversely, now suppose that $U \subseteq S$ and $U \cap T = \emptyset$, and let $x \in U$. Since $x \in U$ and $U \subseteq S$, $x \in S$. Moreover, since $x \in U$ and $U \cap T = \emptyset$, $x \notin T$. Since $x \in S$ and $x \notin T$, we conclude that $x \in S \setminus T$. In summary, we have shown that $x \in S \setminus T$ for all $x \in U$, and so $U \subseteq S \setminus T$.

■

Corollary 1.1.15 Properties of the set operations.

1. *$S \cap S = S \cup S = S \cup \emptyset = S \setminus \emptyset = S$ and $S \cap \emptyset = S \setminus S = \emptyset \setminus S = \emptyset$ for all sets S.*

2. *Commutativity.*

   $S \cap T = T \cap S$ and $S \cup T = T \cup S$ for all sets S and T.

3. *Associativity.*

   $(S \cap T) \cap U = S \cap (T \cap U)$ and $(S \cup T) \cup U = S \cup (T \cup U)$ for all sets S, T, and U.

Proof.

1. Since $S \subseteq S$, $S \subseteq S \cap S$ and $S \cup S \subseteq S$ by (1) and (2) of Proposition **??** Set operations and containment, respectively. Of course $S \cap S \subseteq S \subseteq S \cup S$, so that $S \cap S = S = S \cup S$.

   Similarly, since $S \subseteq S$ and $\emptyset \subseteq S$, $S \cup \emptyset \subseteq S$ by (2) of Proposition **??** Set operations and containment. Of course $S \subseteq S \cup \emptyset$, and so $S \cup \emptyset = S$.

   $\emptyset \subseteq S \cap \emptyset \subseteq \emptyset$, and so $S \cap \emptyset = \emptyset$. Moreover, together with the observation that $S \subseteq S$, this implies by (1) of Proposition **??** Set operations and containment that $S \subseteq S \setminus \emptyset$. Of course, $S \setminus \emptyset \subseteq S$, and so $S \setminus \emptyset = S$.

   Similarly, $\emptyset \subseteq \emptyset \setminus S \subseteq \emptyset$, and so $\emptyset \setminus S = \emptyset$.

   Finally, note that any element of $S \setminus S$ is both an element of S and not an element of S, so that no such element can exist. Thus $S \setminus S = \emptyset$.

2. Since $S \cap T \subseteq T$ and $S \cap T \subseteq S$, (1) of Proposition **??** Set operations and containment implies that $S \cap T \subseteq T \cap S$. By symmetry, $T \cap S \subseteq S \cap T$, so that $S \cap T = T \cap S$.

   Similarly, since $S \subseteq T \cup S$ and $T \subseteq T \cup S$, (2) of Proposition **??** Set operations and containment implies that $S \cup T \subseteq T \cup S$. By symmetry, $T \cup S \subseteq S \cup T$, so that $S \cup T = T \cup S$.

3. We note that

   $$\begin{aligned} (S \cap T) \cap U &\subseteq S \cap T \subseteq T \\ (S \cap T) \cap U &\subseteq U, \end{aligned}$$

   so that $(S \cap T) \cap U \subseteq T \cap U$ by (1) of Proposition **??** Set operations and containment. Similarly,

   $$(S \cap T) \cap U \subseteq S \cap T \subseteq T,$$

   so that $(S \cap T) \cap U \subseteq S \cap (T \cap U)$ by another application of (1) of Proposition **??** Set operations and containment.

   Conversely, we note that

   $$\begin{aligned} S \cap (T \cap U) &\subseteq S \\ S \cap (T \cap U) &\subseteq T \cap U \subseteq T, \end{aligned}$$

   so that $S \cap (T \cap U) \subseteq S \cap T$ by (1) of Proposition **??** Set operations and containment. Similarly,

   $$S \cap (T \cap U) \subseteq T \cap U \subseteq U,$$

   so that $S \cap (T \cap U) \subseteq (S \cap T) \cap U$ by another application of (1) of Proposition **??** Set operations and containment. Thus

   $$(S \cap T) \cap U = S \cap (T \cap U)\,.$$

■

In Subsection **??** Indexing in the next section, we will generalize the first two ways (**intersection** and **union**) of constructing sets to arbitrary sets of sets; this generalization will mirror the similarities between conjunction $\wedge$ and universal quantification $\forall$ and between disjunction $\vee$ and existential quantification $\exists$, respectively.

While sets and set theory form the foundations on which we will build our understanding of analysis, these topics are not the focus of this text. As such, we will leave behind the question of precise definitions of sets in favor of developing set-theoretic ideas which are more relevant to the world of analysis.

1.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Elements of sets. Determine whether or not the given object is an element of the given set. Write your answer using the notation $\in$ or $\notin$.

1. Is 3 an element of the set $\{1, 2, 3\}$?

2. Is 5 an element of the set $\{3, 9, 12\}$?

3. Is 8 an element of the set $\{0, 2, 4, \ldots, 12\}$?

4. Is 50 an element of the set $\{0, 1, 4, 9, 16, \ldots\}$?

Roster notation. Write down the specified set in roster notation.

5. Write down the set E of even numbers between 1 and 15 in roster notation.

6. Write down the set L of letters in your full name in roster notation.

7. Write down the set C of colors in the rainbow in roster notation.

Set containment. Let S, T, and U be the sets defined by

$$S = \{0, 1, 2\} \qquad T = \{3, 4\} \qquad U = \{1, 2, 3\}.$$

Determine whether or not the stated set containment is true or false.

8. Is $S \subseteq U$?

9. Is $S \cap T \subsetneq U$?

10. Is $U \subseteq S \cup T$?

Power sets. Write down the power sets of the given sets in roster notation.

11. Write down the power set $\mathcal{P}(S)$ of the set $S = \{1\}$ in roster notation.

12. Write down the power set $\mathcal{P}(T)$ of the set $T = \{1, 2\}$ in roster notation.

13. Write down the power set $\mathcal{P}(U)$ of the set $U = \{1, 2, 3\}$ in roster notation.

Set Operations. Compute the given set operations. Write your answer in roster or set-builder notation, or as a well-known set.

14. Compute $S \cap T$, where

$$S = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers and

$$T = \{n \in \mathbb{N} \mid n = 2k + 1 \text{ for some } k \in \mathbb{N}\} = \{1, 3, 5, 7, \ldots\}$$

is the set of odd integers.

15. Compute $S \cup T$, where $S = \{0, 3, 4, 9\}$ and $T = \{1, 3, 6\}$.

16. Compute $S \setminus T$, where

$$S = \{n \in \mathbb{N} \mid n = 5k \text{ for some } k \in \mathbb{N}\} = \{0, 5, 10, 15, \ldots\}$$

is the set of natural numbers divisible by 5, and

$$T = \{n \in \mathbb{N} \mid n = 2k \text{ for some } k \in \mathbb{N}\} = \{0, 2, 4, 6, \ldots\}$$

is the set of even natural numbers.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.1.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. The distributive properties of set operations.

(a) Prove that $S \cap (T \cup U) = (S \cap T) \cup (S \cap U)$ for all sets S, T, and U.

(b) Prove that $S \cup (T \cap U) = (S \cup T) \cap (S \cup U)$ for all sets S, T, and U.

Taken together, these results are called the **distributive properties** of the set operations.

2. De Morgan's laws.

(a) Prove that $S \setminus (T \cap U) = (S \setminus T) \cup (S \setminus U)$ for all sets S, T, and U.

(b) Prove that $S \setminus (T \cup U) = (S \setminus T) \cap (S \setminus U)$.

Taken together, these results are called **De Morgan's laws**.

3. Power sets. Determine whether the following statement is true or false:

The power set of a set contains at least two distinct elements.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Hint. Revisit the proofs of Lemma **??** Power sets are non-empty.

4. **Set operations and containment.** Let S and T be sets.

   (a) Prove that $S \cap T = S$ if and only if $S \subseteq T$.

   (b) Prove that $S \cup T = S$ if and only if $T \subseteq S$.

   (c) Prove that $S \setminus T = S$ if and only if $S \cap T = \emptyset$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.2 Maps and Functions

In this section, we introduce the fundamental notion of a **map** between sets. Informally, a map is an assignment to each element of one set an element of another. You likely have been exposed to maps in previous math courses in the form of real-valued functions of one real variable. We will treat maps in far greater generality, which will allow us to perform mathematical analysis in a wider array of spaces.

1.2.1 Introduction to Maps

First, however, we define the **Cartesian product** of two sets. This relies on a notion of an ordered pair of objects x and y, which we denote (x, y). Like formalized set theory, rigorous constructions of ordered pairs are, in this context, unnecessarily complicated and annoying to work with. As such, we will take them to be well-defined. The salient properties of an ordered pair (x, y) are its **entries** x and y, which are called its first and second entries, respectively.

Definition 1.2.1 Cartesian product. The **Cartesian product** $X \times Y$ of two sets X and Y is the set which contains all ordered pairs whose first entry is an element of X and whose second entry is an element of Y; that is,

$$X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}.$$

◆

Example 1.2.2 Cartesian products. The sets $X = \{1, 2\}$ and $Y = \{3, 4, 5\}$ have Cartesian product

$$X \times Y = \{(1,3), (1,4), (1,5), (2,3), (2,4), (2,5)\}.$$

Note that $X \times Y \neq X \cup Y$, since the elements of these sets are totally different kinds of objects. Moreover, if $X \neq Y$, then $X \times Y \neq Y \times X$; in the language of Subsection **??** Finitary Operations, the Cartesian product $\times$ is **non-commutative**. ▲

Recall that we would like a map between two sets to be an association of the elements of one set to the elements of the other. We will model this association via ordered pairs. To that end, a map between two sets will just be a subset of their Cartesian product which satisfies certain existence and uniqueness properties.

Definition 1.2.3 Map. A subset $f \subseteq X \times Y$ of the Cartesian product $X \times Y$ of two sets X and Y is called a **map** from X (which is called the **domain** of f and whose elements are called **inputs**) to Y (which is called the **codomain** of f and whose elements are called **outputs**) if it satisfies the following conditions:

1. *Uniqueness.*

   For all inputs $x \in X$, there is an output $y \in Y$ so that $(x, y) \in f$.

2. *Uniqueness (vertical line test).*

   For all inputs $x \in X$ and outputs $y_1, y_2 \in Y$, if $(x, y_1), (x, y_2) \in f$, then $y_1 = y_2$.

So f is a map from X to Y if and only if for each input $x \in X$, there is a unique output $y \in Y$ so that $(x, y) \in f$. We will write $f\colon X \to Y$ to indicate that f is a map from X to Y, and we will write $f\colon x \mapsto y$ to indicate that $(x, y) \in f$. The set of maps from X to Y is denoted $Y^X \subseteq X \times Y$.

We will call a map $f\colon X \to Y$ whose codomain Y is a number system like the natural numbers $Y = \mathbb{N}$, the integers $Y = \mathbb{Z}$, the rational numbers $Y = \mathbb{Q}$, the real numbers $Y = \mathbb{R}$, or the complex numbers $Y = \mathbb{C}$ a Y-valued **function** on X. We will investigate the natural numbers $\mathbb{N}$ in Section **??** Peano Arithmetic and the Natural Numbers [Skip], and Chapter **??** Constructing the Number Systems will be dedicated to constructing the other number systems with which we will become acquainted throughout the course of this text. ◆

So a map between sets is just a subset of their Cartesian product such that each element of the domain appears in precisely one pair with an element of the codomain. The above existence and uniqueness properties of a map implies that there is an unambiguous association between the elements of the domain and the elements of the codomain.

Example 1.2.4 Maps. The following are examples of maps between sets:

- Let $X = \{1, 2\}$ and $Y = \{\text{red}, \text{blue}\}$, and consider the following subsets of $X \times Y$:

  $$\begin{aligned} f_1 &= \{(1, \text{red}), (2, \text{red})\} \\ f_2 &= \{(1, \text{red}), (2, \text{blue})\} \\ f_3 &= \{(1, \text{blue}), (2, \text{red})\} \\ f_4 &= \{(1, \text{blue}), (2, \text{blue})\}. \end{aligned}$$

  As defined above, f_1, f_2, f_3, and f_4 are maps from X to Y. In fact, these are the only such maps; that is,

  $$Y^X = \{f_1, f_2, f_3, f_4\}.$$

- The empty set $\emptyset\colon \emptyset \to Y$ is the unique map from the empty set $\emptyset$ to any set Y; that is,

  $$Y^\emptyset = \{\emptyset\}.$$

  Note that none of the elements of the codomain Y are associated with elements of the empty domain $\emptyset$. However, this does not imply that $\emptyset$ is not a map; in general, not all elements of the codomain of a map need be associated with elements of the domain, since ***the existence property applies only to the domain***.

- Let $U \subseteq X$ be a subset of a set X. The **inclusion map** $\iota_U^X\colon U \to X$ of U in X is the map from U to X defined by

  $$\iota_U^X = \{(u, x) \in U \times X \mid u = x\}.$$

That is, $\iota_U^X : u \mapsto u$ for all $u \in U$.

The inclusion map ι_X^X of X in itself is called the **identity map** on X, and is denoted $\mathrm{id}_X : X \to X$.

▲

Remark 1.2.5 Passive and active characterizations of maps. It may be difficult to see how the above definition of a map is a generalization of the notion of a real-valued function of a real variable. In fact, the above definition might seem more closely related to the graph of such a function, since the uniqueness property of maps is just a generalization of the vertical line test for functions. We will refer to this perspective on maps as subsets of a Cartesian product satisfying certain abstract properties as the **passive characterization** of what maps *are*. In contrast, we often like to think of a map as an active process, which takes an input and transforms it into the unique output with which it is associated. We will refer to this perspective as the **active characterization** of what maps *do*. When it benefits us to favor one perspective over another, we will emphasize either the active or passive characterization over the other. The following definitions allow us to convert between these two equivalent characterizations of maps.

Definition 1.2.6 Image; pre-image. Let $f\colon X \to Y$ be a map from a set X to a set Y.

1. The **image** $f(x)$ of an input $x \in X$ under f is the unique output so that $(x, f(x)) \in Y$. The **image** $f(U)$ of a subset $U \subseteq X$ is the subset of the codomain Y defined by

$$f(U) = \{f(x) \in Y \mid x \in U\}.$$

   The image $f(X)$ of X itself is called the **image** of f, and is denoted $\mathrm{im}(f) = f(X)$.

2. The **pre-image** $f^{-1}(V)$ of a subset $V \subseteq Y$ of the codomain Y under f is the set of all inputs whose image lies in V; that is,

$$f^{-1}(V) = \{x \in X \mid f(x) \in V\}.$$

   The **pre-image** $f^{-1}(y)$ of an output $y \in Y$ under f is just the pre-image of the singleton set $\{y\}$ under f; that is,

$$f^{-1}(y) = f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\}$$

◆

Any map between sets induces a map from any subset of its domain to its codomain via a process known as **restriction**. In light of the active characterization of what maps do, this may seem almost trivial. However, we will formalize this notion of restriction in terms of the passive characterization of what maps are in order to prove rigorously our intuitions are correct. You may be better served by conceptualizing the restriction of a map to be the same active process of transforming inputs into outputs but with a smaller domain, rather than the passive definition below.

Definition 1.2.7 Restriction. The **restriction** $f|_U$ of a map $f\colon X \to Y$ between sets X and Y to a subset $U \subseteq X$ of its domain X is the subset of the Cartesian product $U \times Y$ defined by

$$f|_U = \{(u, y) \in U \times Y \mid f(u) = y\} = \{(u, y) \in U \times Y \mid (u, y) \in f\}$$

$$= (U \times Y) \cap f.$$

◆

As our intuition about the active characterization of what maps do tells us, the restriction of a map to a subset of its domain is itself a map from the chosen subset to the domain of the original map.

Proposition 1.2.8 Restrictions are maps. *Let $f\colon X \to Y$ be a map from a set X to a set Y. For any subset $U \subseteq X$, the restriction $f|_U$ of f to U is a map from U to Y.*

Proof. Let $u \in U$. Then $u \in X$ is an input to f, and so $(u, y) \in f$ for some output $y \in Y$. We conclude that

$$(x, y) \in \{(u, y) \in U \times Y \mid (u, y) \in f\} = f|_U\,,$$

so that $f|_U$ satisfies the existence condition.

Let $u \in U$ and $y_1, y_2 \in U$, and suppose that $(u, y_1), (u, y_2) \in f|_U$. Since

$$f|_U = (U \times Y) \cap f \subseteq f,$$

we have that $(u, y_1), (u, y_2) \in f$, and so $y_1 = y_2$; that is, $f|_U$ satisfies the uniqueness condition. Since $f|_U$ satisfies the existence and uniqueness conditions, it is a map from U to Y. ■

We will shortly see another way to construct restrictions as compositions with inclusion maps.

1.2.2 Composition and Invertibility

Viewed through the perspective of the active characterization of what maps do, it may seem obvious that we can combine two maps between sets by first acting on it with one map and then acting on its image by the other map. This new process yields the **composition** of the two maps, and it may be intuitively clear that this composition is itself a map. However, we will formalize this notion of composition in the passive characterization, so that we may prove this fact rigorously. Again, it may better serve you to conceptualize the composition of maps in the context of the active characterization of what maps do rather than the passive characterization of what maps are in order to better inform your intuition about composition.

Definition 1.2.9 Composition. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between sets X, Y, and Z. The **composition** $g \circ f$ of g with f is the map from X to Y defined by the formula

$$(g \circ f)(x) = g(f(x)).$$

◆

We aren't restricted to the composition of just two maps. In fact, we can compose any finite number of maps with compatible domains and codomains by iterating the above construction.

Proposition 1.2.10 Associativity of composition. *Let $f\colon W \to X$, $g\colon X \to Y$, and $h\colon Y \to Z$ be maps between sets W, X, Y, and Z. Then*

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. Note that

$$\begin{aligned}((h \circ g) \circ f)(w) &= (h \circ g)(f(w)) = h(g(f(w))) \\ &= h((g \circ f)(w)) = (h \circ (g \circ f))(w)\end{aligned}$$

for all $w \in W$. Since $(h \circ g) \circ f$ and $h \circ (g \circ f)$ agree on all inputs $w \in W$,

$$(h \circ g) \circ f = h \circ (g \circ f).$$

■

The above notion of composition introduces a natural question about maps between sets:

> To what degree can the action of a map on its domain be undone?

This question of map **invertibility** is fundamental, and will recur throughout this text.

Definition 1.2.11 Invertibility. Let $f\colon X \to Y$ be a map from a set X to a set Y.

1. *Left-invertibility.*

   A map $g\colon Y \to X$ from Y to X is a **left inverse** to f if $g \circ f = \mathrm{id}_X$. f is called **left-invertible** if such a left inverse exists.

2. *Right-invertibility.*

   A map $g\colon Y \to X$ from Y to X is a **right inverse** to f if $f \circ g = \mathrm{id}_Y$. f is called **right-invertible** if such a right inverse exists.

3. *Invertibility.*

   A map $g\colon Y \to X$ from Y to X is an **inverse** to f if it is both a left inverse and a right inverse to f; that is, g is an inverse to f if both $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. f is called **invertible** if such an inverse exists.

◆

Example 1.2.12 Inverses. The following are examples of left inverses, right inverses, and inverses of maps.

- Let $\emptyset \subsetneq U \subseteq X$ be a non-empty subset of a set X, and fix an element $u \in U$. Consider the map $c\colon X \to U$ from X to U defined piecewise by the formula
  $$c(x) = \begin{cases} x & x \in U \\ u & x \notin U. \end{cases}$$
  Note that
  $$\left(c \circ \iota_U^X\right)(u) = c\left(\iota_U^X(u)\right) = c(u) = u = \mathrm{id}_U(u)$$
  for all $u \in U$, and so $c \circ \iota_U^X = \mathrm{id}_U$; that is, c is a left inverse to the inclusion map $\iota_U^X : U \to X$ of U in X.

- Fix sets X and Y, and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi\colon X \times Y \to X$ be the **projection map** defined by the formula $\pi(x, y) = x$.

  Fix an element $y \in Y$, and consider the map $c\colon X \to X \times Y$ from X to $X \times Y$ defined by the formula $c(x) = (x, y)$. Then
  $$(\pi \circ c)(x) = \pi(c(x)) = \pi(x, y) = x = \mathrm{id}_X(x)$$

for all $x \in X$, and so $\pi \circ c = \mathrm{id}_X$; that is, c is a right inverse to the projection map π.

- The identity map $\mathrm{id}_X \colon X \to X$ on a set X is its own inverse, since $\mathrm{id}_X \circ \mathrm{id}_X = \mathrm{id}_X$. This is actually a special case of the first example above, where we take $U = X$. More generally, $f \circ \mathrm{id}_X = f$ for all maps f with domain X, and $\mathrm{id}_X \circ g = g$ for all maps g with codomain X.

▲

We now give alternative characterizations of left-invertibility, right-invertibility, and invertibility. Intuitively, a map is left-invertible if it does not associate two distinct inputs with the same output, and a map is right-invertible if each output is associated with at least one input. These properties mirror the totality and uniqueness conditions for maps, and will turn out to together imply the existence of a two-sided inverse, which is both a left and right inverse.

Definition 1.2.13 Injectivity; surjectivity; bijectivity. Let $f \colon X \to Y$ be a map from a set X to a set Y.

1. *Injectivity.*

   f is called **injective** (or **one-to-one**) if for all inputs $x_1, x_2 \in X$, if $f(x_1) = f(x_2)$, then $x_1 = x_2$. In this case, f is also called an **injection** of X into Y.

2. *Surjectivity.*

   f is called **surjective** (or **onto**) if for all outputs $y \in Y$ there is an input $x \in X$ so that $y = f(x)$. In this case, f is also called a **surjection** of X onto Y.

3. *Bijectivity.*

   f is called **bijective** if it is both injective and surjective. In this case, f is also called a **bijection** or a **one-to-one correspondence** between X and Y.

◆

Example 1.2.14 Injections, surjections, and bijections. The following are examples of injective, surjective, and bijective functions.

1. Let $U \subseteq X$ be a subset of a set X, and consider $u_1, u_2 \in U$. If $\iota_U^X(u_1) = \iota_U^X(u_2)$, then
   $$u_1 = \iota_U^X(u_1) = \iota_U^X(u_2) = u_2.$$
   So the inclusion map $\iota_U^X : U \to X$ is injective.

2. Fix sets X and Y, and suppose that $Y \neq \emptyset$ is non-empty. Let $\pi \colon X \times Y \to X$ be the **projection map** defined by the formula $\pi(x, y) = x$.

   Let $x \in X$, and note that since $Y \neq \emptyset$ is non-empty, there is some element $y \in Y$. We note that $\pi(x, y) = x$. Since each output $x \in X$ is the image of some input under π, π is surjective.

3. The identity map $\mathrm{id}_X \colon X \to X$ on a set X is injective by virtue of being an inclusion map. id_X is also surjective, since $\mathrm{id}_X(x) = x$ for all $x \in X$. Thus id_X is bijective.

▲

It is no accident that the examples for left- and right- invertibility coincided with the examples for injectivity and surjectivity. It turns out that injectivity is essentially equivalent to left-invertibility, and surjectivity is equivalent to right-invertibility. This is the content of the following theorem, whose proof relies heavily on Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

Theorem 1.2.15 Invertibility. *Let $f\colon X \to Y$ be a map from a set X to a set Y.*

1. *If f is left-invertible, then f is injective.*
2. *If $X \neq \emptyset$ is non-empty and f is injective, then f is left-invertible.*
3. *If f is right-invertible, then f is surjective.*

Proof.

1. Suppose that f is left-invertible, so that $g \circ f = \mathrm{id}_X$ for some map $g\colon Y \to X$. That f is injective follows from the injectivity of the identity map id_X and (d) of Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

2. Conversely, now suppose that $X \neq \emptyset$ is non-empty and f is injective. Then for each output $y \in Y$, there is at most one input $x \in X$ so that $y = f(x)$. Specifically, for each $y \in \mathrm{im}(f)$, there is a unique input $x \in X$ so that $y = f(x)$. Denote by $g(y) = x$ this unique input.

   Since $X \neq \emptyset$ is non-empty, it contains an element $x \in X$. Define $g(y) = x$ for all $y \in Y \setminus \mathrm{im}(f)$. We have defined a map $g\colon Y \to X$. Note that

   $$(g \circ f)(x) = g(f(x)) = x = \mathrm{id}_X(x)$$

   for all inputs $x \in X$, and so $g \circ f = \mathrm{id}_X$; that is, g is a left inverse to f, and so f is left-invertible.

3. Finally, suppose that f is right-invertible, so that $f \circ g = \mathrm{id}_Y$ for some map $g\colon Y \to X$. That f is surjective follows from the surjectivity of the identity map id_Y and (e) of Problem 1.2.5.4 Composition and injectivity/surjectivity/bijectivity.

■

Proving the remaining direction of the equivalence between right invertibility and surjectivity requires a strong assumption about set theory called the Axiom of choice, which we do not yet have the language to express in its simplest form. We will discuss about the Axiom of choice in Subsection **??** Products and Choice, when we generalize the Cartesian product of two sets using indexing maps.

Corollary 1.2.16 Invertibility. *A map is invertible if and only if it is bijective.*

Proof. Let $f\colon X \to Y$ be a map from a set X to a set Y. If f is invertible, it is both left- and right-invertible, and so f is both injective and surjective by (1) and (3) of Theorem **??** Invertibility. Hence f is bijective.

Conversely, now suppose that f is bijective, so that f is both injective and surjective. If $X = \emptyset$, then

$$\begin{aligned} Y &= \mathrm{im}(f) = f(X) = f(\emptyset) \\ &= \{f(x) \in Y \mid x \in \emptyset\} = \emptyset \end{aligned}$$

by the surjectivity of f. Thus $f\colon \emptyset \to \emptyset$ is the unique map from $\emptyset$ to itself, the identity map $\mathrm{id}_\emptyset\colon \emptyset \to \emptyset$. In particular, f is invertible.

So we may assume that $X \neq \emptyset$ is non-empty. Since f is injective, it is left-invertible by (2) of Theorem **??** Invertibility. So $g \circ f = \mathrm{id}_X$ for some map $g\colon Y \to X$; that is, $g(f(x)) = x$ for all inputs $x \in X$. It suffices to show that g is also a right inverse to f.

To that end, consider an output $y \in Y$. Since f is surjective, $y = f(x)$ for some input $x \in X$. Then

$$(f \circ g)(y) = f(g(x)) = f(g(f(x))) = f(x) = y = \mathrm{id}_Y(y).$$

Hence $f \circ g = \mathrm{id}_Y$, and so g is an inverse to f; that is, f is invertible. ■

We will investigate the invertibility and bijectivity of maps and the consequences these properties have on the domains and codomains of such maps more in Section **??** Cardinality and Countability [SKIP] at the end of this chapter, when we study **cardinality** and the size of sets. For now, however, we will move on to an important convention which is ubiquitous throughout mathematics: the notions of **indexed sets** and **indexed families**.

1.2.3 Indexing

Considered as an association between sets, a map can be thought to index its image by the elements of its domain. That is, a map can transfer information from its domain to its image. Just what this transferred information looks like can vary tremendously, but suffice it to say that this notion of indexing is fundamental and will recur throughout this text.

Definition 1.2.17 Indexed family/set. Consider a map $x\colon \mathcal{J} \to X$ from a set $\mathcal{J}$, called the **indexing set** and whose elements are called **indices**, to a set X. The image of such an index $j \in \mathcal{J}$ under such an **(indexed) family** x is often denoted $x_j = x(j)$, and x is often denoted $(x_j)_{j\in\mathcal{J}}$. We say that $\mathrm{im}(x)$ is **indexed** by $\mathcal{J}$, and we write

$$\{x_j\}_{j\in\mathcal{J}} = \{x_j \in X \mid j \in \mathcal{J}\} = \mathrm{im}(x).$$

◆

We can also use this notion of indexing to generalize the intersection and union of sets to act on arbitrary collections of sets. This generalization of the intersection and union of two sets relies on the natural similarities between logical conjunction $\wedge$ and universal quantification $\forall$ and between logical disjunction $\vee$ and existential quantification $\exists$, respectively.

Definition 1.2.18 Indexed set operations. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

1. *Intersection.*

   The **intersection** $\bigcap_{j\in\mathcal{J}} U_j$ of this indexed family of subsets is the subset of X defined by

   $$\bigcap_{j\in\mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for all } j \in \mathcal{J}\}.$$

2. *Union.*

   The **union** $\bigcup_{j\in\mathcal{J}} U_j$ of this indexed family of subsets is the subset of X defined by

   $$\bigcup_{j\in\mathcal{J}} U_j = \{x \in X \mid x \in U_j \text{ for some } j \in \mathcal{J}\}.$$

◆

These generalizations of the intersection and union of two sets extend naturally to generalizations of each of the results proven in the previous section.

Theorem 1.2.19 Analogous properties of indexed intersections and unions. *Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.*

1. *Analogously to Proposition* **??** *Set operations and containment, for any subset $V \subseteq X$,*

$$V \subseteq \bigcap_{j\in\mathcal{J}} U_j$$

if and only if $V \subseteq U_j$ for all indices $j \in \mathcal{J}$. Moreover,

$$\bigcup_{j\in\mathcal{J}} U_j \subseteq V$$

if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

2. *Reindexing.*

Analogously to (2) of Corollary **??** *Properties of the set operations,*

$$\bigcap_{k\in\mathcal{K}} U_{f(k)} = \bigcap_{j\in\mathcal{J}} U_j$$

and

$$\bigcup_{k\in\mathcal{K}} U_{f(k)} = \bigcup_{j\in\mathcal{J}} U_j$$

for any set $\mathcal{K}$ and surjection $f\colon \mathcal{K} \to \mathcal{J}$.

3. *Distributivity.*

Analogously to Problem 1.1.5.1 The distributive properties of set operations,

$$V \cup \bigcap_{j\in\mathcal{J}} U_j = \bigcap_{j\in\mathcal{J}} (V \cup U_j)$$

and

$$V \cap \bigcup_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \cap U_j)$$

for any subset $V \subseteq X$.

4. *De Morgan's laws.*

Analogously to Problem 1.1.5.2 De Morgan's laws,

$$V \setminus \bigcap_{j\in\mathcal{J}} U_j = \bigcup_{j\in\mathcal{J}} (V \setminus U_j)$$

and

$$V \setminus \bigcup_{j\in\mathcal{J}} U_j = \bigcap_{j\in\mathcal{J}} (V \setminus U_j)$$

for any subset $V \subseteq X$.

You will have an opportunity to prove the above results in the problems for this section.

The topics introduced in this section are central not only to the study of real analysis but to all of modern mathematics. In the next section, we will introduce another central object of study, the set $\mathbb{N}$ of **natural numbers**.

1.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Cartesian products. Write down the Cartesian products of the given sets in roster notation.

1. Write down the Cartesian product $S \times T$ of the sets $S = \{0, 1\}$ and $T = \{0, 1, 2\}$ in roster notation.

2. Write down the Cartesian product $U \times V$ of the sets $U = \{0\}$ and $V = \{1, 2, 3, 4\}$ in roster notation.

3. Write down the Cartesian product $X \times Y$ of the sets $X = \{0, 1, 2, 3, 4, 5\}$ and $Y = \emptyset$ in roster notation.

Is it a map? Consider the sets $X = \{0, 2, 4, 6, 8\}$ and $Y = \{0, 1, 2, 3, 4\}$. Determine whether or not the given object is a map from X to Y.

4. Determine whether or not

$$f = \{(0,0), (2,1), (4,2), (6,3), (8,4)\}$$

is a map from X to Y.

5. Determine whether or not

$$g = \{(0,1), (2,1), (4,4), (8,1)\}$$

is a map from X to Y.

6. Determine whether or not

$$h = \{(0,1), (2,1), (4,4), (4,3), (6,2), (8,1)\}$$

is a map from X to Y.

7. Determine whether or not

$$j = \{(0,0), (2,0), (4,0), (6,0), (8,0)\}$$

is a map from X to Y.

8. **Images and pre-images.** Let $f_4 \colon \{1, 2\} \to \{\text{red}, \text{blue}\}$ be the map from $\{1, 2\}$ to $\{\text{red}, \text{blue}\}$ defined in Example **??** Maps; that is,

$$f_4 = \{(1, \text{blue}), (2, \text{blue})\}.$$

Compute the given images and pre-images. Express your answer in roster notation if necessary.

(a) Compute the images $f_4(1)$ and $f_4(2)$.

(b) Compute the image $\operatorname{im}(f_4)$.

(c) Compute the pre-images f_4^{-1} (red) and f_4^{-1} (blue).

Determining injectivity/surjectivity/bijectivity. Determine whether or not the given map is injective, surjective, and/or bijective.

9. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 0)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

10. Let $X = \{0, 1\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

11. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 1), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

12. Let $X = \{0, 1, 2\}$ and $Y = \{0, 1, 2\}$, and consider the following map $f\colon X \to Y$ from X to Y:

$$f = \{(0, 0), (1, 2), (2, 1)\}.$$

Determine whether or not f is injective, surjective, and/or bijective.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.2.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Non-commutativity of Cartesian products.** Give examples of sets X and Y so that $X \times Y \neq Y \times X$.

2. **Existence of maps.** Determine whether the following statement is true or false:

> For any sets X and Y, there is a map from X to Y.

If the statement is true, prove it. If the statement is false, disprove it by providing a counterexample.

Hint. Consider your answers to Exercise Group 1.2.4.1–3 Cartesian products.

3. Non-commutativity of composition.

(a) Give examples of maps f and g so that the composition $g \circ f$ is well-defined but the composition $f \circ g$ is not.

(b) Give examples of maps f and g so that the compositions $g \circ f$ and $f \circ g$ are well-defined but $g \circ f \neq f \circ g$.

4. Composition and injectivity/surjectivity/bijectivity. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between sets X, Y, and Z.

(a) Prove that if f and g are injective, then so too is their composition $g \circ f\colon X \to Z$.

(b) Prove that if f and g are surjective, then so too is their composition $g \circ f\colon X \to Z$.

(c) Prove that if f and g are bijective, then so too is their composition $g \circ f\colon X \to Z$.

(d) Prove that if the composition $g \circ f\colon X \to Z$ is injective, then so too is f.

Hint. Prove the desired result by contraposition.

(e) Prove that if the composition $g \circ f\colon X \to Z$ is surjective, then so too is g.

Hint. Prove the desired result by contraposition.

Note that the above results are used in the proof of Theorem **??** Invertibility. In order to avoid circular reasoning, you should prove the above statements by using only the definitions of injectivity, surjectivity, and bijectivity (and not the alternative characterizations given by Theorem **??** Invertibility and Corollary **??** Invertibility).

5. Indexed intersections and unions. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X indexed by a set $\mathcal{J}$.

(a) Prove that for any subset $V \subseteq X$,

$$V \subseteq \bigcap_{j\in\mathcal{J}} U_j$$

if and only if $V \subseteq U_j$ for all indices $j \in \mathcal{J}$, and

$$\bigcup_{j\in\mathcal{J}} U_j \subseteq V$$

if and only if $U_j \subseteq V$ for all indices $j \in \mathcal{J}$.

(b) Prove that

$$\bigcap_{k\in\mathcal{K}} U_{f(k)} = \bigcap_{j\in\mathcal{J}} U_j$$

and

$$\bigcup_{k\in\mathcal{K}} U_{f(k)} = \bigcup_{j\in\mathcal{J}} U_j$$

for any set $\mathcal{K}$ and surjection $f\colon \mathcal{K} \to \mathcal{J}$.

(c) Prove that

$$V \cup \bigcap_{j \in \mathcal{J}} U_j = \bigcap_{j \in \mathcal{J}} (V \cup U_j)$$

and

$$V \cap \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \cap U_j)$$

for any subset $V \subseteq X$.

(d) Prove that

$$V \setminus \bigcap_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \setminus U_j)$$

and

$$V \setminus \bigcup_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (V \setminus U_j)$$

for any subset $V \subseteq X$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.3 Peano Arithmetic and the Natural Numbers [Skip]

In this section, we formally introduce the **natural number system** $\mathbb{N}$. Also called the **counting numbers** and the **whole numbers**, the natural numbers form the simplest of the number systems; in fact, many people would date the advent of mathematics itself to the invention/discovery of these numbers.

1.3.1 Axiomatizing the Natural Numbers

There are many nonequivalent formalizations and models of the natural number system, but the standard axiomatization is due to Giuseppe Peano in the 19th century. **Peano arithmetic** is **non-constructive**; it gives a description of the natural number system but not a model. We won't introduce any specific model of the natural number system, but, rest assured, there are several. For our purposes, $\mathbb{N}$ itself can be any set which conforms to the following characterization:

Definition 1.3.1 Natural numbers. The **natural number system** $\mathbb{N}$ is a nonempty set with an element $0 \in \mathbb{N}$, called **zero**, along with functions $\operatorname{succ}\colon \mathbb{N} \to \mathbb{N}$ and $\operatorname{add}, \operatorname{mult}\colon \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, called the **successor function** and the **addition** and **multiplication operations** and a subset $L \subseteq \mathbb{N} \times \mathbb{N}$ of the Cartesian product $\mathbb{N} \times \mathbb{N}$ of $\mathbb{N}$ with itself, called the **standard ordering**, which satisfy the following properties:

1. The **successor** $\operatorname{succ}(n)$ of any natural number $n \in \mathbb{N}$ is non-zero; that is $\operatorname{succ}(n) \neq 0$ for all natural numbers $n \in \mathbb{N}$.

2. The successor function $\operatorname{succ}\colon \mathbb{N} \to \mathbb{N}$ is injective; that is, for all natural numbers $m, n \in \mathbb{N}$, if $\operatorname{succ}(m) = \operatorname{succ}(n)$, then $m = n$.

3. $\operatorname{add}(n, 0) = n$ for all natural numbers $n \in \mathbb{N}$.

4. $\operatorname{add}(m, \operatorname{succ}(n)) = \operatorname{succ}(\operatorname{add}(m, n))$ for all natural numbers $m, n \in \mathbb{N}$.

5. $\operatorname{mult}(n, 0) = 0$ for all natural numbers $n \in \mathbb{N}$.

6. $\operatorname{mult}(m, \operatorname{succ}(n)) = \operatorname{add}(\operatorname{mult}(m, n), m)$ for all natural numbers $n \in \mathbb{N}$.

7. For all natural numbers $l, m \in \mathbb{N}$, $(l, m) \in L$ if and only if $\operatorname{add}(l, n) = m$ for some natural number $n \in \mathbb{N}$.

8. *Axiom of induction.*

   For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:

   (a) $0 \in U$.
   (b) For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $\operatorname{succ}(n) \in U$.

Typically, for all natural numbers $m, n \in \mathbb{N}$ we write $m + n = \operatorname{add}(m, n)$ and $mn = m \cdot n = \operatorname{mult}(m, n)$, and we write $m \leq n$ if and only if $(m, n) \in L$. The successor of 0 is denoted $1 = \operatorname{succ}(0)$. Likewise, the successor of 1 is denoted $2 = \operatorname{succ}(1)$, the successor of 2 is denoted $3 = \operatorname{succ}(2)$, and so on. We see that

$$\operatorname{succ}(n) = \operatorname{succ}(\operatorname{add}(n, 0)) = \operatorname{add}(n, \operatorname{succ}(0)) = n + 1$$

for all natural numbers $n \in \mathbb{N}$. With these new notations, we may restate the Peano axioms as follows:

1. $n + 1 \neq 0$ for all $n \in \mathbb{N}$.

2. For all natural numbers $m, n \in \mathbb{N}$, if $m + 1 = n + 1$, then $m = n$.

3. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$.

4. $m + (n + 1) = (m + n) + 1$ for all natural numbers $m, n \in \mathbb{N}$.

5. $n \cdot 0 = 0$ for all natural numbers $n \in \mathbb{N}$.

6. $m(n + 1) = mn + m$ for all natural numbers $n \in \mathbb{N}$.

7. For all natural numbers $l, m \in \mathbb{N}$, $l \leq m$ if and only if $l + n = m$ for some natural number $n \in \mathbb{N}$.

8. For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:

   (a) $0 \in U$.
   (b) *Axiom of induction.*
   For all natural numbers $n \in \mathbb{N}$, if $n \in U$, then $n + 1 \in U$.

◆

Under the usual (and original) interpretation of the natural number system $\mathbb{N}$ as characterizing quantities of discrete physical objects, the above rules (with the possible exception of the axiom of induction) may seem obviously true but also somewhat arbitrary. It is a fact, though we will not prove it here, that these axioms are independent of each other; it is impossible to derive one of the above statements from only the others. We hope that this, along with the results in this section about arithmetic, convince you that the above axiomatization is a good choice, in that it is efficient in the above sense and that it can be used to prove many arithmetical truths.

First, however, we introduce some ubiquitous terminology and notation.

A family $(x_n)_{n\in\mathcal{N}}$ of objects x_n indexed by some subset $\mathcal{N} \subseteq \mathbb{N}$ is called a **sequence**. If the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n \text{ and } n \leq b\}$$

for some natural numbers $a, b \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^{b}$. Similarly, if the index set $\mathcal{N}$ is of the form

$$\mathcal{N} = \{n \in \mathbb{N} \mid a \leq n\}$$

for some natural number $a \in \mathbb{N}$, then we denote this family by $(x_n)_{n=a}^{\infty}$. These notations also apply to intersections and unions of sequences of sets indexed by subsets of $\mathbb{N}$ of the above form.

Lemma 1.3.2 0 is minimal. *For all natural numbers $n \in \mathbb{N}$, if $n \leq 0$, then $n = 0$.*

Proof. We first observe that $\{0\} \cup \mathrm{im}\,(\mathrm{succ}) = \mathbb{N}$. Indeed, note that $0 \in \{0\} \subseteq \{0\} \cup \mathrm{im}\,(\mathrm{succ})$. Moreover, if $n \in \{0\} \cup \mathrm{im}\,(\mathrm{succ})$, then

$$\mathrm{succ}\,(n) \in \mathrm{im}\,(\mathrm{succ}) \subseteq \{0\} \cup \mathrm{im}\,(\mathrm{succ})\,.$$

Thus $\{0\} \cup \mathrm{im}\,(\mathrm{succ}) = \mathbb{N}$ by the axiom of induction.

Let $n \in \mathbb{N}$ be a natural number, and suppose that $n \leq 0$ for some natural number $n \in \mathbb{N}$. Then $n + m = 0$ for some natural number $m \in \mathbb{N}$. Suppose for a contradiction that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Then

$$0 = n + m = n + (p + 1) = (n + p) + 1,$$

which contradicts the first Peano axiom. Thus $m \notin \mathrm{im}\,(\mathrm{succ})$. However, $m \in \mathbb{N} = \{0\} \cup \mathrm{im}\,(\mathrm{succ})$, and so $m \in \{0\}$; that is, $m = 0$, and so

$$n = n + 0 = n + m = 0.$$

■

Proposition 1.3.3 Principle of strong induction. *For all subsets $U \subseteq \mathbb{N}$, $U = \mathbb{N}$ if and only if it satisfies the following conditions:*

1. $0 \in U$.
2. *For all natural numbers $n \in \mathbb{N}$, if $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, then $n + 1 \in U$.*

Proof. Clearly $\mathbb{N}$ satisfies the above conditions, and so it suffices to show that any subsets which satisfies these conditions is equal to $\mathbb{N}$. To that end, let $U \subseteq \mathbb{N}$ be a subset which satisfies the above conditions, and consider the subset

$$V = \{n \in \mathbb{N} \mid k \in U \text{ for all } k \in \mathbb{N} \text{ so that } k \leq n\}\,.$$

Since the only $k \in \mathbb{N}$ so that $k \leq 0$ is $k = 0$ itself, and $0 \in U$, we have that $0 \in V$. Moreover if $n \in V$, then $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n$, and so $n + 1 \in U$ by hypothesis. It remains to conclude that $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$.

To that end, let $k \in \mathbb{N}$, and suppose that $k \leq n + 1$. Then $k + m = n + 1$ for some natural number $m \in \mathbb{N}$. If $m = 0$, then $k = n + 1 \in U$. On the other hand, if $m \neq 0$, then the argument in the proof of Lemma **??** 0 is minimal implies that $m = p + 1$ for some natural number $p \in \mathbb{N}$. Thus

$$(k + p) + 1 = k + (p + 1) = k + m = n + 1,$$

and so the injectivity of the successor function implies that $k + p = n$; that is, $k \leq n$ and so $k \in U$. Since $k \in U$ for all $k \in \mathbb{N}$ so that $k \leq n + 1$, we conclude that $n + 1 \in V$.

In summary, we have shown that $0 \in V$ and that for all natural numbers $n \in \mathbb{N}$, if $n \in V$, then $n + 1 \in V$. Thus $V = \mathbb{N}$ by the axiom of induction. One can check that $\mathbb{N} = V \subseteq U \subseteq \mathbb{N}$, and so $U = \mathbb{N}$. ■

The axiom of induction and Proposition **??** Principle of strong induction form the basis for ubiquitous methods of proof called **proof by induction** and **proof by strong induction**, respectively. You will have the opportunity to practice constructing such proofs in the problems for this section.

1.3.2 Properties of Arithmetic

The axioms of Peano arithmetic actually imply a great deal about the properties of the natural number system. We will now state and prove a great deal of results about the successor function and the addition and multiplication operators. Many of these results will give us practice in the fundamental skill of proof by induction, and we will use these results implicitly for the rest of this text.

Proposition 1.3.4 Associativity of addition. $(l + m) + n = l + (m + n)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$ to show that $(l + m) + n = l + (m + n)$.

Base case. If $n = 0$, then

$$(l + m) + n = (l + m) + 0 = l + m = l + (m + 0) = l + (m + n)\,.$$

Inductive step. Now suppose for the inductive hypothesis that $(l + m) + n = l + (m + n)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned}(l + m) + (n + 1) &= ((l + m) + n) + 1 = (\ell + (m + n)) + 1\\ &= \ell + ((m + n) + 1) = \ell + (m + (n + 1))\,.\end{aligned}$$

We conclude that $(l + m) + n = l + (m + n)$ for all natural numbers $n \in \mathbb{N}$. ■

One consequence of Proposition **??** Associativity of addition is that we may unambiguously parse expressions with multiple sums without the need of parentheses; for example,if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression $l + m + n$ is well-defined, because

$$(l + m) + n = l + (m + n)\,.$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

Proposition **??** Associativity of addition implies that the result of adding a finite number of natural numbers depends only on the order of the numbers from left to right, and not on the iterative structure of the summands induced by parentheses. In fact, the order of the summands does not matter either. In order to show this, it helps to notice the following two results about addition.

Lemma 1.3.5 Addition by 0**.** $n + 0 = 0 + n = n$ *for all natural numbers* $n \in \mathbb{N}$.

Proof. $n + 0 = n$ for all natural numbers $n \in \mathbb{N}$ by the Peano axioms, and so it suffices to show $0 + n = n$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0 + n = 0 + 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $0 + n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$0 + (n + 1) = (0 + n) + 1 = n + 1.$$

We conclude that $0 + n = n$ for all natural numbers $n \in \mathbb{N}$. ■

Lemma 1.3.6 Towards commutativity of addition. $1 + n = n + 1$ *for all natural numbers* $n \in \mathbb{N}$.

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1 + n = 1 + 0 = 0 + 1 = n + 1$$

by Lemma **??** Addition by 0.

Inductive step. Now suppose for the inductive hypothesis that $1 + n = n + 1$ for some natural number $n \in \mathbb{N}$, and note that

$$1 + (n + 1) = (1 + n) + 1 = (n + 1) + 1$$

by Proposition **??** Associativity of addition. We conclude that $1 + n = n + 1$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.7 Commutativity of addition. $m + n = n + m$ *for all natural numbers* $m, n \in \mathbb{N}$.

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$m + n = m + 0 = m = 0 + m = n + m$$

by Lemma **??** Addition by 0.

Inductive step. Now suppose for the inductive hypothesis that $m + n = n + m$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} m + (n + 1) &= (m + n) + 1 = (n + m) + 1 = 1 + (n + m) \\ &= (1 + n) + m = (n + 1) + m \end{aligned}$$

by Proposition **??** Associativity of addition and Lemma **??** Towards commutativity of addition. We conclude that $m + n = n + m$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition **??** Associativity of addition and Proposition **??** Commutativity of addition together imply that, when adding natural numbers, neither the order of the summands nor the order of the addition operations performed matters. In order to see that analogous properties also hold for multiplication, we will require the following supporting results. Along the way, we will also prove the distributivity of multiplication over addition.

Lemma 1.3.8 Multiplication by 1**.** *$n \cdot 1 = n$ and $1n = n$ for all natural numbers $n \in \mathbb{N}$.*

Proof. For all natural numbers $n \in \mathbb{N}$,

$$n \cdot 1 = n\,(0+1) = (n \cdot 0) + n = 0 + n = 0$$

by Lemma **??** Addition by 0. Thus it suffices to prove that $1n = n$ for all natural numbers $n \in \mathbb{N}$. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$1n = 1 \cdot 0 = 0 = n.$$

Inductive step. Now suppose for the inductive hypothesis that $1n = n$ for some natural number $n \in \mathbb{N}$, and note that

$$1\,(n+1) = 1n + 1 = n + 1.$$

We conclude that $1n = n$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.9 Left distributivity of multiplication over addition. *$l\,(m+n) = lm + ln$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$l\,(m+n) = l\,(m+0) = lm = lm + 0 = lm + l \cdot 0 = lm + ln.$$

Inductive step. Now suppose for the inductive hypothesis that $l\,(m+n) = lm + ln$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} l\,(m+(n+1)) &= l\,((m+n)+1) = l\,(m+n) + l \\ &= (lm + ln) + l = lm + (ln + l) = lm + l\,(n+1) \end{aligned}$$

by Proposition **??** Associativity of addition. We conclude that $l\,(m+n) = lm + ln$ for all natural numbers $n \in \mathbb{N}$. ■

Proposition 1.3.10 Associativity of multiplication. *$(lm)\,n = l\,(mn)$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(lm)\,n = (lm)\,0 = 0 = l \cdot 0 = l\,(m \cdot 0) = l\,(mn)\,.$$

Inductive step. Now suppose for the inductive hypothesis that $(lm)\,n = l\,(mn)$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned} (lm)\,(n+1) &= (lm)\,n + lm = l\,(mn) + lm = \\ &= l\,(mn + m) = l\,(m\,(n+1)) \end{aligned}$$

by Proposition **??** Left distributivity of multiplication over addition. We conclude that $(lm)\,n = l\,(mn)$ for all natural numbers $n \in \mathbb{N}$. ■

Just as with addition, one consequence of Proposition **??** Associativity of multiplication is that we may unambiguously parse expressions with multiple products without the need of parentheses; for example,if $l, m, n \in \mathbb{N}$ are natural numbers, then the expression lmn is well-defined, because

$$(lm)\,n = l\,(mn)\,.$$

Going forward, we will make use of these expressions wherever it is possible to do so without obscuring the relevant arguments.

Proposition 1.3.11 Right distributivity of multiplication over addition. *$(l+m)\,n = ln + mn$ for all natural numbers $l, m, n \in \mathbb{N}$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$(l+m)\,n = (l+m)\cdot 0 = 0 = 0 + 0 = l\cdot 0 + m\cdot 0 = ln + mn.$$

Inductive step. Now suppose for the inductive hypothesis that $(l+m)\,n = ln + mn$ for some natural number $n \in \mathbb{N}$, and note that

$$\begin{aligned}(l+m)\,(n+1) = (l+m)\,n + (l+m) = ln + l + mn + m&\\ = l\,(n+1) + m\,(n+1)&\end{aligned}$$

by repeated applications of Proposition **??** Associativity of addition and Proposition **??** Commutativity of addition. We conclude that $(l+m)\,n = ln + mn$ for all natural numbers $n \in \mathbb{N}$. ∎

Lemma 1.3.12 Multiplication by 0. *$0n = 0$ for all natural numbers $n \in \mathbb{N}$.*

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$0n = 0 \cdot 0 = 0.$$

Inductive step. Now suppose for the inductive hypothesis that $0n = 0$ for some natural number $n \in \mathbb{N}$, and note that

$$0\,(n+1) = 0n + 0 = 0 + 0 = 0.$$

We conclude that $0n = 0$ for all natural numbers $n \in \mathbb{N}$. ∎

Proposition 1.3.13 Commutativity of multiplication. *$mn = nm$ for all natural numbers $m, n \in \mathbb{N}$.*

Proof. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then

$$mn = m \cdot 0 = 0 = 0 \cdot m = nm$$

by Lemma **??** Multiplication by 0.

Inductive step. Now suppose for the inductive hypothesis that $mn = nm$ for some natural number $n \in \mathbb{N}$, and note that

$$m\,(n+1) = mn + m = nm + 1m = (n+1)\,m$$

by Lemma **??** Multiplication by 1 We conclude that $mn = nm$ for all natural numbers $n \in \mathbb{N}$. ∎

Lemma 1.3.14 Cancellation of addition. *For all natural numbers $n \in \mathbb{N}$, addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$.*

Proof. Let $l, m \in \mathbb{N}$ be natural numbers. We proceed by induction on $n \in \mathbb{N}$.

Base case. If $n = 0$ and $l + n = m + n$, then

$$l = l + 0 = l + n = m + n = m + 0 = m$$

Inductive step. Now suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that if $l + n = m + n$, then $l = m$. Note that if $l + (n + 1) = m + (n + 1)$, then

$$(l + n) + 1 = l + (n + 1) = m + (n + 1) = (m + n) + 1,$$

and so $l + n = m + n$ by the injectivity of the successor function. Thus $l = m$ by the inductive hypothesis. In summary, we have shown that if $l + (n + 1) = m + (n + 1)$, then $l = m$. We conclude that for all natural numbers $n \in \mathbb{N}$, if $l + n = m + n$, then $l = m$. ■

A similar cancellation law exists for multiplication, but its proof will require some properties of the comparison of natural numbers via its standard ordering.

1.3.3 Properties of Comparison

The final component of Peano's axiomatization of the natural numbers $\mathbb{N}$ is that of comparison. The standard ordering $\leq$ on the natural numbers has several important and familiar properties which we will now derive from first principles.

Proposition 1.3.15 Comparison is a total order. *Comparison of natural numbers has the following properties:*

1. *Reflexivity.*

   $n \leq n$ for all natural numbers $n \in \mathbb{N}$.

2. *Antisymmetry.*

   For all natural numbers $m, n \in \mathbb{N}$, if $m \leq n$ and $n \geq m$, then $m = n$.

3. *Transitivity.*

   For all natural numbers $l, m, n \in \mathbb{N}$, if $l \leq m$ and $m \leq n$, then $l \leq n$.

4. *Totality.*

   For all natural numbers $m, n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

Proof.

1. For all natural numbers $n \in \mathbb{N}$, $n + 0 = n$, so that $n \leq n$.

2. Let $m, n \in \mathbb{N}$ be natural numbers, and suppose that $m \leq n$ and $n \leq m$. Then $n = m + p$ and $m = n + q$ for some natural numbers $p, q \in \mathbb{N}$. Note that

   $$m + 0 = m = n + q = m + p + q,$$

   and so $0 = p + q$ by Lemma **??** Cancellation of addition; that is, $q \leq 0$. Lemma **??** 0 is minimal now implies that $q = 0$, so that

   $$m = n + q = n + 0 = n.$$

3. Let $l, m, n \in \mathbb{N}$ be natural numbers, and suppose that $l \leq m$ and $m \leq n$. Then $l + p = m$ and $m + q = n$ for some natural numbers $p, q \in \mathbb{N}$, and

so

$$l + (p + q) = (l + p) + q = m + q = n.$$

We conclude that $l \leq n$.

4. Let $m \in \mathbb{N}$ be a natural number. We proceed by induction on $n \in \mathbb{N}$.

   Base case. If $n = 0$, then $0 + m = m$ by Lemma **??** Addition by 0, and so $n = 0 \leq m$.

   Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that either $m \leq n$ or $n \leq m$. If $m \leq n$, then $m + p = n$ for some natural number $p \in \mathbb{N}$, and so

   $$m + (p + 1) = (m + p) + 1 = n + 1;$$

   that is, $m \leq n + 1$.

   Now suppose that $n \leq m$, so that $n + p = m$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

   $$m + 1 = (n + p) + 1 = (n + 0) + 1 = n + 1,$$

   and so $m \leq n + 1$. On the other hand, if $p \neq 0$, then $p = q + 1$ for some natural number $q \in \mathbb{N}$ by an argument in the proof of Lemma **??** 0 is minimal. Then

   $$(n + 1) + q = n + 1 + q = n + q + 1 = n + p = m,$$

   and so $n + 1 \leq m$. We conclude that for all natural numbers $n \in \mathbb{N}$, either $m \leq n$ or $n \leq m$.

■

Corollary 1.3.16 Cancellation of multiplication. *For all natural numbers $n \in \mathbb{N}$, if $n \neq 0$, then addition by n is injective; that is, for all natural numbers $l, m, n \in \mathbb{N}$, if $ln = mn$, then either $l = m$ or $n = 0$.*

Proof. By induction on $n \in \mathbb{N}$.

Base case. If $n = 0$, then $n = 0$, and so the desired conclusion holds for all natural numbers $l, m \in \mathbb{N}$.

Inductive step. Now suppose for the inductive hypothesis that there is a natural number $n \in \mathbb{N}$ so that for all natural numbers $l, m \in \mathbb{N}$, if $ln = mn$, then either $n = 0$ or $l = m$. Let $l, m \in \mathbb{N}$ be natural numbers, and suppose that $l\,(n + 1) = m\,(n + 1)$. Then $n + 1 \neq 0$, and so it suffices to show that $l = m$.

To that end, note that (4) of Proposition **??** Comparison is a total order implies that either $l \leq m$ or $m \leq l$. Without loss of generality, we may suppose that $l \leq m$, so that $m = l + p$ for some natural number $p \in \mathbb{N}$. We observe that

$$\begin{aligned} 0 + (ln + l) &= ln + l = l\,(n + 1) = m\,(n + 1) \\ &= (l + p)\,(n + 1) = ln + l + pn + p = (pn + p) + (ln + l) \end{aligned}$$

by Lemma **??** Addition by 0, Proposition **??** Left distributivity of multiplication over addition, Proposition **??** Right distributivity of multiplication over addition, Proposition **??** Associativity of addition, and Proposition **??** Commutativity of addition.

In particular, Lemma **??** Cancellation of addition implies that $0 = pn + p$, so that $p \leq 0$. We conclude by Lemma **??** 0 is minimal that $p = 0$, so that

$$l = m + p = m + 0 = m.$$

We conclude that for all natural numbers $l, m, n \in \mathbb{N}$, either if $ln = mn$, then either $l = m$ or $n = 0$. ■

Lemma 1.3.17 Comparison and arithmetic. *Let $l, m, n, p \in \mathbb{N}$ be natural numbers.*

1. *If $l \leq n$ and $m \leq p$, then $l + m \leq n + p$.*
2. *If $l \leq n$ and $m \leq p$, then $lm \leq np$.*

Proof.

1. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

   $$(l + m) + (q + r) = l + m + q + r = l + q + m + r = n + p.$$

   We conclude that $l + m \leq n + p$.

2. If $l \leq n$ and $m \leq p$, then $n = l + q$ and $p = m + r$ for some natural numbers $q, r \in \mathbb{N}$, and so

   $$\begin{aligned} lm + (lr + qm + qr) &= lm + lr + qm + qr = l\,(m + r) + q\,(m + r) \\ &= (l + q)\,(m + r) = np. \end{aligned}$$

   We conclude that $lm \leq np$.

■

For ease of reference, we collect the above results (along with some immediate corollaries) in the following theorem.

Theorem 1.3.18 Properties of Peano arithmetic. *Addition, multiplication, and comparison of natural numbers have the following properties:*

1. *Associativity of addition.*

   $(l + m) + n = l + (m + n)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

2. *Commutativity of addition.*

   $m + n = n + m$ *for all natural numbers* $m, n \in \mathbb{N}$.

3. *Additive identity.*

   $n + 0 = 0 + n = 0$ *for all natural numbers* $n \in \mathbb{N}$.

4. *Associativity of multiplication.*

   $(lm)\,n = l\,(mn)$ *for all natural numbers* $l, m, n \in \mathbb{N}$.

5. *Commutativity of multiplication.*

   $mn = nm$ *for all natural numbers* $m, n \in \mathbb{N}$.

6. *Multiplicative annihilation.*

   $n \cdot 0 = 0n = 0$ *for all natural numbers* $n \in \mathbb{N}$.

7. *Multiplicative identity.*

   $n \cdot 1 = 1n = n$ *for all natural numbers* $n \in \mathbb{N}$.

8. *Distributivity of multiplication over addition.*

   $l\,(m+n) = lm + ln$ *and* $(l+m)\,n = ln + mn$ *for all natural numbers* $l, m, n \in \mathbb{N}$*.*

9. *Reflexivity of comparison.*

   $n \leq n$ *for all natural numbers* $n \in \mathbb{N}$*.*

10. *Antisymmetry of comparison.*

    For all natural numbers $m, n \in \mathbb{N}$*, if* $m \leq n$ *and* $n \leq m$*, then* $m = n$*.*

11. *Transitivity of comparison.*

    For all natural numbers $l, m, n \in \mathbb{N}$*, if* $l \leq m$ *and* $m \leq n$*, then* $l \leq n$*.*

12. *Totality of comparison.*

    For all natural numbers $m, n \in \mathbb{N}$*, either* $m \leq n$ *or* $n \leq m$*.*

13. *Compatibility of comparison and addition/multiplication.*

    For all natural numbers $l, m, n, p \in \mathbb{N}$*, if* $l \leq n$ *and* $m \leq p$*, then*

    $$l + m \leq n + p$$

    and

    $$lm \leq np$$

Many of these properties of the arithmetic and comparison of natural numbers also extend to the arithmetic of the other classes of numbers we will consider, namely the integers, rational numbers, real numbers, and complex numbers.

In the next sections, we will generalize the properties of addition and multiplication and the order-theoretic properties of comparison of natural numbers and arrive at fundamental notions of **finitary operations** and **relations**, **partial** and **total orders**, and the Well-ordering principle.

1.3.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.3.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.4 Finitary Operations and Relations

In this section, we will develop the notion of (finitary) operations and relations on sets. Intuitively, a **finitary operation** on a set is an assignment of an element of that set to some finite number of other elements, and a **finitary relation** on a set is a designation of certain groups of some finite number of elements as associated.

1.4.1 Products and Choice

Previously, we have used the mathematical convention of indexing to generalize the intersection and union of two sets. We may also use indexing to generalize the Cartesian product of two sets to construct products of arbitrary indexed families of sets. In this section, we will develop this **generalized Cartesian product**, and use it to introduce operations and relations on a set. Instead of containing ordered pairs of elements, our new Cartesian product consists of indexed families of elements.

Definition 1.4.1 Cartesian product. Let $(U_j)_{j\in\mathcal{J}}$ be a family of subsets $U_j \subseteq X$ of a set X, called **(Cartesian) factors**, indexed by a set $\mathcal{J}$.

1. *Cartesian product.*

   The **Cartesian product** $\prod_{j\in\mathcal{J}} U_j$ of this indexed family is the set

   $$\prod_{j\in\mathcal{J}} U_j = \left\{ (u_j)_{j\in\mathcal{J}} \in X^{\mathcal{J}} \;\middle|\; u_j \in U_j \text{ for all } j \in \mathcal{J} \right\}$$

   of families $(u_j)_{j\in\mathcal{J}}$ of elements $u_j \in X$ with the property that $u_j \in U_j$ for all indices $j \in \mathcal{J}$.

2. *Projection.*

   For each index $k \in \mathcal{J}$, **projection onto the kth factor** is the map $\pi_k \colon \prod_{j\in\mathcal{J}} U_j \to U_k$ is the map from the Cartesian product $\prod_{j\in\mathcal{J}} U_j$ to the kth factor U_k defined by the formula

   $$\pi^{U_k}_{\prod_{j\in\mathcal{J}} U_j} (u_j)_{j\in\mathcal{J}} = u_k.$$

◆

Convention 1.4.2 Empty and singleton products.

- *Empty product.*

  What does a family indexed by the empty set look like? There is a unique map from the empty set $\emptyset$ to a set X, and so there is only one such family

  $$(x_j)_{j\in\emptyset}$$

of elements x_j of X indexed by the empty set $\emptyset$. This unique indexed family is usually denoted $()$. By the above remark,

$$\prod_{j\in\emptyset} U_j = \{()\}\,.$$

- *Singleton product.*

  Similarly, a family $(x_j)_{j\in\{j\}}$ of elements $x_j \in X$ of a set S indexed by a singleton set $\{j\}$ is uniquely determined by the image x_j of the index j. Such an indexed family is denoted

  $$(x_j) = (x_j)_{j\in\{j\}}\,.$$

  We usually act under the convention $(x_j) = x_j$, so that

  $$\prod_{j\in\{j\}} U_j = U_j.$$

It may seem intuitive or obvious that the Cartesian product of an indexed family of nonempty sets is itself nonempty. After all, one can specify an indexed family of elements of these sets just by choosing an element of each nonempty set. This notion of a choice of elements of nonempty sets gives the following axiom its name.

Axiom 1.4.3 Axiom of choice. *Let $(U_j)_{j\in\mathcal{J}}$ be an indexed family of subsets $U_j \subseteq X$ of a set X. If $U_j \neq \emptyset$ for all indices $j \in \mathcal{J}$, then the Cartesian product*

$$\prod_{j\in\mathcal{J}} U_j \neq \emptyset$$

is also non-empty.

We will take the Axiom of choice as true for the remainder of this text. While seemingly too specific to be of much use, this axiom can be used to prove many useful results. In particular, we can use it to complete the characterization of invertibility given in Theorem **??** Invertibility.

Corollary 1.4.4 Invertibility. *Let $f\colon X \to Y$ be a map from a set X to a set Y.*

1. *Left-invertibility.*

   f is left-invertible if and only if f is injective or $X = \emptyset$ is empty.

2. *Right-invertibility.*

   f is right-invertible if and only if f is surjective.

3. *Invertibility.*

   f is invertible if and only if f is bijective.

Proof.

1. This follows directly from (1) and (2) of Theorem **??** Invertibility.

2. The forward implication is (3) of Theorem **??** Invertibility. Thus we may suppose that f is surjective. Then for each output $y \in Y$, the pre-image

   $$f^{-1}(y) = \{x \in X \mid f(x) = y\} \neq \emptyset$$

   is a non-empty subset of X, and so the Axiom of choice implies that the

Cartesian product

$$\prod_{y \in Y} f^{-1}(y) \neq \emptyset$$

is a non-empty subset of X^Y. Choose an element $g = (x_y)_{y \in Y} \in \prod_{y \in Y} f^{-1}(y)$, and note that for all $y \in Y$, $g(y) = x_y \in f^{-1}(y)$, so that

$$(f \circ g)(y) = f(g(y)) = y = \mathrm{id}_Y(y);$$

that is, $f \circ g = \mathrm{id}_Y$ so that g is a right inverse to f. Hence f is right-invertible.

3. This is (3) of Theorem **??** Invertibility.

■

One particularly useful kind of Cartesian product is obtained by taking all factors to be the same set. The resulting **Cartesian powers** will be used to define both finitary operations and finitary relations.

Definition 1.4.5 Cartesian power. Let X be a set and $n \in \mathbb{N}$ be a natural number. The nth **Cartesian power** X^n of X is the Cartesian product

$$X^n = \prod_{k=1}^{n} X = \{(x_k)_{k=1}^{n} \mid x_k \in X \text{ for all } 1 \leq k \leq n\}.$$

◆

We will now move on to study the Cartesian powers of sets. Maps from the Cartesian power of a set to the set itself will be called **finitary operations** on that set, and subsets of the Cartesian power of a set will be called **finitary relations** on that set.

1.4.2 Finitary Operations

Informally, a **finitary operation** on a set is an assignment of its elements to all finite sequence of a certain length in that set. That is, a finitary operation on a set is a map from some finite Cartesian power of that set to the set itself. We will view a finitary operation (through the active characterization of what maps do) as a process which combines some fixed number of elements of a set into a new element.

Definition 1.4.6 Finitary operation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary operation** of **arity** n (or more simply a n**-ary operation**) is a map $f: X^n \to X$. ◆

Example 1.4.7 Finitary operations. The following are examples of finitary operations:

- *Nullary operation.*

  A 0-ary operation (or more simply, a **nullary operation**) on a set X is just a map $f: \{\emptyset\} \to X$ from $X^0 = \{\emptyset\}$ to X. Such a map $f: \{\emptyset\} \to X$ is uniquely determined by the image $f(\emptyset) \in X$ of the unique element $\emptyset \in X^0$. In this way, we may view the nullary operations on X as synonymous with the elements of X.

- *Unary operation.*

  A 1-ary operation (or more simply, a **unary operation**) on a set X is just a map $f: X \to X$. One example of such a unary operation on the natural numbers $\mathbb{N}$ is the successor function $\mathrm{succ}: \mathbb{N} \to \mathbb{N}$.

- *Binary operation.*

  A 2-ary operation (or more simply, a **binary operation**) on a set X is a map $f\colon X^2 \to X$. Two examples of such binary operations on the natural numbers $\mathbb{N}$ are the addition and multiplication operations $\mathrm{add}, \mathrm{mult}\colon \mathbb{N}^2 \to \mathbb{N}$.

  Binary operations are usually denoted with symbols such as $+$, $\times$, $\cdot$, $*$, and $\star$. For a given binary operation $*\colon X \times X \to X$, we write

  $$x_1 * x_2 = *\,(x_1, x_2)$$

  for all elements $x_1, x_2 \in X$.

▲

In many ways, algebra is the study of finitary operations on sets. Particularly emphasized in this study is that of binary operation on sets. We won't delve too deep into the study of binary operations, but we will introduce some nice properties which some binary operations may satisfy.

Definition 1.4.8 Commutativity; associativity. Let $*\colon X^2 \to X$ be a binary operation on a set X.

1. *Commutativity.*

   $*$ is called **commutative** if

   $$x_1 * x_2 = x_2 * x_1$$

   for all elements $x_1, x_2 \in X$.

2. *Associativity.*

   $*$ is called **associative** if

   $$(x_1 * x_2) * x_3 = x_1 * (x_2 * x_3)$$

   for all elements $x_1, x_2, x_3 \in X$.

◆

Example 1.4.9 Commutative/associative binary operations. The following are examples of commutativity and associativity in binary operations:

- *Arithmetic of natural numbers.*

  Addition $+\colon \mathbb{N}^2 \to \mathbb{N}$ and multiplication $\cdot\colon \mathbb{N}^2 \to \mathbb{N}$ are commutative and associative binary operations on $\mathbb{N}$ by (1), (2), (4), and (5) of Theorem **??** Properties of Peano arithmetic.

- *Set operations.*

  Intersection $\cap\colon \mathcal{P}(X)^2 \to \mathcal{P}(X)$ and union $\cup\colon \mathcal{P}(X)^2 \to \mathcal{P}(X)$ are commutative and associative binary operations on the power set $\mathcal{P}(X)$ of a set X by (2) and (3) of Corollary **??** Properties of the set operations.

- *Composition of maps.*

  Composition $\circ\colon \left(X^X\right)^2 \to X^X$ is an associative but non-commutative binary operation on the set X^X of maps from a set X to itself by Proposition **??** Associativity of composition and Problem 1.2.5.3 Non-commutativity of composition.

- *Exponentiation of natural numbers.*

Let $e \colon \mathbb{N}^2 \to \mathbb{N}$ be the **exponentiation** function defined by the formula $e(m,n) = m^n$; that is, e is defined recursively by the formulae $e(m,0) = 1$ and $e(m,n+1) = e^{m,n} \cdot m$. Then e is a binary operation on $\mathbb{N}$, but e is neither commutative nor associative.

▲

Associative binary operations can be extended recursively to act on finite sequences of any length. For example, given a finite sequence $(n_j)_{j=1}^m$ of natural numbers $n_j \in \mathbb{N}$, we write

$$\sum_{j=1}^{m} n_j = n_1 + n_2 + \cdots + n_m \qquad \prod_{j=1}^{m} n_j = n_1 n_2 \cdots n_m.$$

As in the case of finitary operations, we will introduce finitary relations of all arities, but we will emphasize binary relations in order to study congruences and order relations, which are ubiquitous throughout both classical and modern mathematics and which will recur throughout this text.

1.4.3 Finitary Relations

A **finitary relation** on a set is just a subset of some finite Cartesian power of the set. This subset is usually interpreted as distinguishing its elements certain finite sequences of elements of the set as important in some way.

Definition 1.4.10 Finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. A **finitary relation** of **arity** n (or more simply a n**-ary relation**) is a subset $R \subseteq X^n$. ◆

We will view a finitary relation on a set as elevating specific finite sequences of elements of that set as distinguished.

Example 1.4.11 Finitary relations. The following are examples of finitary relations:

- *Unary relation.*

  In light of Convention **??** Empty and singleton products that $X^1 = X$, a 1-ary relation (or more simply, a **unary relation**) on a set X is just a subset of X.

- *Binary relation.*

  A 2-ary relation (or more simply, a **binary relation**) on a set X is a subset $R \subseteq X^2$. Two examples of such binary relations are the containment relation $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X and the standard ordering $\leq$ of the natural numbers $\mathbb{N}$.

  Just as there is a special notation for binary operations, there is a special relation for binary relations as well. Specifically, for a given binary relation $R \subseteq X^2$ on a set X, we write $x_1 \mathrel{R} x_2$ to mean that $(x_1, x_2) \in R$.

▲

Definition 1.4.12 Pullback of a finitary relation. Let X and Y be sets and $n \in \mathbb{N}$ be a natural number. The **pullback** R_f of an n-ary relation R on Y along a map $f \colon X \to Y$ is the n-ary relation on X defined by

$$R_f = \left\{ (x_k)_{k=1}^n \in X^n \mid (f(x_k))_{k=1}^n \in R \right\}.$$

◆

Example 1.4.13 Restriction of a finitary relation. Let X be a set and $n \in \mathbb{N}$ be a natural number. The **restriction** $R|_U$ of an n-ary relation R on X to a subset $U \subseteq X$ is the pullback of R along the inclusion map $\iota_U^X : U \to X$; that is,

$$\begin{aligned} R|_U = R_{\iota_U^X} &= \left\{ (u_k)_{k=1}^n \in U^n \;\middle|\; \left(\iota_U^X (u_k)\right)_{k=1}^n \in R \right\} = R \cap U^n \\ &= \{(x_k)_{k=1}^n \in X^n \mid x_1, \ldots, x_n \in U \text{ and } (x_k)_{k=1}^n \in R\} = R \cap U^n. \end{aligned}$$

▲

One particularly nice type of binary relation is the **equivalence relation**. This is a generalization of the notion of equality, which has many of the salient properties of the equality relation. We will see that an equivalence relation partitions a set into disjoint subsets known as **equivalence classes**. When we consider more sophisticated structures than just sets, we will use equivalence relations to construct **quotient** objects, as is done in algebraic settings such as group theory or ring theory.

Definition 1.4.14 Equivalence relation. Let $\equiv$ be a binary relation on a set X.

1. *Reflexivity.*

   The binary relation $\equiv$ is called **reflexive** if $x \equiv x$ for all elements $x \in X$.

2. *Symmetry.*

   The binary relation $\equiv$ is called **symmetric** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $x_2 \equiv x_1$.

3. *Transitivity.*

   The binary relation $\equiv$ is called **transitive** if for all elements $x_1, x_2, x_3 \in X$, if $x_1 \equiv x_2$ and $x_2 \equiv x_3$, then $x_1 \equiv x_3$.

A binary relation $\equiv$ is called an **equivalence relation** if it is reflexive, symmetric, and transitive. ◆

Lemma 1.4.15 Pullbacks of equivalence relations. *Let $f : X \to Y$ be a map from a set X to a set Y, and consider a binary relation $\equiv$ on Y.*

1. *Reflexivity.*

   If $\equiv$ is reflexive, then its pullback $\equiv_f$ along f is reflexive.

2. *Symmetry.*

   If $\equiv$ is symmetric, then its pullback $\equiv_f$ along f is symmetric.

3. *Transitivity.*

   If $\equiv$ is transitive, then its pullback $\equiv_f$ along f is transitive.

In particular, if $\equiv$ is an equivalence relation on Y, then its pullback $\equiv_f$ along f is an equivalence relation on X.

Proof.

1. Let $x \in X$, and note that $f(x) \equiv f(x)$ by the reflexivity of $\equiv$, and so $x \equiv_f x$; that is, $\equiv_f$ is reflexive.

2. Let $x_1, x_2 \in X$, and suppose that $x_1 \equiv_f x_2$. Then $f(x_1) \equiv f(x_2)$, and so $f(x_2) \equiv f(x_1)$ by the symmetry of $\equiv$. In particular, $x_2 \equiv_f x_1$, and so $\equiv_f$ is symmetric.

3. Let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \equiv_f x_2$ and $x_2 \equiv_f x_3$. Then $f(x_1) \equiv f(x_2)$ and $f(x_2) \equiv f(x_3)$, and so $f(x_1) \equiv f(x_3)$ by the transitivity of $\equiv$. In particular, $x_1 \equiv_f x_3$, and so $\equiv_f$ is transitive.

■

One of the most important properties of equivalence relations is that they partition sets into disjoint sets called **equivalence classes**.

Definition 1.4.16 Equivalence class; quotient set. Let $\equiv$ be an equivalence relation on a set X.

1. *Equivalence class.*

   The **equivalence class** $[x]_{\equiv}$ of an element $x \in X$ is the subset of X defined by
   $$[x]_{\equiv} = \{y \in X \mid x \equiv y\};$$
   that is, $[x]_{\equiv}$ is the set of elements which are equivalent to x under the relation $\equiv$.

2. *Quotient.*

   The **quotient** $X/{\equiv}$ of X by $\equiv$ is the set of equivalence classes; that is,
   $$X/{\equiv} = \{[x]_{\equiv} \in \mathcal{P}(X) \mid x \in X\}.$$
   The **quotient map** $\pi_{\equiv} \colon X \to X/{\equiv}$ is the map from X to the quotient $X/{\sim}$ defined by the formula $\pi_{\equiv}(x) = [x]_{\equiv}$.

   If the set X is equipped with additional structure, then this structure might also descend to the quotient $X/{\equiv}$. In settings in which this is possible, we might distinguish between the **quotient set** $X/{\equiv}$ and more structured quotient objects.

◆

As mentioned above, an equivalence relation on a given set partitions that set into pairwise disjoint equivalence classes.

Proposition 1.4.17 Equivalence partition. *The equivalence classes of an equivalence relation $\equiv$ on a set X form a* ***partition*** *of X; that is, two equivalence classes are either equal or disjoint, and*

$$X = \bigcup_{c \in X/\equiv} c.$$

Proof. Each equivalence class $c \in X/{\equiv}$ is a subset of X, and so

$$\bigcup_{c \in X/\equiv} c \subseteq X.$$

In order to show the reverse containment, let $x \in X$, and note that $x \equiv x$ by reflexivity; that is, $x \in [x]_{\equiv}$, and so

$$x \in [x]_{\equiv} = \bigcup_{c \in X/\equiv} c.$$

We conclude that

$$X = \bigcup_{c \in X/\equiv} c.$$

It remains to show that two equivalence classes are either equal or disjoint. To that end, consider elements $x_1, x_2 \in X$. First suppose that $x_1 \equiv x_2$, and let $x \in [x_2]_\equiv$. Then $x_2 \equiv x$, and so transitivity implies that $x_1 \equiv x$; that is, $x \in [x_1]_\equiv$.

In summary, we have shown that if $x_1 \equiv x_2$, then $[x_2]_\equiv \subseteq [x_1]_\equiv$. Symmetry implies that if $x_1 \equiv x_2$, then $x_2 \equiv x_1$, and so $[x_1]_\equiv \subseteq [x_2]_\equiv$; that is, if $x_1 \equiv x_2$, then $[x_1]_\equiv = [x_2]_\equiv$.

Conversely, now suppose that the equivalence classes $[x_1]_\equiv$ and $[x_2]_\equiv$ are not disjoint, so that $x \in [x_1]_\equiv \cap [x_2]_\equiv$ for some element $x \in X$. Since $x \in [x_1]_\equiv$ and $x \in [x_2]_\equiv$, $x \equiv x_1$ and $x \equiv x_2$. We conclude by symmetry and transitivity that $x_1 \equiv x_2$. ■

Equivalence relations are ubiquitous throughout mathematics, and we will work with them for the remainder of this text.

While this section has dealt with finitary operations and relations in broad generality, the vast majority of mathematics relevant to this text deals with *binary* operations and relations. In the next section, we will see another use for binary relations; that of **ordering** sets.

1.4.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

1. **Indexed Cartesian products.** Let $U_1 = \{1,2\}$, $U_2 = \{1,2,3\}$, and $U_3 = \{0,1\}$, and consider the Cartesian product $U = \prod_{k=1}^{3} U_k$.

   (a) Write down the Cartesian product $U = \prod_{k=1}^{3} U_k$ in roster notation.

   (b) For each index $k \in \{1,2,3\}$, let $\pi_k \colon U \to U_k$ be the projection map onto the kth factor. Compute $\pi_1(2,3,0)$, $\pi_2(1,1,1)$, and $\pi_3(2,2,0)$.

2. **Exponentiation of natural numbers.** Let $e\colon \mathbb{N}^2 \to \mathbb{N}$ be the **exponentiation** operation defined by the formula

$$e(m,n) = m^n;$$

   that is, e is defined recursively by the formulae $e(m,0) = 1$ and $e(m,n+1) = e^{m,n} \cdot m$.

   (a) Compute $e(1,2)$ and $e(2,1)$. Is exponentiation a commutative binary operation?

   (b) Compute $e(e(2,1),2)$ and $e(2,e(1,2))$. Is exponentiation an associative binary operation?

Equivalence classes. Let $X = \{0,1,2,3,\ldots,10\}$. Write down all the equivalence classes for the given equivalence relation on X in roster notation.

3. Consider the equivalence relation F on X defined by the condition that for all $x, y \in X$, $x \mathrel{F} y$ if and only if the English words for x and y start with the same letter. Write down the equivalence classes in roster notation.

4. Consider the equivalence relation L on X defined by the condition that for all $x, y \in X$, $x \mathrel{L} y$ if and only if the English words for x and y end with the same letter. Write down all the equivalence classes in roster notation.

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.4.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Revisiting commutativity/associativity of Cartesian products.** Recall from Problem 1.2.5.1 Non-commutativity of Cartesian products that there are sets X and Y so that $X \times Y \neq Y \times X$. Similarly, there are sets X, Y, and Z so that $(X \times Y) \times Z \neq X \times (Y \times Z)$.

(a) Prove that there is a bijection between $X \times Y$ and $Y \times X$.

(b) Prove that there is a bijection between $(X \times Y) \times Z$ and $X \times (Y \times Z)$.

While the sets $X \times Y$ and $Y \times X$ and the sets $(X \times Y) \times Z$ and $X \times (Y \times Z)$ are not equal, they may safely be identified via these bijections.

2. **Cross products.** The **cross product** $\times\colon \left(\mathbb{R}^3\right)^2 \to \mathbb{R}^3$ is the binary operation on 3-dimensional real coordinate space

$$\mathbb{R}^3 = \{(x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R}\}$$

defined by the formula

$$(x_1, x_2, x_3) \times (y_1, y_2, y_3) = (x_2 y_3 - x_3 y_2, x_3 y_1 - x_1 y_3, x_1 y_2 - x_2 y_1)$$

For this problem, you may assume all standard properties of the arithmetic of real numbers, even though we will not formally define $\mathbb{R}$ until Section **??** Constructing the Real Numbers.

(a) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is commutative. Justify your answer.

(b) Determine whether or not the cross product $\times$ on $\mathbb{R}^3$ is associative. Justify your answer.

3. **Generating equivalence relations.** Let X be a set.

(a) Prove that the intersection $R = \bigcap_{j \in \mathcal{J}} R_j$ of any indexed family $(R_j)_{j \in \mathcal{J}}$ of equivalence relations R_j on X is an equivalence relation on X.

(b) Prove that for any binary relation R on X, there is a unique equivalence relation S on X which satisfies the following conditions:

(a) S is a **refinement** of R; that is, $R \subseteq S$.

(b) S is **coarser** than any equivalence relation which **refines** R; that is, $S \subseteq T$ for any equivalence relation T on X so that $R \subseteq T$.

The unique equivalence relation on X constructed in (b) above is called the equivalence relation **generated** by R.

4. **Universal property of quotients.** Let $\equiv$ be an equivalence relation on a set X. A map $f\colon X \to Y$ from X to a set Y is called **$\equiv$-invariant** if for all elements $x_1, x_2 \in X$, if $x_1 \equiv x_2$, then $f(x_1) = f(x_2)$.

Prove that for every $\equiv$-invariant map $f\colon X \to Y$, there is a map $\overline{f}\colon X/{\equiv} \to Y$ so that $\overline{f} \circ \pi_{\equiv} = f$.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.5 Introduction to Order Theory

In this section, we will investigate another class of binary relations, which can be used to establish a notion of an **ordering** on a set. Touchstone examples will be the standard ordering $\leq$ of the natural numbers, which is a **total order**, and set containment $\subseteq$, which generally is only a **partial order**.

1.5.1 Preorders and Partial Orders

We have seen that a finitary relation on a set can be viewed as elevating its elements as distinguished finite sequences in that set. Some binary relations on a set may be interpreted in this way as establishing an **ordering**, with the first entries of each distinguished ordered pair preceding the second entries in this ordering.

What are the salient properties of the standard ordering of the natural numbers? We will see that, in order to establish a similar notion of ordering on a set, a given binary operation should satisfy similar properties. However, even if a binary operation satisfies some, but not all, of the desired properties, we will see that we can still interpret it as a weaker type of ordering relation. However, all ordering relations should, at the very least, satisfy reflexivity and transitivity. Such a reflexive and transitive binary operation is called a **preorder**.

Definition 1.5.1 Preorder. A binary relation $\precsim$ on a set X is called a **preorder** if it is reflexive and transitive; that is, $\precsim$ is a preorder if it satisfies the following conditions:

- *Reflexivity.*

  $x \precsim x$ for all elements $x \in X$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \precsim x_2$ and $x_2 \precsim x_3$, then $x_1 \precsim x_3$.

If $\precsim$ is a preorder on a set X, then the ordered pair $(X, \precsim)$ is called a **preordered set** (or a **proset**). In this case, one may refer to X as the **ground set** of the preordered set $(X, \precsim)$. ◆

Lemma 1.5.2 Preorders from pullbacks. *The pullback $\leq_f$ of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ along a function $f\colon X \to \mathbb{N}$ on a set X is a preorder on X.*

Proof. First let $x \in X$, and note that $f(x) \leq f(x)$ by (1) of Proposition **??** Comparison is a total order. In particular, $x \leq_f x$, and so $\leq_f$ is reflexive.

Now let $x_1, x_2, x_3 \in X$, and suppose that $x_1 \leq_f x_2$ and $x_2 \leq_f x_3$. Then $f(x_1) \leq f(x_2)$ and $f(x_2) \leq f(x_3)$, and so $f(x_1) \leq f(x_3)$ by (3) of Proposition **??** Comparison is a total order. In particular, $x_1 \leq_f x_3$, and so $\leq_f$ is transitive. Since $\leq_f$ is both reflexive and transitive, it is a preorder on X. ■

It turns out that, at least relative to the standard ordering $\leq$ of the natural numbers $\mathbb{N}$, the defining properties of a preorder are still too weak to be of much use in the general case. After all, an equivalence relation is just a symmetric preorder, and there is no way of meaningfully assigning a notion of an ordering based on an equivalence relation. This is because an equivalence relation is symmetric; in order to consider an ordering based on a binary operation, we would want to avoid any symmetry at all. Therefore, we must require that an ordering relation satisfy stronger assumptions.

Definition 1.5.3 Partial order. A binary relation $\preceq$ on a set X is called a **partial order** if it satisfies the following conditions:

- *Reflexivity.*

  $x \preceq x$ for all elements $x \in X$.

- *Anti-symmetry.*

  For all elements $x_1, x_2 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_1$, then $x_1 = x_2$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \preceq x_2$ and $x_2 \preceq x_3$, then $x_1 \preceq x_3$.

If $\preceq$ is a partial order on a set X, then the ordered pair $(X, \preceq)$ is called a **partially ordered set** (or a **poset**). In this case, one may refer to X as the **ground set** of the partially ordered set $(X, \preceq)$. ◆

Example 1.5.4 Partial orders. The following are examples of partial orders on sets:

- *Standard ordering on the natural numbers.*

  The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a partial order on $\mathbb{N}$ by (1), (2), and (3) of Proposition **??** Comparison is a total order.

- *Set containment.*

  Containment $\subseteq$ is a partial order on the power set $\mathcal{P}(X)$ of a set X by Definition **??** Set equality and Lemma **??** Transitivity of set containment.

▲

At the end of this section, we will explore a property of partial orders called **totality** which distinguishes between our touchstone examples of the standard ordering $\leq$ of natural numbers and the containment $\subseteq$ of sets.

1.5.2 Boundedness and Extrema

Partial orders satisfy many of the properties of the standard ordering of the natural numbers. We may use these properties for inspiration in generalizing the definitions of many of the usual notions associated with an ordering, such as upper and lower bounds and maximal and minimal elements, to preordered sets. However, we will see that anti-symmetry (and sometimes even stronger assumptions) is needed in order to recover many of these desired results.

Definition 1.5.5 Boundedness. Let $U \subseteq X$ be a subset of a preordered ordered set $(X, \precsim)$.

- *Upper bound.*

  An element $x \in X$ is called an **upper bound** for U if $u \precsim x$ for all elements $u \in U$. If such an upper bound for U exists, then U is called **bounded from above** in X. Otherwise, if no such upper bound for U exists, then U is called **unbounded from above**.

- *Lower bound.*

  An element $x \in X$ is called a **lower bound** for U if $x \precsim u$ for all elements $u \in U$. If such a lower bound for U exists, then U is called **bounded from below** in X. Otherwise, if no such upper bound for U exists, then U is called **unbounded from below**.

A subset $U \subseteq X$ is called **bounded** in X if it is both bounded from above in X and bounded from below in X, and U is called **unbounded** in X otherwise if it is either unbounded from above in X or unbounded from below in X. ◆

Lemma 1.5.6 Unique extrema. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. At most one element of U is an upper bound for U in X, and at most one element of U is a lower bound for U in X.*

Proof. Let $u_1, u_2 \in U$, and suppose first that u_1 and u_2 are upper bounds for U in X, so that $u \preceq u_1$ and $u \preceq u_2$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry.

Similarly, now suppose that u_1 and u_2 are lower bounds for U in X, so that $u_1 \preceq u$ and $u_2 \preceq u$ for all elements $u \in U$. In particular, $u_1 \preceq u_2$ and $u_2 \preceq u_1$, and so $u_1 = u_2$ by anti-symmetry. ■

We observe that the above proof of Lemma **??** Unique extrema is strongly dependent on the anti-symmetry of the partial order in question. Indeed, there is no analogous result for preordered sets.

Definition 1.5.7 Extrema. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Maximum.*

  If there is an element of U which is an upper bound for U in X, then this unique element is called the **maximum** of U and is denoted $\max(U)$.

- *Minimum.*

  If there is an element of U which is a lower bound for U in X, then this unique element is called the **minimum** of U and is denoted $\min(U)$.

Together, maxima and minima are called **extrema**. ◆

The uniqueness of extrema in partially ordered sets is one way in which partial orders resemble the standard ordering of the natural number system. However, unlike in the standard ordering of the natural number system, there is in general a distinction between extrema, which are defined above, and **extremal elements**, which are defined below.

Definition 1.5.8 Extremal element. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Maximal element.*

  An element $u \in U$ is called **maximal** in U if for all elements $v \in U$, if $u \preceq v$, then $v = u$.

- *Minimal element.*

  An element $u \in U$ is called **minimal** in U if for all elements $v \in U$, if $v \preceq u$, then $v = u$.

Together, maximal and minimal elements are called **extremal elements**. ◆

Example 1.5.9 Extrema and extremal elements. The following are examples of extrema and extremal elements:

- *Standard ordering on the natural numbers.*

  The minimum of the natural numbers $\mathbb{N}$ is $\min(\mathbb{N}) = 0$, which is the unique minimal element. However, $\mathbb{N}$ has no maximum or maximal element.

- *Set containment.*

  Consider set containment $\subseteq$ on the power set $\mathcal{P}(X)$ of a set X. $\mathcal{P}(X)$ has minimum and maximum $\min(\mathcal{P}(X)) = \emptyset$ and $\max(\mathcal{P}(X)) = X$, which are also the unique minimal and maximal elements, respectively.

- Now consider set containment $\subseteq$ on the set X of proper non-empty subsets of the set $\{1, 2, 3\}$; that is,

  $$X = \{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}.$$

  Then X has no maximum or minimum, but $\{1\}$, $\{2\}$, and $\{3\}$ are minimal elements, and $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3\}$ are maximal elements.

▲

Proposition 1.5.10 Extrema are extremal elements. *Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$. If U has a maximum, then $\max(U)$ is the unique maximal element of U. If U has a minimum, then $\min(U)$ is the unique minimal element of U.*

Proof. First suppose that U has a maximum. Let $u \in U$, and suppose that $\max(U) \preceq u$. Of course $u \preceq \max(U)$, and so anti-symmetry implies that $u = \max(U)$; that is, $\max(U)$ is maximal in U.

On the other hand, now consider a maximal element $u \in U$. Since $u \preceq \max(U)$, the maximality of u implies that $u = \max(U)$.

Now suppose that U has a minimum. Let $u \in U$, and suppose that $u \preceq \min(U)$. Of course $\min(U) \preceq u$, and so anti-symmetry implies that $u = \min(U)$; that is, $\min(U)$ is minimal in U.

On the other hand, now consider a minimal element $u \in U$. Since $\min(U) \preceq u$, the minimality of u implies that $u = \min(U)$. ■

The distinction between extrema and extremal elements is one way in which more general partial orders differ from the standard ordering of the natural numbers. However, this subtlety does not arise in all partial orders. We will shortly explore a class of partial orders called **total orders** for which the aforementioned distinction between extrema and extremal elements does not exist.

1.5.3 Comparability and Total Orders

The fact that the converse of Proposition **??** Extrema are extremal elements does not hold in general stems from the fact that in a partially ordered set, not all pairs of elements are comparable. That is, given two elements of a partially ordered set, it is not necessarily true that one will precede the other.

Definition 1.5.11 Comparability. Two elements $x_1, x_2 \in X$ in a partially ordered set $(X, \preceq)$ are called **comparable** if either $x_1 \preceq x_2$ or $x_2 \preceq x_1$, and **incomparable** otherwise. ◆

Any two natural numbers are comparable in the standard ordering. However, this is not true for partially ordered sets in general. As an example, consider the subsets $S = \{0, 1\}$ and $T = \{1, 2\}$ of the natural numbers $\mathbb{N}$. One can check that $S \not\subseteq T$ and $T \not\subseteq S$, so that S and T are incomparable in the power set $\mathcal{P}(\mathbb{N})$ of the natural numbers $\mathbb{N}$. So partially ordered sets may have elements which are incomparable. This is the reason why partial orders are called "partial"; partial orders in which all elements are comparable are called **total**.

Definition 1.5.12 Total order. A binary relation $\leq$ on a set X is called a **total order** if it is anti-symmetric and transitive and each pair of elements is comparable; that is, $\leq$ is a total order if it satisfies the following conditions:

- *Antisymmetry.*

  For all elements $x_1, x_2 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_1$, then $x_1 = x_2$.

- *Transitivity.*

  For all elements $x_1, x_2, x_3 \in X$, if $x_1 \leq x_2$ and $x_2 \leq x_3$, then $x_1 \leq x_3$.

- *Totality.*

  For all elements $x_1, x_2 \in X$, either $x_1 \leq x_2$ or $x_2 \leq x_1$.

If $\leq$ is a total order on a set X, then the ordered pair $(X, \leq)$ is called a **totally ordered set**. ◆

Example 1.5.13 Total order. The standard ordering $\leq$ on the natural numbers $\mathbb{N}$ is a total order by (2), (3), and (4) of Proposition **??** Comparison is a total order. ▲

Lemma 1.5.14 Total orders are partial orders. *Every total order on a set X is a partial order on X.*

Proof. Let $\leq$ be a total order on a set X. Since $\leq$ is antisymmetric and transitive, it remains to show that $\leq$ is reflexive. To that end, let $x \in X$, and note that totality implies that either $x \leq x$ or $x \leq x$; that is $\leq$ is reflexive (and hence a partial order on X). ■

The careful reader might object that the above definition of total orders neglected to require reflexivity. In fact, this property follows from the other properties of total orders.

Proposition 1.5.15 Extremal elements are extrema in totally ordered sets. *Let $\leq$ be a total order on a set X.*

1. *If X contains a maximal element $e \in X$, then $e = \max(X)$.*

2. *If X contains a minimal element $e \in X$, then $e = \min(X)$.*

Proof.

1. Suppose that $e \in X$ is a maximal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $e \leq x$, then maximality implies that $x = e$, so that $x \leq e$ by reflexivity. Thus $e \leq x$ for all $x \in X$, and so $\max(X) = e$.

2. Now suppose that $e \in X$ is a minimal element, and let $x \in X$. Totality implies that $x \leq e$ or $e \leq x$. If $x \leq e$, then minimality implies that $x = e$, so that $e \leq x$ by reflexivity. Thus $x \leq e$ for all $x \in X$, and so $\min(X) = e$.

■

We end with an important property of the standard ordering $\leq$ on the natural numbers $\mathbb{N}$ related to the axiom of induction.

Theorem 1.5.16 Well-ordering principle. *Every non-empty subset of the natural numbers $\mathbb{N}$ has a minimum.*

Proof. Let $U \subseteq \mathbb{N}$ be a subset of the natural numbers $\mathbb{N}$, and suppose that U does not have a minimum. We proceed by strong induction on $n \in \mathbb{N}$ to show that $n \notin U$.

Base case. Suppose that $n = 0$, and suppose for a contradiction that $n \in U$. Since $0 \leq n$ for all natural numbers $n \in \mathbb{N}$ by Lemma **??** 0 is minimal and (2) of Proposition **??** Extremal elements are extrema in totally ordered sets, then $n = \min(U)$. This contradiction implies that $n \notin U$.

Inductive step. Suppose for the inductive hypothesis that there is some natural number $n \in \mathbb{N}$ so that $k \notin U$ for all natural numbers $k \in \mathbb{N}$ so that $0 \leq k \leq n$. Suppose for a contradiction that $n + 1 \in U$, and let $u \in U$. Totality implies that either $u \leq n$ or $n \leq u$. The inductive hypothesis implies that $u \not\leq n$, so that $n \leq u$.

So $u = n + p$ for some natural number $p \in \mathbb{N}$. If $p = 0$, then

$$n = n + 0 = n + p = u \in U,$$

and so $p \neq 0$. Thus $p = q + 1$ for some natural number $q \in \mathbb{N}$, and so

$$n + 1 \leq n + q + 1 = n + p = u.$$

In summary, we have shown that $n+1 \leq u$ for all $u \in U$, so that $\min(U) = n+1$. This contradiction implies that $n + 1 \notin U$.

We conclude that $n \notin U$ for all natural numbers $n \in \mathbb{N}$, and so $U = \emptyset$. ■

The Well-ordering principle is sometimes taken to be an axiom and sometimes derived from the axiom of induction as in the above proof. In fact, these results are logically equivalent, as one can derive the axiom of induction from the Well-ordering principle.

We will revisit ordering relations in Chapter **??** Constructing the Number Systems, when we extend the standard ordering on the natural numbers to many of the extensions of the natural number system which we construct.

1.5.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Identifying partial and total orders. Determine whether or not the given preorder is a partial order. If the preorder is a partial order, determine whether or not it is total.

1. Let R be the preorder on the set $\{1, 2, 3, 4\}$ defined by

$$\{(4,2), (4,3), (2,1), (3,1), (4,1), (1,3), (2,3), (4,4), (3,3), (2,2), (1,1)\}.$$

Determine whether or not R is a partial order on $\{1, 2, 3, 4\}$. If so, determine whether or not it is total.

2. Let $\precsim$ be the preorder on the set P of all people (living or dead) defined so that for all people $p, q \in P$, $p \precsim q$ if and only if either p is either q or an ancestor of q. Determine whether or not $\precsim$ is a partial order on P. If so, determine whether or not it is total.

3. Let $\precsim$ be the preorder on the set C of all circles in the Cartesian plane so that for all circles $c, d \in C$, $c \precsim d$ if and only if the diameter of c is at most the diameter of d. Determine whether or not $\precsim$ is a partial order on C. If so, determine whether or not it is total.

Bounds. Determine whether or not the following subsets of the partially ordered set $\mathcal{P}(\mathbb{N})$ are bounds for the given element.

4. Which of the following are upper bounds for the set $\{1, 2, 3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

5. Which of the following are lower bounds for the set $\{1, 2, 3\}$ in the partially ordered set $\mathcal{P}(\mathbb{N})$?

A. $\emptyset$

B. $\{1, 2\}$

C. $\{1, 2, 3\}$

D. $\mathbb{N} \setminus \{1, 2\}$

E. $\mathbb{N} \setminus \{0\}$

F. $\mathbb{N}$

Boundedness. Determine the boundedness of the following subsets of the totally ordered set $\mathbb{N}$.

6. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from below?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

7. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded from above?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

8. Which of the following subsets of the totally ordered set $\mathbb{N}$ are bounded?

A. $\emptyset$

B. $\{1, 2\}$

C. $\mathbb{N} \setminus \{1, 2\}$

D. $\mathbb{N} \setminus \{0\}$

E. $\mathbb{N}$

Short answers and/or in-depth solutions to these exercises are available. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.5.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Order by divisibility.** Consider the binary relation $\mid$ on the natural numbers $\mathbb{N}$ defined so that for all natural numbers $m, n \in \mathbb{N}$, $m \mid n$ if and only if $mk = n$ for some natural number $k \in \mathbb{N}$; that is, $m \mid n$ if and only if n is a multiple of m.

(a) Prove that $\mid$ is a partial order on the set $\mathbb{N}$ of natural numbers.

(b) Is the binary relation $\mid$ a total order on $\mathbb{N}$? Justify your answer.

(c) What are the extrema of the partially ordered set $(\mathbb{N}, |)$?

(d) We will abuse notation and denote the restriction of $|$ to the set $\mathbb{N} \setminus \{0,1\}$ by $|$ as well. What are the extremal elements of the partially ordered set $(\mathbb{N} \setminus \{0,1\}, |)$? Are these extremal elements extrema?

2. **Extremal subsets.** Find the extremal elements of the partially ordered set $\mathcal{P}(\mathbb{N}) \setminus \{\emptyset, \mathbb{N}\}$.

3. **Opposite of an order.** The **opposite** R^{op} of a binary relation R on a set X is the binary relation defined by

$$R^{\mathrm{op}} = \left\{(x_2, x_1) \in X^2 \,\middle|\, (x_1, x_2) \in R\right\}.$$

(a) Prove that the opposite of a preorder is a preorder.

(b) Prove that the opposite of a partial order is a partial order.

(c) Prove that the opposite of a total order is a total order.

4. **Orders from pullbacks.**

(a) Prove that the pullback of a preorder is a preorder.

Hint. You may find it useful to review the proof of Lemma **??** Preorders from pullbacks

(b) Prove that the pullback $\preceq_f$ of a partial order $\preceq$ on a set Y along a map $f\colon X \to Y$ is a partial order on X if and only if f is injective.

(c) Is the pullback of a total order along an injective map necessarily a total order? If so, prove it. If not, provide a counterexample.

In-depth solutions to these problems are available. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

1.6 Cardinality and Countability [SKIP]

In this section, we define a notion of the size of a set called **cardinality**. The cardinality of a set is a generalization of the number of elements of a set which contains only finitely many elements based on the notion of bijections. First, we will develop these notions for finite sets and then extend them to infinite sets. At first glance, it may seem that there is essentially only one kind of infinite set. We will show that this is not the case; in fact, there are infinitely many "sizes" of infinite sets with respect to this notion of cardinality.

1.6.1 Finiteness

Before we brave the unintuitive mess which is the realm of infinite sets, we will rigorously develop a notion of size for sets which contain only finitely many elements. Like in the infinite case, these finite cardinalities will be based on bijections.

Proposition 1.6.1 Injections of finite sets. *For any natural numbers $m, n \in \mathbb{N}$, there exists an injective map $\{1, \ldots, m\} \to \{1, \ldots, n\}$ if and only if $m \leq n$.*

Proof. If $m \leq n$, then $\{1, \ldots, m\} \subseteq \{1, \ldots, n\}$, and so the inclusion map $\iota_{\{1,\ldots,m\}}^{\{1,\ldots,n\}} \colon \{1, \ldots, m\} \to \{1, \ldots, n\}$ is an injective map from $\{1, \ldots, m\}$ to $\{1, \ldots, n\}$. So it suffices to show the reverse implication.

To that end, consider the subset

$$S = \{m \in \mathbb{N} \mid \text{There are } n \in \mathbb{N} \text{ and } f\colon \{1, \ldots, m\} \to \{1, \ldots, n\} \text{ so that } m \not\leq n \text{ and } f \text{ is injective}\}$$

of the natural numbers $\mathbb{N}$. If suffices to show that $S = \emptyset$. To that end, we first note that $0 \leq n$ for all natural numbers $n \in \mathbb{N}$, so that $0 \notin S$.

Now let $m \in \mathbb{N}$ be a natural number, and suppose that $m + 1 \in S$. Then there is a natural number $n \in \mathbb{N}$ so that $m + 1 \not\leq n$ and an injective function $f\colon \{1, \ldots, m+1\} \to \{1, \ldots, n\}$. We note that $n \neq 0$, since there are no functions from a non-empty set $\{1, \ldots, m+1\}$ to the empty set $\emptyset$. Thus $n = p + 1$ for some natural number $p \in \mathbb{N}$.

If $m \leq p$, then $m + 1 \leq p + 1 = n$, and so $m \not\leq p$. It therefore suffices to construct an injective function $\{1, \ldots, m\} \to \{1, \ldots, p\}$. To that end, fix a bijection $g\colon \{1, \ldots, p+1\} \to \{1, \ldots, p+1\}$ so that $g\left(f\left(m+1\right)\right) = p + 1$, and let $h\colon \{1, \ldots, m\} \to \{1, \ldots, p\}$ be the function defined by the formula

$$h\left(j\right) = g\left(f\left(j\right)\right).$$

Since g and f are injective, h is injective by (a) of `[provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections]`. We conclude that $n \in S$.

In summary, we have shown that $0 \notin S$ and for all natural numbers $n \in \mathbb{N}$, if $n + 1 \in S$, then $n \in S$. These conclusions are equivalent to $0 \in \mathbb{N} \setminus S$ and for all natural numbers $n \in \mathbb{N}$, if $n \in \mathbb{N} \setminus S$, then $n + 1 \in \mathbb{N} \setminus S$. We conclude that $\mathbb{N} \setminus S = \mathbb{N}$ by the axiom of induction, and so $S = \emptyset$. ■

Corollary 1.6.2 Finite cardinality is well-defined. *For any set X, if there are natural numbers $m, n \in \mathbb{N}$ and bijections $f\colon \{1, \ldots, m\} \to X$ and $g\colon \{1, \ldots, n\} \to X$, then $m = n$.*

Proof. Under the given assumptions, $g^{-1} \circ f\colon \{1, \ldots, m\} \to \{1, \ldots, n\}$ and $f^{-1} \circ g\colon \{1, \ldots, n\} \to \{1, \ldots, m\}$ are injective functions by (a) of `[provisional cross-reference: problem-compositions-of-injections-surjections-and-bijections]`. Proposition **??** Injections of finite sets now implies that $m \leq n$ and $n \leq m$, and so $m = n$ by (2) of Proposition **??** Comparison is a total order. ■

Definition 1.6.3 Finiteness; cardinality. A set X is called **finite** if there is a natural number $n \in \mathbb{N}$ and a bijection $\{1, \ldots, n\} \to X$, in which case n is uniquely defined by this property and is called the **cardinality** of X and denoted $|X| = n$. A set X is called **infinite** if no such number and bijection exist. ◆

Proposition 1.6.4 Maps and cardinality. *Let $f\colon X \to Y$ be a map from a set X to a set Y.*

1. *If f is injective and Y is finite, then X is finite and $|X| \leq |Y|$.*

2. *If f is surjective and X is finite, then Y is finite and $|Y| \leq |X|$.*

3. *If f is bijective and either X or Y is finite, then both X and Y are finite and $|X| = |Y|$.*

Proof. ■

Text after the last subdivision.

1.6.2 Cardinal Comparison

Text before the first subdivision.

Text after the last subdivision.

1.6.3 Countability

Text before the first subdivision.

Text after the last subdivision.

Text after the last subsection.

1.6.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

1.6.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these reading questions are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

Armed with a solid understanding of the theory of sets, maps, and relations, we now have much of the necessary background to begin studying real analysis. First, however, we will construct the standard number systems (with a focus on the real numbers $\mathbb{R}$) as extensions of the natural numbers $\mathbb{N}$ and catalog their properties.

Chapter 2

Constructing the Number Systems

In this section, we provide rigorous constructions of well-known number systems including the **integers** $\mathbb{Z}$, the **rational numbers** $\mathbb{Q}$, the **real numbers** $\mathbb{R}$, and the **complex numbers** $\mathbb{C}$. These number systems, particularly the real and complex number systems, provide the backdrop for much of the discipline of analysis.

2.1 The Integers and the Rational Numbers [SKIP]

Text before the first subsection.

Text after the last subsection.

Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

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Problems

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less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

2.2 Constructing the Real Numbers

In this section, we construct the **real number system** $\mathbb{R}$ by **Dedekind cuts**, and we explore the resultant properties of arithmetic and comparison. However, we will find the existence of such a formal construction and its properties more useful than the particular details of the construction itself, and we will shortly leave behind the perspective suggested by our construction.

2.2.1 Cutting the Rational Numbers

So far, our progression through the standard number systems $\mathbb{N} \subsetneq \mathbb{Z} \subsetneq \mathbb{Q}$ has proceeded in order of containment. Every time we have expanded our perspective to consider larger number systems, it has resulted in only the acquisition of new desirable properties. This is an artefact of our focus on the analytical rather than say the number-theoretical; there are for example incredibly interesting properties of the natural numbers $\mathbb{N}$ and the integers $\mathbb{Z}$ which are lost in the jump to the rational numbers $\mathbb{Q}$.

However, we have yet to see any deficiencies with the rational number system $\mathbb{Q}$ which would motivate seeking an extension. These deficiencies exist, and we will see in Section **??** The Complete Ordered Field and later in Section **??** Completeness and Compactness the benefits in working in the **real numbers** $\mathbb{R}$. As a brief example, we note that $\mathbb{Q}$ can be written as a particular type of disjoint union $\mathbb{Q} = X \sqcup Y$, where

$$X = \left\{ q \in \mathbb{Q} \,\middle|\, q < 0 \text{ or } q^2 < 2 \right\}$$
$$Y = \left\{ q \in \mathbb{Q} \,\middle|\, q > 0 \text{ and } q^2 > 2 \right\}.$$

are defined by strict inequalities. This decomposition of $\mathbb{Q}$ is an example of a **Dedekind cut**, and we will see that such a decomposition of $\mathbb{R}$ cannot exist.

Definition 2.2.1 Dedekind cut. A proper, non-empty subset $\emptyset \subsetneq X \subsetneq \mathbb{Q}$ of the rational numbers $\mathbb{Q}$ is called a **Dedekind cut** if it satisfies the following conditions:

1. For all rational numbers $q_1, q_2 \in \mathbb{Q}$, if $q_1 \leq q_2$ and $q_2 \in X$, then $q_1 \in X$.

2. X does not have a maximum; that is, for all elements $q_1 \in X$, there is an element $q_2 \in X$ so that $q_1 < q_2$.

The set of Dedekind cuts is denoted $\mathbb{R}$. ◆

Example 2.2.2 Dedekind cuts. For each rational number $q \in \mathbb{Q}$, consider the subset

$$X_q = \{p \in \mathbb{Q} \mid p < q\}$$

of the rational numbers $\mathbb{Q}$. We observe that X_q is a proper and non-empty subset of $\mathbb{Q}$, since it contains $q - 1$ but not q itself. Moreover, for all rational numbers $p_1, p_2 \in \mathbb{Q}$, if $p_1 \leq p_2$ and $p_2 \in X_q$, then

$$p_1 \leq p_2 < q,$$

and so $p_1 \in X_q$. Finally, for all elements $p_1 \in X$, we observe that

$$p < p + \frac{q-p}{2} < p + (q-p) = q,$$

so that $p + \frac{q-p}{2} \in X_q$. In summary, we have shown that associated to each rational number $q \in \mathbb{Q}$ is a Dedekind cut X_q. ▲

We will see that the above construction realizes the rational numbers $\mathbb{Q}$ as a subset of the set of Dedekind cuts. The Dedekind cuts which arise in this manner will be called **rational**. We will shortly see that not every Dedekind cut is rational.

Lemma 2.2.3 Rational Dedekind cuts. *A Dedekind cut X is rational if and only if its complement $\mathbb{Q} \setminus X$ has a minimum.*

Proof. Let X be a Dedekind cut, and suppose that $X = X_q$ for some rational number $q \in \mathbb{Q}$. Note that $X = X_q$ has complement

$$\mathbb{Q} \setminus X = \mathbb{Q} \setminus X_q = \{p \in \mathbb{Q} \mid p \notin X_q\} = \{p \in \mathbb{Q} \mid p \geq q\}.$$

This set contains q, since $q \geq q$. Moreover, $p \geq q$ for all elements $p \in \mathbb{Q} \setminus X_q$. Thus $\min(\mathbb{Q} \setminus X) = q$.

Conversely, now suppose that $\mathbb{Q} \setminus X$ has a minimum element, say $q = \min(\mathbb{Q} \setminus X) \in \mathbb{Q}$. Let $p \in \mathbb{Q}$, and suppose that $p \geq q$. If $p \in X$, then that would imply that $q \in X$. This contradiction implies that $p \notin X$. On the other hand, if $p < q$, then $p \notin \mathbb{Q} \setminus X$, since $\min(\mathbb{Q} \setminus X) = q$. Thus

$$X = \{p \in \mathbb{Q} \mid p < q\} = X_q$$

is rational. ■

Proposition 2.2.4 Irrational Dedekind cuts. *Not all Dedekind cuts are rational.*

Proof. Consider the subset

$$X = \left\{q \in \mathbb{Q} \,\middle|\, q < 0 \text{ or } q^2 < 2\right\}$$

of the rational numbers $\mathbb{Q}$. We observe that X is a non-empty and proper subset of $\mathbb{Q}$, since it contains 1 but not 2. Let $p_1, p_2 \in \mathbb{Q}$ be rational numbers, and suppose that $p_1 \leq p_2$ and $p_2 \in X$. If $p_2 < 0$, then

$$p_1 \leq p_2 < 0,$$

and so $p_1 \in X$. On the other hand, if $p_2^2 < 2$, then either $p_1 < 0$ or $0 \leq p_1 \leq p_2$, in which case $p_1^2 \leq p_2^2 < 2$. In either case, we conclude that $p_1 \in X$.

Finally, let $q \in X$. If $q \leq 0$, then let $p = 1$, and note that $p \in X$ and $q < p$. On the other hand, suppose that $q > 0$ and $q^2 < 2$, and consider the rational number

$$p = \frac{2q+2}{2+q} \in \mathbb{Q}.$$

Note that

$$p = \frac{2q+2}{2+q} = \frac{2q+q^2}{2+q} + \frac{2-q^2}{2+q} = q + \frac{2-q^2}{2+q} > q,$$

and

$$\begin{aligned}
2 - p^2 &= \frac{2\left(2+q\right)^2}{\left(2+q\right)^2} - \frac{\left(2q+2\right)^2}{\left(2+q\right)^2} \\
&= \frac{4-2q^2}{\left(2+q\right)^2} = \frac{2\left(2-q^2\right)}{\left(2+q\right)^2} > 0,
\end{aligned}$$

so that $p^2 < 2$; that is, $p \in X$. We conclude that X is a Dedekind cut, and so it suffices to show that X is not rational.

To that end, we note that X has complement

$$\mathbb{Q} \setminus X = \left\{ q \in \mathbb{Q} \,\middle|\, q \geq 0 \text{ and } q^2 \geq 2 \right\}.$$

In fact, since $q^2 \neq 2$ for all rational numbers $q \in \mathbb{Q}$ (this fact is proven in any introductory number theory course),

$$\mathbb{Q} \setminus X = \left\{ q \in \mathbb{Q} \,\middle|\, q \geq 0 \text{ and } q^2 > 2 \right\}.$$

We will show that $\mathbb{Q} \setminus X$ has no minimum element. Indeed, let $q \in \mathbb{Q} \setminus X$, so that $q \geq 0$ and $q^2 > 2$. We again consider the rational number

$$p = \frac{2q+2}{2+q} \in \mathbb{Q}.$$

We see that $p \geq 0$. Moreover, now that $q^2 > 2$, we observe that

$$p = \frac{2q+2}{2+q} = \frac{2q+q^2}{2+q} + \frac{2-q^2}{2+q} = q + \frac{2-q^2}{2+q} < q$$

and

$$\begin{aligned}
2 - p^2 &= \frac{2\left(2+q\right)^2}{\left(2+q\right)^2} - \frac{\left(2q+2\right)^2}{\left(2+q\right)^2} \\
&= \frac{4-2q^2}{\left(2+q\right)^2} = \frac{2\left(2-q^2\right)}{\left(2+q\right)^2} < 0,
\end{aligned}$$

so that $p^2 > 2$. In particular, $p < q$ and $p \in \mathbb{Q} \setminus X$. Thus $\mathbb{Q} \setminus X$ has no minimum element, and so Lemma **??** Rational Dedekind cuts implies that X is not rational. ■

Once we define the arithmetic of Dedekind cuts, we will see that the Dedekind cut

$$X = \left\{ q \in \mathbb{Q} \,\middle|\, q < 0 \text{ or } q^2 < 2 \right\}$$

from the above proof of Proposition **??** Irrational Dedekind cuts represents the **irrational** real number $\sqrt{2}$. First, however, we observe that the rational numbers $\mathbb{Q}$ inject into $\mathbb{R}$.

Proposition 2.2.5 Rational numbers as real numbers. *The map* $\iota\colon \mathbb{Q} \to \mathbb{R}$ *defined by the formula* $\iota\left(q\right) = X_q$ *is injective.*

Proof. Let $q_1, q_2 \in \mathbb{Q}$ be rational numbers, and suppose that $\iota\left(q_1\right) = \iota\left(q_2\right)$. As shown in the proof of Lemma **??** Rational Dedekind cuts, $\min\left(X_q\right) = q$ for all rational numbers $q \in \mathbb{Q}$. Therefore,

$$\begin{aligned}
q_1 &= \min\left(X_{q_1}\right) = \min\left(\iota\left(q_1\right)\right) \\
&= \min\left(\iota\left(q_2\right)\right) = \min\left(X_{q_2}\right) = q_2.
\end{aligned}$$

Thus $\iota\colon \mathbb{Q} \to \mathbb{R}$ is injective. ■

Going forward, we will often exploit the injection $\iota\colon \mathbb{Q} \to \mathbb{R}$ to pretend that $\mathbb{Q} \subseteq \mathbb{R}$ by identifying a rational number with its associated rational Dedekind cut. We will now construct a total order $\leq$ on $\mathbb{R}$ which extends the standard order on $\mathbb{Q}$.

2.2.2 Comparison of Dedekind Cuts

Our two perspectives on the set $\mathbb{Q}$ (as equivalence classes of fractions and as rational Dedekind cuts) yield two natural orderings: the standard ordering $\leq$ and the partial order of set containment. In fact, these orderings agree.

Corollary 2.2.6 Ordering Dedekind cuts by containment. *The standard ordering $\leq$ on the rational numbers $\mathbb{Q}$ is the pullback of set containment $\subseteq$ on the set $\mathbb{R}$ of Dedekind cuts.*

Proof. Let $q_1, q_2 \in \mathbb{Q}$ be rational numbers. We note that

$$\begin{aligned} X_{q_1} &= \{p \in \mathbb{Q} \mid p < q_1\} \\ X_{q_2} &= \{p \in \mathbb{Q} \mid p < q_2\}. \end{aligned}$$

First suppose that $q_1 \leq q_2$. If $p \in X_{q_1}$, then

$$p < q_1 \leq q_2$$

and so $p \in X_{q_2}$; that is,

$$\iota(q_1) = X_{q_1} \subseteq X_{q_2} = \iota(q_2),$$

so that $q_1 \subseteq_\iota q_2$.

Conversely, now suppose that $q_1 \subseteq_\iota q_2$. Then

$$X_{q_1} = \iota(q_1) \subseteq \iota(q_2) = X_{q_2}.$$

If $q_1 > q_2$, then

$$q_2 < \frac{q_1 + q_2}{2} < q_1,$$

and so $\frac{q_1+q_2}{2} \in X_{q_1} \setminus X_{q_2}$. This would imply that $X_{q_1} \not\subseteq X_{q_2}$, and so we must have $q_1 \leq q_2$.

In summary, we have shown that $q_1 \leq q_2$ if and only if $q_1 \subseteq_\iota q_2$, and so $\leq$ and $\subseteq_\iota$ are the same binary relation on $\mathbb{Q}$. ■

Corollary **??** Ordering Dedekind cuts by containment implies that comparing rational numbers is the same as comparing their associated rational Dedekind cuts by containment. This gives us the template for how to compare Dedekind cuts in general: we compare by set containment.

Theorem 2.2.7 Comparison of Dedekind cuts. *Set containment $\subseteq$ is a total order on the set $\mathbb{R}$ of Dedekind cuts.*

Proof. Since set containment is a partial order, it suffices to show only totality. To that end, let $X, Y \in \mathbb{R}$ be Dedekind cuts, and suppose that $X \not\subseteq Y$. Then there is some rational number $q \in X$ so that $q \notin Y$.

Let $p \in Y$. If $q \leq p$, then we would have $q \in Y$, and so we must have $p \leq q$. But this implies that $p \in X$, and so $Y \subseteq X$.

In summary, we have shown that for all Dedekind cuts $X, Y \in \mathbb{R}$, either $X \subseteq Y$ or $Y \subseteq X$. Thus set containment $\subseteq$ is a total order on the set $\mathbb{R}$ of Dedekind cuts. ■

Just as with the integers $\mathbb{Z}$ and rational numbers $\mathbb{Q}$, we can use the standard ordering on $\mathbb{R}$ to distinguish Dedekind cuts by their **sign**: either **positive**, **negative**, or **zero**.

Definition 2.2.8 Sign of a Dedekind cut. Recall that

$$X_0 = \{q \in \mathbb{Q} \mid q < 0\}$$

is the rational Dedekind cut associated to the rational number 0.

- *Positivity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **positive** if it strictly contains X_0; that is, X is positive if $X \supsetneq X_0$.

- *Negativity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **negative** if it is strictly contained in X_0; that is, X is negative if $X \subsetneq X_0$.

- *Non-negativity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **non-negative** if it contains X_0; that is, X is non-negative if $X \supseteq X_0$.

- *Non-positivity.*

  A Dedekind cut $X \in \mathbb{R}$ is called **non-positive** if it is contained in X_0; that is, X is positive if $X \subseteq X_0$.

In particular, the **sign** of a Dedekind cut is either positive, negative, or zero. ◆

This classification of Dedekind cuts by sign will be useful in formalizing the multiplication of Dedekind cuts. We now proceed in that direction, extending the usual operations of arithmetic on $\mathbb{Q}$ to binary operations on $\mathbb{R}$ and determining their properties.

2.2.3 Addition of Dedekind Cuts

Finally, we begin to define arithmetic for Dedekind cuts. We emphasize that the utility of what follows is neither the details of the construction nor the methods of proof, but rather the very fact that the operations exist with the same properties we would expect. That is to say, we will eventually lose sight of the particularities of the construction of the real number system by Dedekind cuts and treat real numbers as objects which obey the rules of arithmetic.

Lemma 2.2.9 Addition of Dedekind cuts. *Given two Dedekind cuts $X, Y \in \mathbb{R}$, the set*

$$\{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}$$

is a Dedekind cut.

Proof. Denote by Z the set

$$Z = \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}.$$

Since X and Y are Dedekind cuts, they are non-empty subsets of $\mathbb{Q}$, and so there are rational numbers $p \in X$ and $q \in Y$. Thus $p + q \in Z$, and so Z is also non-empty.

On the other hand, X and Y are proper subsets of $\mathbb{Q}$, and so there are rational numbers $r \in \mathbb{Q} \setminus X$ and $s \in \mathbb{Q} \setminus Y$. We observe that that $r > p$ for all

$p \in X$ and $s > q$ for all $q \in Y$. In particular, for all $t \in Z$ there are rational numbers $p \in X$ and $q \in Y$ so that $t = p + q$, and so

$$r + s > p + q = t$$

In particular, $r + s \notin Z$, and so Z is also a proper subset of $\mathbb{Q}$.

Now let $r, s \in \mathbb{Q}$ be rational numbers, and suppose that $s \leq r$ and $r \in Z$. We wish to show that $s \in Z$. To that end, we see that $r = p + q$ for some rational numbers $p \in X$ and $q \in Y$. Since $s - r \leq 0$,

$$q + (s - r) \leq q + 0 = q \in Y,$$

and so $q + (s - r) \in Y$. This implies that

$$s = r + (s - r) = p + (q + (s - r)) \in Z.$$

It remains to show that Z does not have a maximum. To that end, let $r_1 \in Z$, so that $r_1 = p_1 + q_1$ for some rational numbers $p_1 \in X$ and $q_1 \in Y$. Since X and Y are Dedekind cuts, they do not have maxima, and so $p_1 < p_2$ and $q_1 < q_2$ for some rational numbers $p_2 \in X$ and $q_2 \in Y$. We conclude that $r_2 = p_2 + q_2 \in Z$, and that

$$r_1 = p_1 + q_1 < p_2 + q_2 = r_2.$$

Thus Z does not have a maximum, and so is a Dedekind cut. ■

Definition 2.2.10 Addition of Dedekind cuts. The **sum** $X + Y$ of two Dedekind cuts $X, Y \in \mathbb{R}$ is the Dedekind cut defined by

$$X + Y = \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\}.$$

This defines a binary operation $+$ on the set $\mathbb{R}$ of Dedekind cuts. ◆

We first observe that addition of rational Dedekind cuts mirrors the addition of the corresponding rational numbers.

Lemma 2.2.11 Addition of rational Dedekind cuts. $X_p + X_q = X_{p+q}$ *for all rational numbers* $p, q \in \mathbb{Q}$.

Proof. Let $r \in \mathbb{Q}$ be a rational number, and suppose first that $r \in X_p + X_q$. Then $r = s + t$ for some rational numbers $s \in X_p$ and $t \in X_q$, so that $s < p$ and $t < q$. We observe that

$$r = s + t < p + q,$$

and so $r \in X_{p+q}$.

Conversely, now suppose that $r \in X_{p+q}$, so that $r < p + q$. Consider the rational numbers

$$s = p + \frac{r - (p + q)}{2} \qquad t = q + \frac{r - (p + q)}{2}.$$

Since $\left(\frac{r-(p+q)}{2}\right) < 0$, $s < p$ and $t < q$, and so $s \in X_p$ and $t \in X_q$. We conclude that

$$\begin{aligned} r &= p + q + (r - (p + q)) = p + \frac{r - (p + q)}{2} + q + \frac{r - (p + q)}{2} \\ &= s + t \in X_p + X_q. \end{aligned}$$

In summary, we have shown that $r \in X_p + X_q$ if and only if $r \in X_{p+q}$, and so $X_p + X_q = X_{p+q}$. ■

Proposition 2.2.12 Addition of Dedekind cuts. *Addition $+$ of Dedekind cuts has the following properties:*

1. *Commutativity.*

   $X + Y = Y + X$ *for all Dedekind cuts* $X, Y \in \mathbb{R}$*.*

2. *Associativity.*

   $(X + Y) + Z = X + (Y + Z)$ *for all Dedekind cuts* $X, Y, Z \in \mathbb{R}$*.*

3. *Identity element.*

   The rational Dedekind cut X_0 associated to 0 is an additive identity element; that is, $X + X_0 = X_0 + X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

4. *Inverses.*

   For all Dedekind cuts $X \in \mathbb{R}$, there is a Dedekind cut $Y \in \mathbb{R}$ so that $X + Y = X_0$.

5. *Compatibility with the standard order.*

   For all Dedekind cuts $W, X, Y, Z \in \mathbb{R}$, if $W \subseteq Y$ and $X \subseteq Z$, then $W + X \subseteq Y + Z$.

Proof.

1. We note that

   $$\begin{aligned} X + Y &= \{p + q \in \mathbb{Q} \mid p \in X \text{ and } q \in Y\} \\ &= \{q + p \in \mathbb{Q} \mid q \in Y \text{ and } p \in X\} = Y + X \end{aligned}$$

   for all Dedekind cuts $X, Y \in \mathbb{R}$, and so addition $+$ of Dedekind cuts is commutative.

2. We note that

   $$\begin{aligned} (X + Y) + Z &= \{s + r \in \mathbb{Q} \mid s \in X + Y \text{ and } r \in Z\} \\ &= \{(p + q) + r \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } r \in Z\} \\ &= \{p + (q + r) \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } r \in Z\} \\ &= \{p + t \in \mathbb{Q} \mid p \in X \text{ and } t \in Y + Z\} = X + (Y + Z)\,. \end{aligned}$$

   for all Dedekind cuts $X, Y, Z \in \mathbb{R}$, and so addition $+$ of Dedekind cuts is associative.

3. Let $p \in \mathbb{Q}$ be a rational number, and suppose first that $p \in X + X_0$. Then $p = q + r$ for some rational numbers $q \in X$ and $r \in X_0$. Since

   $$X_0 = \{s \in \mathbb{Q} \mid s < 0\}$$

   is the set of negative rational numbers,

   $$p = q + r < q + 0 = q \in X,$$

   and so $p \in X$.

   Conversely, now suppose that $p \in X$. Since X does not have a maximum, $r \in X$ for some rational number $r > p$. Let $s = p - r \in X_0$, and note that

   $$p = r + (p - r) = r + s$$

   is the sum of a rational number $r \in X$ and a rational number $p - r \in X_0$. We conclude that $p \in X + X_0$.

In summary, we have shown that $p \in X + X_0$ if and only if $p \in X$, and so $X + X_0 = X$. That $X_0 + X = X$ now follows from (1) above.

4. There are two cases: either X is rational, or X is irrational. If X is rational, say $X = X_q$ for some rational number $q \in \mathbb{Q}$, then we may choose $Y = X_{-q}$, since Lemma **??** Addition of rational Dedekind cuts implies that
$$X + Y = X_q + X_{-q} = X_{q+(-q)} = X_0.$$
So we may suppose that X is irrational. Let
$$Y = \{q \in \mathbb{Q} \mid -q \notin X\}.$$
Since X is a non-empty proper subset of $\mathbb{Q}$, so too is its complement $\mathbb{Q} \setminus X$, which implies that Y is also non-empty and proper. We aim to show first that Y is a Dedekind cut.

To that end, let $p, q \in \mathbb{Q}$ be rational numbers, and suppose that $p < q$ and $q \in Y$. Then $-q \notin X$, and so $-q > r$ for all rational numbers $r \in X$. We observe that
$$-p \geq -q > r$$
for all rational numbers $r \in X$, and so $-p \notin X$; that is, $p \in Y$.

Now let $q \in Y$, so that $-q \notin X$. Since X is irrational, Lemma **??** Rational Dedekind cuts implies that $\mathbb{Q} \setminus X$ does not have a minimum. Thus there is some rational number $r \in \mathbb{Q} \setminus X$ so that $r < -q$. We conclude that $-r \in Y$ and $-r > q$, so that Y does not have a maximum.

We have shown that Y is a Dedekind cut, but it remains to show that $X + Y = X_0$. To that end, let $r \in \mathbb{Q}$ be a rational number, and suppose first that $r \in X + Y$. Then $r = p + q$ for some $p \in X$ and $q \in Y$. Since $-q \notin X$, we must have $p < -q$, so that $r = p + q < 0$; that is, $r \in X_0$.

Conversely, now suppose that $r \in X_0$, so that $r < 0$. In particular, $-r > 0$. If $p + (-r) \in X$ for all rational numbers $p \in X$, then X is not bounded from above, which would imply that $X = \mathbb{Q}$. Thus $p - r \notin X$ for some rational number $p \in X$. We conclude that
$$r = p + (r - p) \in X + Y$$
is the sum of two rational numbers $p \in X$ and $r - p \in Y$.

In summary, we have shown that $r \in X + Y$ if and only if $r \in X_0$, and so $X + Y = X_0$.

5. We observe that if $W \subseteq Y$ and $X \subseteq Z$, then
$$\begin{aligned} W + X &= \{p + q \in \mathbb{Q} \mid p \in W \text{ and } q \in X\} \\ &\subseteq \{p + q \in \mathbb{Q} \mid p \in Y \text{ and } q \in Z\} = Y + Z. \end{aligned}$$

■

All that remains is to define the multiplication of Dedekind cuts and derive relevant results about multiplication itself and its relation to addition and comparison.

2.2.4 Multiplication of Dedekind Cuts

Finally, we define multiplication for Dedekind cuts. We will see that this definition depends in a complicated way on the signs of the two factors. As a result of this case-by-case definition of multiplication, we offer the remaining results of this section without proof.

Lemma 2.2.13 Multiplication of positive Dedekind cuts. *Given two positive Dedekind cuts $X, Y \in \mathbb{R}$, the set*

$$\{pq \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } (p \geq 0 \text{ or } q \geq 0)\}$$

is a Dedekind cut.

Definition 2.2.14 Multiplication of Dedekind cuts. The **product** $X \cdot Y$ of two Dedekind cuts $X, Y \in \mathbb{R}$ is the Dedekind cut defined by the following formulae:

- *Zero.*

  If either $X = X_0$ or $Y = X_0$, then $X \cdot Y = X_0$.

- *Positive times positive.*

  If both X and Y are positive, then

  $$X \times Y = \{pq \in \mathbb{Q} \mid p \in X \text{ and } q \in Y \text{ and } (p \geq 0 \text{ or } q \geq 0)\}.$$

- *Positive times negative.*

  If X is positive and Y is negative, then

  $$X \times Y = -(X \times (-Y)).$$

- *Negative times positive.*

  If X is negative and Y is positive, then

  $$X \times Y = -((-X) \times Y).$$

- *Negative times negative.*

  If both X and Y are negative, then

  $$X \times Y = (-X) \times (-Y).$$

This defines a binary operation $\cdot$ on the set $\mathbb{R}$ of Dedekind cuts. ◆

We end with a compilation of all relevant results about the arithmetic of Dedekind cuts. While one can verify these results, we omit the proofs of those that involve multiplication in order to simplify the narrative of exposition. Indeed, these proofs rely on casework which, while not difficult, would require more time than we can dedicate to this topic.

Theorem 2.2.15 Arithmetic of Dedekind cuts. *The addition and multiplication of Dedekind cuts has the following properties:*

- *Commutativity of addition.*

  $X + Y = Y + X$ for all Dedekind cuts $X, Y \in \mathbb{R}$.

- *Associativity of addition.*

$(X+Y)+Z = X+(Y+Z)$ and $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$ for all Dedekind cuts $X, Y, Z \in \mathbb{R}$.

- *Additive identity element.*

  The rational Dedekind cut X_0 associated to 0 is an additive identity element; that is, $X + X_0 = X_0 + X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

- *Additive inverses.*

  For all Dedekind cuts $X \in \mathbb{R}$, there is a Dedekind cut $-X \in \mathbb{R}$ so that $X + (-X) = (-X) + X = X_0$.

- *Commutativity of multiplication.*

  $X \cdot Y = Y \cdot X$ for all Dedekind cuts $X, Y \in \mathbb{R}$.

- *Associativity of multiplication.*

  $(X \cdot Y) \cdot Z = X \cdot (Y \cdot Z)$ for all Dedekind cuts $X, Y, Z \in \mathbb{R}$.

- *Multiplicative identity element.*

  The rational Dedekind cut X_1 associated to 1 is a multiplicative identity element; that is, $X \cdot X_1 = X_1 \cdot X = X$ for all Dedekind cuts $X \in \mathbb{R}$.

- *Multiplicative inverses.*

  For all Dedekind cuts $X \in \mathbb{R}$, if $X \neq X_0$, then there is a Dedekind cut $X^{-1} \in \mathbb{R}$ so that $X \cdot X^{-1} = X^{-1} \cdot X = X_1$.

- *Distributivity of multiplication over addition.*

  For all Dedekind cuts $X, Y, Z \in \mathbb{R}$, both

  $$X \cdot (Y + Z) = (X \cdot Y) + (X \cdot Z)$$

  and

  $$(X + Y) \cdot Z = (X \cdot Z) + (Y \cdot Z) .$$

- *Compatibility of addition and the standard order.*

  For all Dedekind cuts $W, X, Y, Z \in \mathbb{R}$, if $W \subseteq Y$ and $X \subseteq Z$, then $W + X \subseteq Y + Z$.

- *Compatibility of multiplication and the standard order.*

  For all Dedekind cuts $X, Y \in \mathbb{R}$, if X and Y are non-negative, then so too is their product $X \cdot Y$.

All of the above properties of the arithmetic of Dedekind cuts not involving multiplication have already been proven. Again, we highlight that while proofs of the remaining parts are not too difficult to construct, they rely on significant casework involving signs; we unfortunately don't have time to study these proofs in detail.

In the next section, we will continue to explore the properties of the real number system. In particular, we will characterize $\mathbb{R}$ as the unique **Dedekind complete ordered field**.

2.2.5 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.2.6 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Cancellation of addition.** Prove that for all Dedekind cuts $X, Y, Z \in \mathbb{R}$, if $X + Z \subseteq Y + Z$, then $X \subseteq Y$.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

2.3 The Complete Ordered Field

In the previous section, we gave an explicit and detailed construction of the real number system $\mathbb{R}$. In this section, we give an alternative and non-constructive characterization of $\mathbb{R}$ as the unique **Dedekind complete ordered field**. This will set the stage for abandoning the Dedekind cut perspective in favor of an abstract conception of the properties of the real number system.

2.3.1 The Ordered Field Axioms

We begin by revisiting the properties of the arithmetic of Dedekind cuts collected in Section **??** Constructing the Real Numbers. These properties define an algebraic object called a **field**.

Definition 2.3.1 Field. Let $+$ and $\cdot$ be binary operations on a set F, called **addition** and **multiplication**. The ordered triple $(F, +, \cdot)$ is called a **field** if the binary operations satisfy the following conditions:

- *Commutativity of addition.*

  $x + y = y + x$ for all elements $x, y \in F$.

- *Associativity of addition.*

  $(x + y) + z = x + (y + z)$ for all elements $x, y, z \in F$.

- *Additive identity element.*

There is an element $0 \in F$ so that $x + 0 = 0 + x = x$ for all elements $x \in F$.

- *Additive inverses.*

  For all elements $x \in X$, there is an element $-x \in F$ called the **additive inverse** of x so that $x + (-x) = (-x) + x = 0$.

- *Commutativity of multiplication.*

  $x \cdot y = y \cdot x$ for all elements $x, y \in F$.

- *Associativity of addition.*

  $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all elements $x, y, z \in F$.

- *Multiplicative identity element.*

  There is an element $1 \in F \setminus \{0\}$ so that $x \cdot 1 = 1 \cdot x = x$ for all elements $x \in F$.

- *Multiplicative inverses.*

  For all elements $x \in X$, if $x \neq 0$ then there is an element $x^{-1} \in F$ called the **multiplicative inverse** of x so that $x \cdot x^{-1} = x^{-1} \cdot x = 1$.

- *Distributivity of multiplication over addition.*

  $x \cdot (y + z) = (x \cdot y) + (x \cdot z)$ and $(x + y) \cdot z = (x \cdot z) + y \cdot z$ for all elements $x, y, z \in F$.

If $(F, +, \cdot)$ is a field, then the set F is called its **ground set** or **underlying set**. If the binary operations $+$ and $\cdot$ are clear from context, then we will usually abuse notation and refer to a field by the ground set alone. ◆

Example 2.3.2 Fields. The following are examples of fields:

- *The field of rational numbers.*

  Addition $+$ and multiplication $\cdot$ of rational numbers satisfies the above field axioms, and so $\mathbb{Q}$ is a field.

- *The field of real numbers.*

  Addition $+$ and multiplication $\cdot$ of real numbers satisfies the above field axioms by Theorem **??** Arithmetic of Dedekind cuts, and so $\mathbb{R}$ is a field.

- *The field of rational functions.*

  Let $n \in \mathbb{N}$ be a natural number, and fix real numbers $x_1, \ldots c, x_n \in \mathbb{R}$. A real-valued function $f \colon \mathbb{R} \setminus \{x_1, \ldots, x_n\} \to \mathbb{R}$ is called a **rational function** if it has a formula of the form

  $$f(x) = \frac{p(x)}{q(x)}$$

  for some polynomials $p(x)$ and $q(x)$. Such a rational function $f(x)$ has a **removable discontinuity** at x_k if both $p(x)$ and $q(x)$ have a root at $x = x_k$, and the corresponding factor $(x - x_k)$ appears at least as many times in the factorization of $p(x)$ as it does in the factorization of $q(x)$.

  The set of rational functions without removable discontinuities is denoted $\mathbb{R}(x)$. One can verify that the addition and multiplication of these rational functions satisfies the field axioms, and so $\mathbb{R}(x)$ is a field.

▲

We have seen that the fields $\mathbb{Q}$ and $\mathbb{R}$ of rational and real numbers are equipped with additional structure beyond that of arithmetic: we can compare rational numbers and real numbers by the standard orders $\leq$ on $\mathbb{Q}$ and $\mathbb{R}$, respectively. These are total orders, and we have seen they are compatible with arithmetic in key ways. Generalizing these relationships, we arrive at the notion of an **ordered field**.

Definition 2.3.3 Ordered field. Let $+$ and $\cdot$ be binary operations on a set F, called **addition** and **multiplication**, and consider a binary relation $\leq$ on F, called **comparison**. The ordered quadruple $(F, +, \cdot, \leq)$ is called an **ordered field** if it satisfies the following conditions:

- The ordered triple $(F, +, \cdot)$ is a field.
- The ordered pair $(F, \leq)$ is a totally ordered set.
- *Compatibility of addition and comparison.*

  For all elements $x, y, z \in F$, if $x \leq y$, then $x + z \leq y + z$.
- *Compatibility of multiplication and comparison.*

  For all elements $x, y \in F$, if $0 \leq x$ and $0 \leq y$, then $0 \leq x \cdot y$.

If $(F, +, \cdot, \leq)$ is an ordered field, then the set F is called its **ground set** or **underlying set**. If the binary operations $+$ and $\cdot$ and the binary relation $\leq$ are clear from context, then we will usually refer to an ordered field by the ground set alone. ◆

Example 2.3.4 Ordered fields. The following are examples of ordered fields:

- *The ordered field of rational numbers.*

  Addition $+$, multiplication $\cdot$, and comparison $\leq$ of rational numbers satisfies the above ordered field axioms, and so $\mathbb{Q}$ is an ordered field.
- *The ordered field of real numbers.*

  Addition $+$, multiplication $\cdot$, and comparison $\leq$ of rational numbers satisfies the above ordered field axioms by Theorem **??** Arithmetic of Dedekind cuts, and so $\mathbb{R}$ is an ordered field.

▲

That the above list of examples includes both the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$ begs the question: Why bother with the real numbers $\mathbb{R}$, when the rational numbers $\mathbb{Q}$ are already an ordered field?

2.3.2 Dedekind completeness

So far, we have failed to meaningfully distinguish between the properties enjoyed by the rational number system $\mathbb{Q}$ and the real number system $\mathbb{R}$. We now address this issue via the notion of **Dedekind completeness**. Informally, Dedekind completeness is defined in terms of the set of upper bounds of a subset which is bounded from above (or dually the set of lower bounds of a subset which is bounded from below).

Definition 2.3.5 Supremum; infimum. Let $U \subseteq X$ be a subset of a partially ordered set $(X, \preceq)$.

- *Supremum.*

If the set of upper bounds for U in X has a minimum, then this minimum is called the **supremum** (or **least upper bound**) of U and is denoted $\sup(U)$; that is,

$$\sup(U) = \min(\{x \in X \mid u \leq x \text{ for all } u \in U\}).$$

- *Infimum.*

  If the set of lower bounds for U in X has a maximum, then this maximum is called the **infimum** (or **greatest lower bound**) of U and is denoted $\inf(U)$; that is,

$$\inf(U) = \max(\{x \in X \mid x \leq u \text{ for all } u \in U\}).$$

◆

Example 2.3.6 Suprema and infima. Let $(U_j)_{j\in\mathcal{J}}$ be an indexed family of subsets $U_j \subseteq X$ of a set X. These subsets U_j are elements of the power set $\mathcal{P}(X)$, which is partially ordered by set containment $\subseteq$.

We may reinterpret (a) of Problem 1.2.5.5 Indexed intersections and unions as saying that the indexed intersection $\bigcap_{j\in\mathcal{J}} U_j$ and the indexed union $\bigcup_{j\in\mathcal{J}} U_j$ are the infimum and supremum of $\{U_j\}_{j\in\mathcal{J}}$. ▲

Remark 2.3.7 Suprema of Dedekind cuts. The complement $\mathbb{Q} \setminus X$ of a Dedekind cut $X \in \mathbb{R}$ is exactly the set of upper bounds of X in $\mathbb{Q}$, and so Lemma **??** Rational Dedekind cuts implies that X is rational if and only if it has a supremum in $\mathbb{Q}$.

This is the deficiency of the rational numbers $\mathbb{Q}$ from the analytic perspective: Proposition **??** Irrational Dedekind cuts posits the existence of subsets of $\mathbb{Q}$ which are bounded from above ***and yet do not have suprema***. We will shortly see that this is not the case in the real number system $\mathbb{R}$.

Definition 2.3.8 Dedekind completeness. An ordered field $(F, +, \cdot, \leq)$ is called **Dedekind complete** if every non-empty subset of F which is bounded from above has a supremum. ◆

Theorem 2.3.9 Dedekind completeness of the real numbers. *The real number system* $(\mathbb{R}, +, \cdot, \leq)$ *is Dedekind complete.*

Proof. Let $\emptyset \subsetneq U \subseteq \mathbb{R}$ be a non-empty subset of U which is bounded from above, and consider the indexed union

$$S = \bigcup_{X\in U} X = \{q \in \mathbb{Q} \mid q \in X \text{ for some } X \in U\}.$$

Since U is a non-empty collection of non-empty subsets of $\mathbb{Q}$, S is non-empty. Moreover, since U is bounded from above, there is some Dedekind cut $Y \in \mathbb{R}$ so that $X \subseteq Y$ for all $X \in U$. In particular, if $q \in \mathbb{Q} \setminus Y$, then $q \notin X$ for all $X \in U$, and so $q \notin S$.

Let $p, q \in \mathbb{Q}$ be rational numbers, and suppose that $p \leq q$ and $q \in S$. Since $q \in S$, $q \in X$ for some $X \in U$, and so $p \in X \subseteq S$.

Now let $p \in S$, so that $p \in X$ for some $X \in U$. Since X does not have a maximum, there is some rational number $q \in X \subseteq S$ so that $p < q$. We conclude that S does not have a maximum, and so is a Dedekind cut.

We note that (a) of Problem 1.2.5.5 Indexed intersections and unions implies that for a Dedekind cut $Y \in \mathbb{R}$, $S \subseteq Y$ if and only if $X \subseteq Y$ for all $X \in U$; that is, any upper bound for U in $\mathbb{R}$ contains S, so that

$$S = \min(\{Y \in \mathbb{R} \mid X \subseteq Y \text{ for all } X \in U\}) = \sup(U).$$

■

We will see that Theorem **??** Dedekind completeness of the real numbers has far-reaching consequences for the discipline of real analysis, most notably in the existence of limits for certain objects constructed out of the real numbers $\mathbb{R}$. First, however, we present the following non-constructive characterization of $\mathbb{R}$ as the *unique* Dedekind complete ordered field.

Theorem 2.3.10 Uniqueness of the real numbers. *If $(F,+,\cdot,\leq)$ is a Dedekind complete ordered field, then there is a bijection $\varphi\colon \mathbb{R}\to F$ so that for all real numbers $x,y\in\mathbb{R}$, $\varphi(x+y)=\varphi(x)+\varphi(y)$, $\varphi(x\cdot y)=\varphi(x)\cdot\varphi(y)$, and if $x\leq y$, then $\varphi(x)\leq\varphi(y)$.*

With Theorem **??** Uniqueness of the real numbers, we are now ready to shake off the burden of viewing the real number system $\mathbb{R}$ through the perspective of Dedekind cuts. All relevant properties of the analysis of real numbers can be derived abstractly from the axioms of a Dedekind complete ordered field, and as a consequence, we will see that we no longer need to rely on any specific formal model of $\mathbb{R}$.

In the next section, we will construct various extensions of the real number system $\mathbb{R}$. In particular, we will see the **extended real numbers**, the **complex numbers**, and the **quaternions**.

2.3.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Algebraic structures other than fields. Determine which of the ordered field axioms are satisfied by the given algebraic structure.

1. Which of the ordered field axioms are not satisfied by the set $\mathbb{Z}$ of integers?
2. Which of the ordered field axioms are not satisfied by the set $\mathbb{N}$ of natural numbers?
3. Let X be a set. Which of the ordered field axioms are not satisfied by the ordered quadruple $(\mathcal{P}(X),\cup,\cap,\subseteq)$?

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.3.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be

less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Uniqueness in a field. Let $(F, +, \cdot)$ be a field

(a) Show that the additive identity element $0 \in F$ is unique.

(b) Show that the multiplicative identity element $1 \in F$ is unique.

(c) Show that additive inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x + y = x + z = 0,$$

then $y = z$.

(d) Show that multiplicative inverses are unique; that is, given elements $x, y, z \in F$, show that if

$$x \cdot y = x \cdot z = 1,$$

then $y = z$.

2. Sign in an ordered field. Let $(F, +, \cdot, \leq)$ be an ordered field. As in the case of the rational numbers $\mathbb{Q}$ and the real numbers $\mathbb{R}$, we can define the **sign** of an element of F by comparison with the additive identity element 0:

- *Positivity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **positive** if $0 \leq x$.

- *Negativity.*

  A non-zero element $x \in F \setminus \{0\}$ is called **negative** if $x \leq 0$.

- *Zero.*

  The additive identity element $0 \in F$ is not positive or negative.

(a) Let $x \in F$. Show that if $x \neq 0$, then x and its additive inverse $-x$ have opposite sign.

Hint. Add $-x$ to the inequality defining the sign of x.

(b) More generally, show that for all elements $x, y \in F$, if $x \leq y$ then $-y \leq -x$.

(c) Show that $(-1) \cdot x = -x$ for all elements $x \in F$.

(d) Show that the product of two negative elements is positive.

3. Infima and Dedekind completeness. Let $(F, +, \cdot, \leq)$ be an ordered field, and consider the map $n \colon F \to F$ defined by the formula $n(x) = -x$; that is, n sends an input $x \in F$ to its additive inverse $-x$.

(a) Show that for a subset $U \subseteq F$, U is bounded from above if and only if $n(U)$ is bounded from below, and U is bounded from below if and only if $n(U)$ is bounded from above.

(b) Show that a subset $U \subseteq F$ has a supremum if and only if $n(U)$ has an infimum, and in this case

$$\inf(n(U)) = n(\sup(U)).$$

Similarly, show that U has an infimum if and only if $n(U)$ has a supremum, and in this case

$$\sup(n(U)) = n(\inf(U))$$

(c) Show that the ordered field $(F, +, \cdot, \leq)$ is Dedekind complete if and only if every non-empty subset of F which is bounded from below has an infimum.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

2.4 Beyond the Real Number System

In this section, we address several common extensions to the real number system $\mathbb{R}$. Some of these, like the **complex numbers** and the **quaternions**, see extensions to arithmetic (at the cost of some of the ordered field axioms). Others, like the **extended real numbers**, abandon arithmetic altogether in favor of certain order-theoretic properties.

2.4.1 The Complex Numbers

Despite not playing a large role in the field of real analysis, the **complex number system** is ubiquitous throughout mathematics and merits at least a brief mention. The extension of the real numbers to the complex numbers ameliorates an algebraic insufficiency of $\mathbb{R}$, namely that it does not contain the roots of all polynomials. For example, there is no real number $x \in \mathbb{R}$ so that $x^2 + 1 = 0$.

Definition 2.4.1 Complex number. A **complex number** is an ordered pair $z = (x, y)$ of real numbers $x, y \in \mathbb{R}$, the first of which is called the **real part** and is denoted $\Re(z) = x$ and the second of which is called the **imaginary part** and is denoted $\Im(z) = y$. The set of all complex numbers is denoted $\mathbb{C}$.

The **sum** $z_1 + z_2$ and **product** $z_1 z_2$ of two complex numbers $z_1, z_2 \in \mathbb{C}$ are the complex numbers defined by the formulae

$$\begin{aligned} z_1 + z_2 &= (\Re(z_1) + \Re(z_2), \Im(z_1) + \Im(z_2)) \\ z_1 z_2 &= (\Re(z_1) \cdot \Re(z_2) - \Im(z_1) \cdot \Im(z_2), \Re(z_1) \cdot \Im(z_2) + \Im(z_1) \cdot \Re(z_2)). \end{aligned}$$

The **imaginary unit** i is the complex number $(0, 1)$, and we note that

$$i \cdot i = (0, 1) \cdot (0, 1) = (0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0) = (-1, 0).$$

◆

Lemma 2.4.2 Real numbers as complex numbers. *There is an injective map $\iota\colon \mathbb{R} \to \mathbb{C}$ so that $\iota(x_1 + x_2) = \iota(x_1) + \iota(x_2)$ and $\iota(x_1 x_2) = \iota(x_1) \cdot \iota(x_2)$ for all real numbers $x_1, x_2 \in \mathbb{R}$.*

Proof. Let $\iota\colon \mathbb{R} \to \mathbb{C}$ be the function defined by the formula $\iota(x) = (x, 0)$. Since projection $\pi\colon \mathbb{C} \to \mathbb{R}$ onto the first factor is a left inverse to ι, ι is left-invertible and hence injective by (1) of Theorem **??** Invertibility. Moreover,

$$\iota(x_1 + x_2) = (x_1 + x_2, 0) = (x_1, 0) + (x_2, 0) = \iota(x_1) + \iota(x_2),$$

and

$$\begin{aligned}\iota(x_1x_2) &= (x_1x_2, 0) = (x_1x_2 - 0\cdot 0, x_1\cdot 0 - 0\cdot x_2)\\ &= (x_1, 0)\cdot(x_2, 0) = \iota(x_1)\cdot\iota(x_2)\end{aligned}$$

for all real numbers $x_1, x_2 \in \mathbb{R}$. ■

As in previous extensions of the usual number systems, we will use the injective map $\iota\colon \mathbb{R} \to \mathbb{C}$ defined in Lemma **??** Real numbers as complex numbers to view the real numbers $\mathbb{R}$ as a subset of the complex numbers $\mathbb{C}$. As a first example of this, we will denote the complex number (x, y) as $x + yi$ going forward.

Definition 2.4.3 Complex conjugate. The **conjugate** $\overline{z}$ of a complex number $x + yi \in \mathbb{C}$ is the complex number $\overline{z} = x - yi$. ◆

Lemma 2.4.4 Complex conjugates. *For all complex numbers $z \in \mathbb{C}$, $z\overline{z}$ is a non-negative real number, and $z\overline{z}$ is positive if and only if $z \neq 0$ is non-zero*

Proof. Let $z = x + yi \in \mathbb{C}$ be a complex number, and note that

$$z\overline{z} = (x + yi)(x - yi) = \left(x^2 + y^2\right) + 0i = x^2 + y^2$$

is a sum of squares of real numbers, and hence is non-negative. Moreover, $z\overline{z} = x^2 + y^2$ is positive unless $x = y = 0$, in which case $z = 0$ as well. ■

Definition 2.4.5 Magnitude of a complex number. The **magnitude** $|z|$ of a complex number $z \in \mathbb{C}$ is the non-negative real number defined by $|z|^2 = z\overline{z}$. ◆

Proposition 2.4.6 The field of complex numbers. $(\mathbb{C}, +, \cdot)$ *is a field.*

While the complex numbers $\mathbb{C}$ are a field, they are not an ordered field.

Proposition 2.4.7 Orders on the complex numbers. *There is no total order $\leq$ on the complex numbers $\mathbb{C}$ so that $(\mathbb{C}, +, \cdot, \leq)$ is an ordered field.*

So the addition of the non-real complex numbers to the real number system $\mathbb{R}$ comes at the expense of some of the ordered field axioms. We will see that this is also true of further extensions; for example, multiplication of **quaternions** is not commutative.

2.4.2 Infinity

Definition 2.4.8 Extended real number. An **extended real number** is either a real number or one of two **infinite** elements ∞ or $-\infty$. The set of extended real numbers is denoted $\overline{\mathbb{R}} = \mathbb{R} \cup \{\infty, -\infty\}$.

The standard ordering $\leq$ on the real numbers $\mathbb{R}$ is extended to a total order on $\overline{\mathbb{R}}$ by requiring $-\infty \leq x \leq \infty$ for all real numbers $x \in \mathbb{R}$. ◆

Definition 2.4.9 Interval. Let $a, b \in \overline{\mathbb{R}}$ be extended real numbers so that $a \leq b$.

1. *Closed interval.*

   The **closed interval** $[a, b]$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

   $$[a, b] = \left\{x \in \overline{\mathbb{R}} \;\middle|\; a \leq x \leq b\right\}.$$

2. *Open interval.*

The **open interval** (a, b) from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$(a, b) = \left\{ x \in \overline{\mathbb{R}} \,\middle|\, a < x < b \right\}.$$

3. *Half-closed, half-open interval.*

   The **half-closed, half-open interval** $[a, b)$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$[a, b) = \left\{ x \in \overline{\mathbb{R}} \,\middle|\, a \leq x < b \right\}.$$

4. *Closed interval.*

   The **half-open, half-closed interval** $(a, b]$ from a to b is the subset of the extended real numbers $\overline{\mathbb{R}}$ defined by

$$(a, b] = \left\{ x \in \overline{\mathbb{R}} \,\middle|\, a < x \leq b \right\}.$$

◆

Example 2.4.10 Intervals. The following are examples of intervals:

- *(Extended) real numbers.*

  The real numbers $\mathbb{R}$ and the extended real numbers $\overline{\mathbb{R}}$ are open and closed intervals, respectively; concretely, $\mathbb{R} = (-\infty, \infty)$ and $\overline{\mathbb{R}} = [-\infty, \infty]$.

- *Dedekind cuts.*

  Given a real number $x \in \mathbb{R}$, the Dedekind cut X corresponding to x is $X = \mathbb{Q} \cap (-\infty, x)$.

▲

Theorem 2.4.11 Suprema and infima. *Any subset of the extended real numbers $\overline{\mathbb{R}}$ has a supremum in $\overline{\mathbb{R}}$.*

Proof. Let $U \subseteq \overline{\mathbb{R}}$ be a subset of the extended real numbers $\mathbb{R}$. If $\infty \in U$, then $\sup(U) = \max(U) = \infty$. If $\infty \notin U$ and $U \cap \mathbb{R}$ is not bounded from above, then

$$\sup(U) = \sup(U \cap \mathbb{R}) = \infty.$$

If $\infty \notin U$ and $U \cap \mathbb{R} \neq \emptyset$ is bounded from above, then

$$\sup(U) = \sup(U \cap \mathbb{R}).$$

Finally, if $\infty \notin U$ and $U \cap \mathbb{R} = \emptyset$, then $\sup(U) = -\infty$. ■

We note that the binary operations of addition and multiplication do not fully extend to the set $\overline{\mathbb{R}}$ of extended real numbers. Indeed, there are several **indeterminate forms** like $\infty + (-\infty)$ and $0 \cdot \infty$ which cannot be consistently resolved.

Text after the last subsection.

2.4.3 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

2.4.4 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

Text after the last section.

Part II

Point-Set Topology

Chapter 3

Metric Spaces

In this chapter, we introduce **metric spaces**, which generalize the notion of distance in the physical world, and are the natural setting for beginning our study of analysis. Analysis as a field of mathematics is concerned with the limiting behavior of mathematical objects. In a metric space, it makes sense to talk about the distance between any two points. Armed only with this notion of distance, we will define what it means

- for sequences and functions to **converge** to a **limit**;
- for sets to be **open** and/or closed;
- for maps to be **continuous**; and
- for metric spaces to be **complete**.

However, in spite of the strength and applicability of the results to be proven in this chapter, we will eventually find that we require further generalizations of the above notions, which will occur in the next chapter.

3.1 Introduction to Metric Geometry

To begin, we introduce a generalization of our usual notion of distance which is called a **metric**. The properties of physical distance make it an extremely useful tool for formalizing the analytical notions mentioned above.

3.1.1 Introduction to Metrics

Not all properties of physical distance are generic enough to generalize to other settings in which we might want to measure distance-like quantities. This question of which properties of a specific example (usually Euclidean space) are salient enough to merit requirement in a generalization is ubiquitous throughout mathematical exposition, and will recur frequently throughout this text.

Definition 3.1.1 Metric space. A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X, which is called the **ground set** and whose elements are called **points**, is a **metric** on X if it has the following properties:

1. *Non-degeneracy.*

   For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

2. *Symmetry.*

$\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

3. *Triangle inequality.*

   $\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$ for any points $x_1, x_2, x_3 \in X$.

In this case, the ordered pair (X, ρ) is called a **metric space**. ◆

Remark 3.1.2 Implied metrics. We will see that, in general, there are many metrics on a given set, no one of which may truly be considered canonical. In fact, even though some distinct metrics give identical answers to point-set topological concerns such as convergence and continuity, in most cases even a choice one of these classes of metrics is arbitrary in some respects.

However, sometimes, if a metric $\rho\colon X^2 \to \mathbb{R}$ on a set X is clear from context, previously specified, or implicitly present, some mathematicians will refer to the metric space (X, ρ) by only the underlying set X. We will try to avoid such ambiguities, and we will attempt to always explicitly specify the metric with which we will be working.

Example 3.1.3 Metric spaces. The following are examples of metrics and metric spaces:

1. Let $n \in \mathbb{N}$ be a natural number. The **Euclidean metric** $d\colon (\mathbb{R}^n)^2 \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{R}^n$ is the function defined by the formula

   $$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}.$$

   Similarly, the **Euclidean metric** $d\colon (\mathbb{C}^n)^2 \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{C}^n$ is the function defined by the formula

   $$d(\mathbf{w}, \mathbf{z}) = \sqrt{\sum_{k=1}^{n} |w_k - z_k|^2}.$$

   As defined above, the Euclidean metrics are metrics on $\mathbb{R}^n$ and $\mathbb{C}^n$, respectively. The metric spaces $(\mathbb{R}^n, d)$ and $(\mathbb{C}^n, d)$ are called n-dimensional **real Euclidean space** and n-dimensional **complex Euclidean space**, respectively. 1- and 2-dimensional real Euclidean space $(\mathbb{R}, d)$ and $\left(\mathbb{R}^2, d\right)$ are called the **real line** and the **Euclidean plane**, respectively, and 1-dimensional complex Euclidean space $(\mathbb{C}, d)$ is called the **complex plane**.

2. Consider the real-valued function $\rho\colon (0, \infty)^2 \to \mathbb{R}$ defined by the formula

   $$\rho(x, y) = \left|\ln\left(\frac{x}{y}\right)\right|$$

   As defined above, ρ is a metric on the open interval $(0, \infty)$, and so $((0, \infty), \rho)$ is a metric space.

3. The **discrete metric** $\delta_X\colon X^2 \to \{0, 1\}$ on a set X is the function defined by the formula

   $$\delta_X(x_1, x_2) = \begin{cases} 0 & x_1 = x_2 \\ 1 & x_1 \neq x_2. \end{cases}$$

   As defined above, the discrete metric δ_X is a metric on X, and the metric space (X, δ_X) is called a **discrete metric space**.

▲

The definition of a metric is pretty abstract, and it may be difficult to see how or why requiring only nondegeneracy, symmetry, and the triangle inequality would give a useful generalization of distance, especially given the strange and outlandish properties of the discrete metric described above, which we will explore later in this section. However, requiring the three above properties actually forces a metric to have many of the properties of our usual notion of distance, and so we can often use our knowledge of distance in the physical world to inform our intuition about metric spaces.

As an example, consider for a moment the notion of negative distance. In the real world, it makes no sense to consider negative distances between physical objects, and we would be equally confused by negative distances between points in a metric space. However, the defining axioms of a metric do not a priori explicitly forbid this nonsensical outcome from occurring. At first glance, it might seem like a metric might be allowed to take negative values, even though the three examples of a metric given above are certainly nonnegative. In fact, this is never the case.

Lemma 3.1.4 Metrics are non-negative. *A metric takes only non-negative values.*

Proof. Let ρ be a metric on a set X, and note that

$$\begin{aligned}\rho(x_1,x_2) &= \frac{2\rho(x_1,x_2)}{2} = \frac{\rho(x_1,x_2)+\rho(x_1,x_2)}{2}\\ &= \frac{\rho(x_1,x_2)+\rho(x_2,x_1)}{2} \geq \frac{\rho(x_1,x_1)}{2} = \frac{0}{2} = 0\end{aligned}$$

for all points $x_1, x_2 \in X$. ■

The above argument shows that, just as in the physical world, the distance between any two points in a metric space must be nonnegative. In fact, unlike many of the spaces which are studied by analysts (some of which we will encounter in the following chapters), metric spaces can be surprisingly accessible using only our intuitions about distance in the physical world. For example, the triangle inequality expresses that there are no "shortcuts" between two points in a metric space through a third point; in geometric terms, we require that no leg of a triangle may exceed in length the sum of the lengths of the other two legs. While the triangle inequality establishes an upper bound on the length of a leg of a triangle, the following property of a metric is called the **reverse triangle inequality**, and it establishes a lower bound on the length of a leg of a triangle.

Proposition 3.1.5 Reverse triangle inequality. *Let (X,ρ) be a metric space. Then*

$$\rho(x_1,x_2) \geq |\rho(x_1,x_3)-\rho(x_2,x_3)|$$

for all points $x_1, x_2, x_3 \in X$.

Proof. If $\rho(x_1,x_3) \leq \rho(x_2,x_3)$, then $\rho(x_1,x_3)-\rho(x_2,x_3) \leq 0$, and so

$$\begin{aligned}|\rho(x_1,x_3)-\rho(x_2,x_3)| &= \rho(x_2,x_3)-\rho(x_1,x_3)\\ &\leq \rho(x_2,x_1)+\rho(x_1,x_3)-\rho(x_1,x_3) = \rho(x_1,x_2).\end{aligned}$$

On the other hand, if $\rho(x_1,x_3) \geq \rho(x_2,x_3)$, then $\rho(x_1,x_3)-\rho(x_2,x_3) \geq 0$, and so

$$\begin{aligned}|\rho(x_1,x_3)-\rho(x_2,x_3)| &= \rho(x_1,x_3)-\rho(x_2,x_3)\\ &\leq \rho(x_1,x_2)+\rho(x_2,x_3)-\rho(x_2,x_3) = \rho(x_1,x_2).\end{aligned}$$

■

We have just seen that metric spaces often agree with our geometric intuitions about distance. However, we must be careful not to assume too much about the geometry of metric spaces in general; even though many of the usual properties of distance in the physical world are implied by the defining axioms of a metric space, there is still lots of room for geometric peculiarities. In particular, some of the metrics defined above disagree with our geometric intuitions in confusing ways. This is because they satisfy a stronger form of the triangle inequality than is true in the physical world. We will explore these **ultrametrics** in the reading questions for this section.

3.1.2 Constructing New Metrics

We will now see how to construct new metrics out of given metric spaces. Given a metric space, there is a natural metric on any subset of its ground set. This restriction of the metric structure to a subset is done in the obvious way; the distance between any points in the subset is just their distance when considered as points in the bigger space. In this way, a metric on a set is said to induce a metric on all of its subsets.

Definition 3.1.6 Induced metric. Let $U \subseteq X$ be a subset of a metric space (X, ρ). The **induced metric** on U is the restriction $\rho|_{U^2} : U^2 \to \mathbb{R}$ of ρ to U^2. The metric space $(U, \rho|_U)$ is called a **metric subspace** of the metric space (X, ρ).

In practice, we often write that ρ is a metric on U, even if the domain of ρ is strictly larger than U^2. We will not distinguish between a metric and its restriction to a smaller space. ◆

Example 3.1.7 Discrete metric subspaces. Let $U \subseteq X$ be a subset of a set X. The discrete metric δ_U on U is induced on U by the discrete metric δ_X on X; that is, $\delta_X|_{U^2} = \delta_U$. Thus the discrete metric space (U, δ_U) is a metric subspace of the discrete metric space (X, δ_X). ▲

Thus far, we have been very careful when working in a metric space to specify not only the ground set but also the metric. This is because we can in general find many metrics on a given underlying set. In fact, given one or more metrics on a set, there are several ways of constructing new metrics on that set. In particular, any finite positive pointwise linear combination of metrics is itself a metric.

Proposition 3.1.8 Linear combinations of metrics. *Any positive real linear combination of metrics is a metric.*

Proof. Let X be a set, and consider positive real numbers $\alpha_1, \ldots, \alpha_n \in (0, \infty]$ and metrics $\rho_1, \ldots, \rho_n \colon X^2 \to \mathbb{R}$ on X. Denote by $\rho \colon X^2 \to \mathbb{R}$ the linear combination defined by the formula

$$\rho\,(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k\,(x_1, x_2).$$

First let $x_1, x_2 \in X$ be points, and suppose that $\rho\,(x_1, x_2) = 0$. Then

$$\sum_{k=1}^{n} \alpha_k \rho_k\,(x_1, x_2) = \rho\,(x_1, x_2) = 0.$$

Since $\alpha_k > 0$ and $\rho_k\,(x_1, x_2) \geq 0$ for all indices $1 \leq k \leq n$, we conclude that $\rho_k\,(x_1, x_2) = 0$ for all indices $1 \leq k \leq n$. In particular, $x_1 = x_2$.

Conversely, now suppose that $x_1 = x_2$. Then

$$\rho(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \cdot 0 = 0.$$

Thus ρ is non-degenerate.

Note that

$$\rho(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) = \sum_{k=1}^{n} \alpha_k \rho_k(x_2, x_1) = \rho(x_2, x_1)$$

for all points $x_1, x_2 \in X$, and so ρ is symmetric.

Finally, note that

$$\begin{aligned}\rho(x_1, x_3) &= \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_3) \leq \sum_{k=1}^{n} \alpha_k \left(\rho_k(x_1, x_2) + \rho_k(x_2, x_3)\right) \\ &= \sum_{k=1}^{n} \alpha_k \rho_k(x_1, x_2) + \sum_{k=1}^{n} \alpha_k \rho_k(x_2, x_3) = \rho(x_1, x_2) + \rho(x_2, x_3)\end{aligned}$$

for all points $x_1, x_2, x_3 \in X$, and so ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on X. ■

In particular, the sum of two metrics is a metric, and any positive real multiple of a metric is a metric.

3.1.3 Normed Vector Spaces

The ground sets of many of the metric spaces which we will investigate over the course of this chapter can also be considered as the ground sets of vector spaces. Moreover, in practically all of these cases, the linear-algebraic structure of the vector space is compatible with the analytical structure of the metric, in a way which we will make precise shortly. We now introduce a mathematical structure called a **normed vector space**, the study of which is called **functional analysis** and lies at the intersection of linear algebra and mathematical analysis. Aptly named, a normed vector space is a vector space equipped with the additional structure of a norm. Just as metrics generalize the notion of the distance between two points, norms on vector spaces generalize the related notion of the length of a vector.

Definition 3.1.9 Normed vector space. A real-valued function $\|\cdot\| : V \to \mathbb{R}$ on a real or complex vector space V is a **norm** on V if it has the following properties:

1. *Non-degeneracy.*

   For any vector $\mathbf{v} \in V$, $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

2. *Absolute homogeneity.*

   $\|\alpha \mathbf{v}\| = |\alpha| \, \|\mathbf{v}\|$ for any scalar α and vector $\mathbf{v} \in V$.

3. *Sub-additivity.*

   $\|\mathbf{v}_1 + \mathbf{v}_2\| \leq \|\mathbf{v}_1\| + \|\mathbf{v}_2\|$ for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$.

In this case, the ordered pair $(V, \|\cdot\|)$ is called a **normed vector space**. ◆

Normed vector spaces are ubiquitous throughout mathematics. In fact, many of the vector spaces which you will have encountered in any course in linear algebra have canonical norms associated to them; lots of the vector spaces which are familiar to you have been normed vector spaces all along.

Example 3.1.10 Normed vector spaces. The following are examples of normed vector spaces:

1. Let $n \in \mathbb{N}$ be a natural number, and fix a positive real number $p > 0$. The p**-norm** $\|\cdot\|_p : \mathbb{R}^n \to \mathbb{R}$ on n-dimensional real coordinate space $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_p = \left(\sum_{k=1}^{n} x_k^p\right)^{\frac{1}{p}}.$$

   Similarly, the p**-norm** $\|\cdot\|_p : \mathbb{C}^n \to \mathbb{R}$ on n-dimensional complex coordinate space $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_p = \left(\sum_{k=1}^{n} z_k^p\right)^{\frac{1}{p}}.$$

   If $p \geq 1$, then the p-norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ and $\left(\mathbb{C}^n, \|\cdot\|_p\right)$ are normed vector spaces.

   In particular, the 2-norms $\|\cdot\|_2$ on $\mathbb{R}^n$ and $\mathbb{C}^n$ are called the **Euclidean norms** on $\mathbb{R}^n$ and $\mathbb{C}^n$. Because the Euclidean norms are so ubiquitous, we will often drop the subscripts in our notation, so that $\|\cdot\| = \|\cdot\|_2$.

2. Let $n \in \mathbb{N}$ be a natural number. The families of p-norms on n-dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ can be extended to the case $p = \infty$. The ∞**-norm** $\|\cdot\|_\infty : \mathbb{R}^n \to \mathbb{R}$ on $\mathbb{R}^n$ is the real-valued function defined by the formula

$$\|\mathbf{x}\|_\infty = \max_{1 \leq k \leq n} |x_k|.$$

   Similarly, the ∞**-norm** $\|\cdot\|_\infty : \mathbb{C}^n \to \mathbb{R}$ on $\mathbb{C}^n$ is the real-valued function defined by the formula

$$\|\mathbf{z}\|_\infty = \max_{1 \leq k \leq n} |z_k|.$$

   The ∞-norms are norms on $\mathbb{R}^n$ and $\mathbb{C}^n$, and so $(\mathbb{R}^n, \|\cdot\|_\infty)$ and $(\mathbb{C}^n, \|\cdot\|_\infty)$ are normed vector spaces.

3. Any Hermitian inner product $\langle\,\cdot\,,\,\cdot\,\rangle$ on a real or complex vector space V induces a norm $\|\cdot\| : V^2 \to \mathbb{R}$ on V defined by the formula

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}.$$

   In fact, for all natural numbers $n \in \mathbb{N}$, the norms produced in this manner by the dot product on n-dimensional real and complex coordinate space $\mathbb{R}^n$ and $\mathbb{C}^n$ are precisely the Euclidean norms. For this reason, the dot product is often called the **Euclidean inner product** on $\mathbb{R}^n$ and $\mathbb{C}^n$.

The above examples show that many of our familiar vector spaces have canonical norms readily available. ▲

Remark 3.1.11 p-norms on 1-dimensional vector spaces. Note that

$$\|x\|_p = \left(|x|^p\right)^{\frac{1}{p}} = |x| = \max\left(\{|x|\}\right) = \|x\|_\infty$$

for all real numbers $x \in \mathbb{R}$, and similarly

$$\|z\|_p = \left(|z|^p\right)^{\frac{1}{p}} = |z| = \max\left(\{|z|\}\right) = \|z\|_\infty$$

for all complex numbers $z \in \mathbb{C}$; that is, all of the p norms (including the case $p = \infty$) on the real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$ agree. Such statements do not hold in higher dimensions, although we will see that the p-norms remain related for topological concerns.

Just as in the case of metrics, we would prefer to consider only non-negative vector lengths. It makes no sense to consider negative lengths of physical vectors, and it is just as difficult to make sense of negative values of a norm in a normed vector space. Thankfully, the defining axioms of a norm prevent this from occurring.

Lemma 3.1.12 Norms are non-negative. *A norm takes only non-negative values.*

Proof. Let $\|\cdot\| : V \to \mathbb{R}$ be a norm on a real or complex vector space V, and note that

$$\begin{aligned}\|\mathbf{v}\| &= \frac{2\|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + \|\mathbf{v}\|}{2} = \frac{\|\mathbf{v}\| + |-1|\,\|\mathbf{v}\|}{2} \\ &= \frac{\|\mathbf{v}\| + \|-\mathbf{v}\|}{2} \geq \frac{\|\mathbf{v} - \mathbf{v}\|}{2} = \frac{\|\mathbf{0}\|}{2} = \frac{0}{2} = 0\end{aligned}$$

for all vectors $\mathbf{v} \in V$. ■

Theorem 3.1.13 Normed vector spaces are metric spaces. *A norm $\|\cdot\| : V \to \mathbb{R}$ on a real or complex vector space V induces a metric $\rho\colon V^2 \to \mathbb{R}$ defined by the formula*

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\|.$$

Proof. First, let $\mathbf{v}_1, \mathbf{v}_2 \in V$ be vectors, and suppose that $\rho(\mathbf{v}_1, \mathbf{v}_2) = 0$. Then

$$\|\mathbf{v}_1 - \mathbf{v}_2\| = \rho(\mathbf{v}_1, \mathbf{v}_2) = 0,$$

and so $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$; that is, $\mathbf{v}_1 = \mathbf{v}_2$. Conversely, if $\mathbf{v}_1 = \mathbf{v}_2$, then

$$\rho(\mathbf{v}_1, \mathbf{v}_2) = \|\mathbf{v}_1 - \mathbf{v}_2\| = \|\mathbf{0}\| = 0.$$

Thus ρ is non-degenerate.

Since

$$\begin{aligned}\rho(\mathbf{v}_1, \mathbf{v}_2) &= \|\mathbf{v}_1 - \mathbf{v}_2\| = |-1|\,\|\mathbf{v}_1 - \mathbf{v}_2\| \\ &= \|\mathbf{v}_2 - \mathbf{v}_1\| = \rho(\mathbf{v}_2, \mathbf{v}_1)\end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$, ρ is symmetric.

Finally, since

$$\begin{aligned}\rho(\mathbf{v}_1, \mathbf{v}_3) &= \|\mathbf{v}_1 - \mathbf{v}_3\| = \|(\mathbf{v}_1 - \mathbf{v}_2) + (\mathbf{v}_2 - \mathbf{v}_3)\| \\ &\leq \|\mathbf{v}_1 - \mathbf{v}_2\| + \|\mathbf{v}_2 - \mathbf{v}_3\| = \rho(\mathbf{v}_1, \mathbf{v}_2) + \rho(\mathbf{v}_2, \mathbf{v}_3)\end{aligned}$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in V$, ρ satisfies the triangle inequality. Since ρ is non-degenerate, symmetric, and satisfies the triangle inequality, it is a metric on V. ■

So, given a normed vector space, we may easily convert it into a metric space by defining the distance between two vectors to be the norm of their difference. Many important metric spaces arise as a result of this construction.

Example 3.1.14 ***p*-metrics.** Let $n \in \mathbb{N}$ be a natural number, and fix an extended real number $1 \leq p \leq \infty$. The metrics d_p on n-dimensional real and complex coordinate spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ by the p-norms $\|\cdot\|_p$ are called ***p*-metrics**. These p-metrics are defined by the formula

$$d_p(\mathbf{w}, \mathbf{z}) = \|\mathbf{w} - \mathbf{z}\|_p = \left(\sum_{k=1}^{n} |w_k - z_k|^p \right)^{\frac{1}{p}}.$$

In particular, we note that the metrics $d_2 = d$ induced by the Euclidean norms $\|\cdot\|_2 = \|\cdot\|$ are the Euclidean metrics. ▲

So any norm structure on a real or complex vector space can be reanalyzed as also providing a metric structure. However, a metric on a real or complex vector space need not be induced by a norm. Moreover, there exist metric spaces which cannot be constructed from a normed vector space. So, while normed vector spaces are very useful for defining specific metric spaces, they do not capture all of what can be said about metric spaces, and so will not merit any more special consideration just yet. With this illuminating example of normed vector spaces in mind, we now resume our study of metric spaces in their full generality.

In the next section, we introduce one of the most important ideas in the field of analysis: that of convergence to a limit.

3.1.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Computing distances in metric spaces. Compute the distance between the given points in the indicated metric space.

1. Let d be the Euclidean metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.
- **(b)** Compute $d(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

2. Let d_3 be the 3-metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d_3(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

3. Let d_∞ be the ∞-metric on the real coordinate plane $\mathbb{R}^2$.

- **(a)** Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.
- **(b)** Compute $d_\infty(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

4. Let $\delta_{\mathbb{R}^2}$ be the discrete metric on the real coordinate plane $\mathbb{R}^2$.

(a) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 2)$ and $\mathbf{y} = (-1, 4)$.

(b) Compute $\delta_{\mathbb{R}^2}(\mathbf{x}, \mathbf{y})$, where $\mathbf{x} = (3, 4)$ and $\mathbf{y} = (0, 0)$.

Verifying metric properties. Determine whether or not the given function is a metric.

5. Let $d\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the Euclidean metric on the real numbers $\mathbb{R}$; that is,

$$d(x_1, x_2) = |x_1 - x_2|$$

for all real numbers $x_1, x_2 \in \mathbb{R}$. Verify that d is a metric on $\mathbb{R}$.

6. Let $\delta_X : X \times X \to \mathbb{R}$ be the discrete metric on a set X; that is,

$$\delta_X(x_1, x_2) = \begin{cases} 0 & x_1 = x_2 \\ 1 & x_1 \neq x_2 \end{cases}$$

for all points $x_1, x_2 \in X$. Verify that δ_X is a metric on X.

7. Let $\rho\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be the real-valued function defined by the formula

$$\rho(x_1, x_2) = x_1 - x_2.$$

Determine whether or not ρ is a metric on the real numbers $\mathbb{R}$.

8. **Subspaces of discrete metric spaces.** Let $U \subseteq X$ be a subset of a discrete metric space (X, δ_X). Verify that the metric $\delta_X|_U$ induced on U by δ_X is the discrete metric δ_U on U.

9. **Scalar multiples of metrics.** Give an example of a real number $\alpha \in \mathbb{R}$ and a metric $\rho\colon X \times X \to \mathbb{R}$ so that $\alpha\rho$ is not a metric.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.1.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task.*

1. **From pseudometrics to metrics.** A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X, which is called the **ground set** and whose elements are called **points**, is a **pseudometric** on X if it has the following properties:

(a) *Identity.*

$\rho(x, x) = 0$ for any point $x \in X$.

(b) *Symmetry.*

$\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

(c) *Triangle inequality.*

$\rho(x_1, x_3) \leq \rho(x_1, x_2) + \rho(x_2, x_3)$ for any points $x_1, x_2, x_3 \in X$.

In this case, the ordered pair (X, ρ) is called a **pseudometric space**.

(a) Let ρ be a pseudometric on a set X, and consider the binary relation $\equiv_\rho$ on X defined so that $x_1 \equiv_\rho x_2$ if and only if $\rho(x_1, x_2) = 0$. Prove that $\equiv_\rho$ is an equivalence relation on X.

(b) Prove that a pseudometric ρ descends to a well-defined metric $\overline{\rho}$ on the quotient set $X/{\equiv_\rho}$ defined by the formula

$$\overline{\rho}\left([x_1]_{\equiv_\rho}, [x_2]_{\equiv_\rho}\right) = \rho(x_1, x_2).$$

2. **Norm equivalence.** Two norms $\|\cdot\|_1, \|\cdot\|_2 : V \to \mathbb{R}$ on a real or complex vector space V are called **equivalent** if there are positive real numbers $A, B \in (0, \infty)$ so that

$$A\|\mathbf{v}\|_2 \leq \|\mathbf{v}\|_1 \leq B\|\mathbf{v}\|_2$$

for all vectors $\mathbf{v} \in V$.

(a) Prove that norm equivalence is an equivalence relation on the set of norms on V.

(b) Prove that for a given natural number $n \in \mathbb{N}$, any two p-norms $\|\cdot\|_p$ on n-dimensional real coordinate space $\mathbb{R}^n$ for $1 \leq p \leq \infty$ are all equivalent.

3. **Ultrametrics.** A real-valued function $\rho\colon X^2 \to \mathbb{R}$ on the Cartesian square X^2 of a set X is called an **ultrametric** on X, which is called the **ground set** and whose elements are called **points** if it satisfies the following conditions:

- *Non-degeneracy.*

  For any points $x_1, x_2 \in X$, $\rho(x_1, x_2) = 0$ if and only if $x_1 = x_2$.

- *Symmetry.*

  $\rho(x_1, x_2) = \rho(x_2, x_1)$ for any points $x_1, x_2 \in X$.

- *Strong triangle inequality.*

  $\rho(x_1, x_3) \leq \max(\{\rho(x_1, x_2), \rho(x_2, x_3)\})$ for any points $x_1, x_2, x_3 \in X$.

If a real-valued function $\rho\colon X^2 \to \mathbb{R}$ is an ultrametric on X, then the ordered pair (X, ρ) is called an ultrametric space.

(a) Prove that any ultrametric is a metric.

(b) Fix a prime number $p \in \mathbb{N}$. The **p-adic valuation** $|\cdot|_p : \mathbb{Q} \to \mathbb{Q}$ is a rational-valued function defined by the formula

$$|q| = \begin{cases} 0 & q = 0 \\ p^{-k_p(q)} & q \neq 0, \end{cases}$$

where $k_p(q) \in \mathbb{Z}$ is the unique integer so that $q = p^{k_p(q)}\left(\frac{a}{b}\right)$ for some integers $a, b \in \mathbb{Z}$ which are not divisible by p.

Prove that the p-adic valuation $|\cdot|_p$ induces an ultrametric $\rho_p\colon \mathbb{Q}^2 \to \mathbb{R}$ defined by the formula

$$\rho_p(q_1, q_2) = |q_1 - q_2|_p.$$

This ultrametric is called the **p-adic metric** on $\mathbb{Q}$.

(c) Prove that for all points $x_1, x_2, x_3 \in X$ in an ultrametric space (X, ρ), at least two of the distances $\rho(x_1, x_2)$, $\rho(x_2, x_3)$, and $\rho(x_1, x_3)$ are equal.

The geometry of non-Archimedean metric spaces differs in several ways from our intuitions about Euclidean geometry informed by our experience of distance in the physical world. We will see several of these differences over the course of this chapter.

4. **Reverse triangle inequality for norms.** Prove that

$$|\|\mathbf{v}_1\| - \|\mathbf{v}_2\|| \leq \|\mathbf{v}_1 - \mathbf{v}_2\|$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$ in a normed vector space $(V, \|\cdot\|)$.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.2 Convergence and Limits

In this section, we introduce the most fundamental notion of all of mathematical analysis: that of **convergence**. Convergence is a formalization of limiting behavior and closeness in metric spaces and, as we will see in the next chapter, other related analytical settings. In this section we will be concerned with convergence in the setting of abstract metric spaces, which can be interpreted as a rigorous formalization of the informal phrase "eventually close." In the following section, we will gain a more detailed understanding of limiting behavior in less general example of the Euclidean spaces. Note also that in this chapter, we will be concerned only with the convergence of sequences in metric spaces and maps between metric spaces, but in the next chapter, we will make generalizations which in some ways subsume the discussion in this section.

3.2.1 Sequential Convergence

First, we will investigate the limiting behavior of a sequence in a metric space. Informally, a sequence in a metric space **converges** to a point if the values of the sequence eventually grows arbitrarily close to that point; that is, a sequence converges to a point if the values of the sequence are close to that point for large indices. Such a sequence is called **convergent**, and the point to which it converges (which we will see to be uniquely determined by this property) is called its **limit**.

Definition 3.2.1 Sequential convergence/divergence. A sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) **converges** to a point $x \in X$ if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(x_n, x) < \epsilon$$

for all indices $n \geq n_\epsilon$.

A sequence of points in a metric space is called **convergent** if it converges to a point and **divergent** if it does not converge to any point. ◆

So a sequence converges to a point in a metric space if the values of the sequence can be made arbitrarily close to that point solely by choosing large enough indices, and the sequence diverges if this cannot be done.

Example 3.2.2 Sequential convergence/divergence. Consider the sequence $(x_n)_{n=1}^{\infty}$ of real numbers defined by the formula

$$x_n = \frac{1}{n}.$$

We will show that $(x_n)_{n=1}^{\infty}$ converges to 0 in the Euclidean metric d on $\mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number. By the Archimedean principle, there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\epsilon \cdot n_\epsilon > 1$. For all indices $n \geq n_\epsilon$, we observe that

$$d\left(x_n, 0\right) = \left|\frac{1}{n} - 0\right| = \frac{1}{|n|} = \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon.$$

In summary, we have shown that for all positive real numbers $\epsilon > 0$, there is an index $n_\epsilon \in \mathbb{N}$ so that $d\left(x_n, 0\right) < \epsilon$ for all indices $n \geq n_\epsilon$. Thus $(x_n)_{n=1}^{\infty}$ converges to 0.

Note that, in this example, the data of the sequence, the limiting value, the ground set, and the metric are necessary for convergence. By altering any one of these, we may invalidate the convergence we just proved. For example, if we consider $(x_n)_{n=1}^{\infty}$ as a sequence in the metric subspace $(0, \infty)$ of the real line $\mathbb{R}$ consisting of only positive real numbers, then the sequence diverges, since it does not converge in the Euclidean metric to a positive real number.

Similarly, if we change the metric, then the sequence $(x_n)_{n=1}^{\infty}$ may not converge. We have just proven that this sequence is convergent in the Euclidean metric d by finding a point $0 \in \mathbb{R}$ to which it converges, but it is divergent in the discrete metric $\delta_{\mathbb{R}}$. All this is to say that all of the data of a sequence, a metric space, and a purported limiting value are all necessary for claims of convergence or divergence. ▲

Ideally, we would like "the **limit**" of a convergent sequence in a metric space, which is "the point" to which that sequence converges, to be unique with this property. That is, if a sequence in a metric space converges to a point, then we would like this point to be the only such point to which it converges, which a priori need not be the case. Thankfully, the limits of convergent sequences in metric spaces are indeed unique.

Proposition 3.2.3 Uniqueness of sequential limits in a metric space. *Convergent sequences of points in metric spaces converge to unique points.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ), and suppose that $(x_n)_{n=0}^{\infty}$ converges to two points $x, y \in X$. We will show that $x = y$.

To that end, let ϵ be a positive real number, and note that $\frac{\epsilon}{2}$ is a positive real number. There are thus natural numbers $m_{\frac{\epsilon}{2}}, n_{\frac{\epsilon}{2}} \in \mathbb{N}$ so that $\rho\left(x_n, x\right) < \frac{\epsilon}{2}$ for all indices $n \geq m_{\frac{\epsilon}{2}}$ and $\rho\left(x_n, y\right) < \frac{\epsilon}{2}$ for all indices $n \geq n_{\frac{\epsilon}{2}}$.

Let $p_\epsilon = \max\left(m_{\frac{\epsilon}{2}}, n_{\frac{\epsilon}{2}}\right)$, and note that $p_\epsilon \geq m_{\frac{\epsilon}{2}}$ and $p_\epsilon \geq n_{\frac{\epsilon}{2}}$. We conclude that

$$\rho\left(x, y\right) \leq \rho\left(x, x_{p_\epsilon}\right) + \rho\left(x_{p_\epsilon}, y\right) = \rho\left(x_{p_\epsilon}, x\right) + \rho\left(x_{p_\epsilon}, y\right) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

In summary, we have shown that $\rho\left(x, y\right) < \epsilon$ for all positive real numbers $\epsilon > 0$. Since $\rho\left(x, y\right) \geq 0$ by Lemma **??** Metrics are non-negative, we conclude that $\rho\left(x, y\right) = 0$ and hence $x = y$. ■

So if a sequence in a metric space converges to a point, then that point is unique with this property, and is called the limit of the sequence. Many texts in analysis place limits of both sequences and maps (we will define this shortly) in the spotlight; limits are certainly the most important analytical concept introduced in this text. However, we have tried to emphasize the more abstract notion of convergence over that of a limit. We have just proven that the two notions are equivalent in a metric space, but in the next chapter we will explore an analytical setting in which convergent sequences need not converge to a unique limit.

Definition 3.2.4 Sequential limit. The unique point $x \in X$ to which a convergent sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) converges is called the **limit** of said sequence, and it is denoted

$$\lim_{n\to\infty} x_n = x.$$

◆

Example 3.2.5 Sequential limits. Let $x \in X$ be a point in a metric space (X, ρ), and consider the constant sequence $(x_n)_{n=1}^{\infty}$ of points defined by the formula $x_n = x$. For all positive real numbers $\epsilon > 0$ and all indices $n \geq 0$, we observe that

$$\rho(x_n, x) = \rho(x, x) = 0 < \epsilon.$$

Thus $(x_n)_{n=0}^{\infty} = (x)_{n=0}^{\infty}$ converges to x; that is,

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} x = x.$$

▲

Note that the notation introduced in the above definition is somewhat ambiguous; nowhere does it specify with respect to which metric a sequence converges or in which ground set the values of the sequence are considered to lie. We will try to avoid these ambiguities, but (at least in this text) both the ground set and the metric should always be clear from context. If the expression to the right of the limit operator is a real or complex number, then it's a safe bet to assume that convergence is in the standard Euclidean metric d on the real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$, respectively. Similarly, if the expression to the right of the limit operator is a real or complex coordinate vector, then you should assume that we are working in real or complex coordinate space, respectively, taken with the Euclidean metric. We will properly investigate Euclidean convergence in the section to come.

We next establish a necessary and sufficient condition for sequence convergence in any metric space; that the distances between the points of the sequence and the limiting value converge to zero in the real line. The proof of the following proposition follows directly from the definitions of convergence and limit.

Proposition 3.2.6 Sequential convergence. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ). Then for any point $x \in X$,*

$$\lim_{n\to\infty} x_n = x$$

if and only if

$$\lim_{n\to\infty} \rho(x_n, x) = 0.$$

Proof. We observe that

$$d(\rho(x_n, x), 0) = |\rho(x_n, x) - 0| = |\rho(x_n, x)| = \rho(x_n, x)$$

for all indices $n \in \mathbb{N}$. First suppose that

$$\lim_{n\to\infty} x_n = x,$$

and let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_n, x) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$d(\rho(x_n, x), 0) = \rho(x_n, x) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} \rho(x_n, x) = 0.$$

Conversely, now suppose that

$$\lim_{n\to\infty} \rho(x_n, x) = 0,$$

and let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that $d(\rho(x_n, x), 0) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$\rho(x_n, x) = d(\rho(x_n, x), 0) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} x_n = x.$$

■

The implications of the above result are that ***in order to understand sequential convergence and divergence in any metric space, it suffices to understand the convergence and divergence of sequences of real numbers in the Euclidean metric***. In the next section, we will examine the properties of convergence and divergence in the Euclidean metric, and use the standard ordering on the real numbers to enrich our understanding of limiting behavior in the real numbers.

We have just seen that the question of convergence or divergence in an arbitrary metric space can be reduced to the question of convergence or divergence of a related sequence of real numbers in the Euclidean metric. We will begin our study of these Euclidean limits by generalizing to the metric induced on real coordinate space by an arbitrary p-norm. In real coordinate space, sequential convergence in the Euclidean metric is equivalent to sequential convergence in any of the metrics induced on real coordinate space by a p-norm. In fact, something much stronger and much more useful is true; we can reduce the question of convergence or divergence of any sequence of complex coordinate vectors in such a metric to the question of the convergence or divergence of the component sequences in the Euclidean metric on the real numbers.

Theorem 3.2.7 Sequential convergence in the p-metric. *Let $m \in \mathbb{N}$ be a natural number, and consider a sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ of vectors $\mathbf{x}_n = (x_{j,n})_{j=1}^{m} \in \mathbb{R}^m$.*

For any vector $\mathbf{x} = (x_j)_{j=1}^{m}$ and real number $1 \leq p < \infty$, the sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^m$ if and only if for all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Proof. First suppose that $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^m$. Let $1 \leq j \leq m$, and consider a positive real number $\epsilon > 0$. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$d_p(\mathbf{x}_n, \mathbf{x}) < \epsilon$$

for all indices $n \geq n_\epsilon$. We observe that

$$
\begin{aligned}
d\left(x_{j,n}, x_j\right)^p = \left|x_{j,n} - x_j\right|^p &\leq \sum_{k=1}^{m} \left|x_{k,n} - x_k\right|^p \\
&= \left\|\mathbf{x}_n - \mathbf{x}\right\|_p^p = d_p\left(\mathbf{x}_n, \mathbf{x}\right)^p < \epsilon^p,
\end{aligned}
$$

so that $d\left(x_{j,n}, x_j\right) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular, $\left(x_{j,n}\right)_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Conversely, now suppose that for all indices $1 \leq j \leq m$, $\left(x_{j,n}\right)_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$. Let $\epsilon > 0$ be a positive real number. Then $\frac{\epsilon}{m^{\frac{1}{p}}} > 0$ is a positive real number, and so for each index $1 \leq j \leq m$ there is a natural number $n_{j,\epsilon}$ so that

$$
d\left(x_{j,n}, x_j\right) < \frac{\epsilon}{m^{\frac{1}{p}}}
$$

for all indices $n \geq n_{j,\epsilon}$. Let $n_\epsilon = \max_{1 \leq j \leq m} n_{j,\epsilon}$, and note that

$$
\begin{aligned}
d_p\left(\mathbf{x}_n, \mathbf{x}\right)^p = \sum_{j=1}^{m} \left|x_{j,n} - x_j\right|^p &= \sum_{j=1}^{m} d\left(x_{j,n}, x_j\right)^p \\
< \sum_{j=1}^{m} \left(\frac{\epsilon}{m^{\frac{1}{p}}}\right)^p &= \sum_{j=1}^{m} \frac{\epsilon^p}{m} = \epsilon^p
\end{aligned}
$$

and therefore $d_p\left(\mathbf{x}_n, \mathbf{x}\right) < \epsilon$ for all indices $n \geq n_\epsilon$. We conclude that $\left(\mathbf{x}_n\right)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the p-metric d_p on $\mathbb{R}^n$. ■

Theorem **??** Sequential convergence in the p-metric implies that, in order to know everything about sequential convergence and divergence in the metric induced on real coordinate space by a p-norm, it suffices to know about the convergence and divergence of sequences of real numbers in the Euclidean metric. Moreover, all of the p-metrics induced on real coordinate space by the p-norms are equivalent with respect to the question of sequential convergence and divergence; that is, the convergence of a sequence of real coordinate vectors converges in the metric induced on real coordinate space by a p-norm does not depend on p.

The reduction of problems of sequential convergence in a metric space to that of convergence of one or more sequences of real numbers in the Euclidean metric will be a common theme in this chapter. In some respect, this reduction is possible because

1. given a point, a metric necessarily associates the other points of a space with the positive real numbers $(0, \infty)$; and

2. convergence in the Euclidean metric d can be defined solely in terms of the standard order $\leq$ on the real line $\mathbb{R}$.

In the following section, we will discuss sequential convergence and divergence in the Euclidean metric on the real numbers in greater detail.

3.2.2 Pointwise and Uniform Convergence

Later in this chapter, we will establish metrics on certain collections of maps with codomain a given metric space. However, without yet defining a metric on these sets, there are two related but distinct modes of convergence of a sequence of such maps. A sequence of maps to a metric space is said to **converge**

pointwise to a limit map if the images of each input in the domain converge to their images under the limit map. Stronger than pointwise convergence is the notion of **uniform convergence**, which requires that the speed of pointwise convergence not depend on the given input.

Definition 3.2.8 Pointwise convergence/divergence. Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n\colon X \to Y$ from a set X to a metric space (Y, ρ).

- *Pointwise convergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ **converges pointwise** to a map $f\colon X \to Y$ from X to Y if for all inputs $x \in X$, the sequence $(f_n(x))_{n=0}^{\infty}$ of corresponding outputs $f_n(x) \in Y$ converges to $f(x)$.

  The sequence $(f_n)_{n=0}^{\infty}$ is called **pointwise convergent** if it converges pointwise to a map from X to Y.

- *Pointwise divergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ is called **pointwise divergent** if it does not converge pointwise to any map from X to Y.

◆

Example 3.2.9 Pointwise convergence. Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n\colon [0,1] \to [0,1]$ defined by the formulae $f_n(x) = x^n$. Since

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} x^n = \begin{cases} 0 & 0 \le x < 1 \\ 1 & x = 1, \end{cases}$$

the sequence $(f_n)_{n=0}^{\infty}$ converges pointwise to the function $f\colon [0,1] \to [0,1]$ defined piecewise by the formula

$$f(x) = \begin{cases} 0 & 0 \le x < 1 \\ 1 & x = 1. \end{cases}$$

▲

Because of the uniqueness of limits of sequences in metric spaces, if a sequence of maps to a metric space converges pointwise to a map, then this map is unique with this property. In this case, the map is called the **pointwise limit** of the sequence.

Definition 3.2.10 Pointwise limit. For any pointwise convergent sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n\colon X \to Y$ from a set X to a metric space (Y, ρ), the unique map $f\colon X \to Y$ from X to Y to which $(f_n)_{n=0}^{\infty}$ converges pointwise is called the **pointwise limit** of $(f_n)_{n=0}^{\infty}$, and is denoted

$$\lim_{n\to\infty} f_n = f.$$

◆

While some properties of a pointwise convergent sequence of maps to a metric space are shared with its pointwise limit, pointwise convergence does not in general preserve all desirable properties of maps. In particular, Example **??** Pointwise convergence implies that a pointwise limit of a sequence of continuous maps need not itself be continuous. We will explore continuity in metric spaces and its relationship with convergence later in this chapter. In order to preserve these properties, we will need a stronger notion of convergence. Informally, a sequence of maps to a metric space is said to **converge**

uniformly it converges pointwise at roughly the same rate at each point in the domain.

Definition 3.2.11 Uniform convergence. Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n\colon X \to Y$ from a set X to a metric space (Y, ρ).

- *Uniform convergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ **converges uniformly** to a map $f\colon X \to Y$ from X to Y if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

  $$\rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$

  for all inputs $x \in X$ and indices $n \geq n_\epsilon$.

  The sequence $(f_n)_{n=0}^{\infty}$ is called **uniformly convergent** if it converges uniformly to a map from X to Y.

- *Uniform divergence.*

  The sequence $(f_n)_{n=0}^{\infty}$ is called **uniformly divergent** if it does not converge uniformly to any map from X to Y.

◆

Example 3.2.12 Uniform convergence/divergence. The following are examples of uniformly convergent and divergent sequences of maps:

- Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n\colon\ [0,1] \to [0,1]$ defined by the formulae $f_n\left(x\right) = x^n$, and let $f\colon\ [0,1] \to [0,1]$ be the function defined piecewise by the formula

  $$f\left(x\right) = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1. \end{cases}$$

  We observe that

  $$\begin{aligned} d\left(f_n\left(x\right), f\left(x\right)\right) &= \left|f_n\left(x\right) - f\left(x\right)\right| = \left|x^n - 0\right| = \left|x^n\right| = x^n \\ &> \left(\frac{1}{2^{\frac{1}{n}}}\right)^n = \frac{1}{2} \end{aligned}$$

  for all positive integers $n \geq 1$ and inputs $\frac{1}{\sqrt[n]{2}} \leq x < 1$. We conclude that $(f_n)_{n=0}^{\infty}$ does not converge uniformly to f.

- Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n\colon \mathbb{R} \to [-1,1]$ defined by the formulae

  $$f_n\left(x\right) = \frac{\sin\left(nx\right)}{n},$$

  and let $f\colon \mathbb{R} \to [-1,1]$ be the function defined by the formula $f\left(x\right) = 0$. For all positive real numbers $\epsilon > 0$, the Archimedean principle implies that $n_\epsilon \cdot \epsilon > 1$ for some positive integer $n_\epsilon \geq 1$. We observe that

  $$\begin{aligned} d\left(f_n\left(x\right), f\left(x\right)\right) &= \left|f_n\left(x\right) - f\left(x\right)\right| = \left|\frac{\sin\left(nx\right)}{n} - 0\right| = \left|\frac{\sin\left(nx\right)}{n}\right| \\ &= \frac{\left|\sin\left(nx\right)\right|}{n} \leq \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon \end{aligned}$$

  for all indices $n \geq n_\epsilon$ and inputs $x \in \mathbb{R}$. We conclude that $(f_n)_{n=0}^{\infty}$ converges uniformly to f.

▲

We remark that pointwise and uniform convergence are distinct concepts; that is, there are sequences of maps to a metric space which converge pointwise but not uniformly. However, the opposite relationship is not true: uniform convergence is a stronger condition on sequences of maps to a metric space than pointwise convergence; that is, a sequence of maps to a metric space converges uniformly only if it converges pointwise.

Proposition 3.2.13 Pointwise and uniform convergence. *If a sequence of maps from a set to a metric space converges uniformly, then it converges pointwise.*

Proof. Suppose that a sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n\colon X \to Y$ from a set X to a metric space (Y, ρ) converges uniformly to a map $f\colon X \to Y$ from X to Y. Then for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$ and inputs $x \in X$. In particular, for each input $x \in X$,

$$\rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$, and so the sequence $(f_n\left(x\right))_{n=0}^{\infty}$ converges to $f\left(x\right)$. We conclude that $(f_n)_{n=0}^{\infty}$ converges pointwise to f. ■

Taken with the uniqueness of the pointwise limit of a sequence of maps to a metric space, Proposition **??** Pointwise and uniform convergence immediately implies that if a sequence of maps to a metric space converges uniformly to a map, then this map is unique with this property. In this case, the map is called the **uniform limit** of the sequence.

Definition 3.2.14 Uniform limit. For any uniformly convergent sequence $(f_n)_{n=0}^{\infty}$ of maps $f_n\colon X \to Y$ from a set X to a metric space (Y, ρ), the unique map $f\colon X \to Y$ from X to Y to which $(f_n)_{n=0}^{\infty}$ converges uniformly is called the **uniform limit** of $(f_n)_{n=0}^{\infty}$, and is denoted

$$\operatorname*{unif\,lim}_{n\to\infty} f_n = f.$$

◆

Example 3.2.15 Uniform limit. Consider the sequence $(f_n)_{n=0}^{\infty}$ of functions $f_n\colon \mathbb{R} \to [-1, 1]$ defined by the formulae

$$f_n\left(x\right) = \frac{\sin\left(nx\right)}{n},$$

and let $f\colon \mathbb{R} \to [-1, 1]$ be the function defined by the formula $f\left(x\right) = 0$. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f as shown in Example **??** Uniform convergence/divergence, f is the uniform limit of $(f_n)_{n=0}^{\infty}$; that is,

$$\operatorname*{unif\,lim}_{n\to\infty} f_n = f.$$

We might also write that

$$\operatorname*{unif\,lim}_{n\to\infty} \frac{\sin\left(nx\right)}{n} = 0.$$

▲

So a sequence of maps to a metric space converges uniformly only if it converges pointwise. The reverse implication, however, is not in general true. In the following sections, we will see that, unlike in the case of pointwise convergence, many of the desirable properties of maps between metric spaces are preserved under uniform convergence.

3.2.3 Map Convergence

Just as we can use a metric to define the convergence of a sequence of points or maps in a metric space, we can similarly consider the convergence of maps between metric spaces. In our definition of sequence convergence, we considered a sequence to converge to a limit if the points of the sequence could be made arbitrarily close to the limit point solely by choosing large enough indices. We will similarly consider a map between metric spaces to **converge** to a limit in the codomain at a point in the domain if we can make outputs close to the limit in the codomain solely by choosing inputs close to the point in the domain.

In the case of sequential convergence, we used the standard ordering on the natural numbers to induce a notion of limiting behavior by considering arbitrarily large indices. For **map convergence**, however, we use the metric in the domain to consider arbitrarily close points to the point in question.

Definition 3.2.16 Map convergence. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y).

- *Map convergence.*

  f **converges** at a point $c \in X$ to a point $l \in Y$ if for all positive real numbers $\epsilon > 0$, there exists a positive real number $\delta > 0$ so that

  $$\rho_Y\left(f\left(x\right), l\right) < \epsilon$$

  for all inputs $x \in X$ so that $0 < \rho_X\left(x, c\right) < \delta$.

  The map f is called **convergent** at a point $c \in X$ if it converges at c to some point $l \in Y$.

- *Map divergence.*

  The map f is called **divergent** at a point $c \in X$ if it does not converge at c to any point $l \in Y$.

◆

Example 3.2.17 Map convergence. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$f\left(x\right) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

As the graph of $y = f\left(x\right)$ suggests, f converges to 0 at 0 in the Euclidean metric d on $\mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number, and let $\delta = \epsilon > 0$. Then

$$\begin{aligned} d\left(f\left(x\right), 0\right) &= \left|x \sin\left(\frac{1}{x}\right) - 0\right| = |x| \cdot \left|\sin\left(\frac{1}{x}\right)\right| \\ &\leq |x| = |x - 0| = d\left(x, 0\right) < \delta = \epsilon \end{aligned}$$

for all inputs $x \in \mathbb{R}$ so that $0 < d\left(x, 0\right) < \delta$. ▲

In analogy to the case of sequential convergence, in order to study map convergence between metric spaces, it suffices to study the convergence of real-valued functions on metric spaces.

Proposition 3.2.18 Map convergence. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For each point $l \in Y$, let $g_l\colon X \to \mathbb{R}$ be the real-valued function on X defined by the formula*

$$g_l(x) = \rho_Y(f(x), l).$$

Then f converges to l at c if and only if g_l converges to 0 at c.

Proof. We first observe that

$$d(g_l(x), 0) = |\rho_Y(f(x), l) - 0| = |\rho_Y(f(x), l)| = \rho_Y(f(x), l)$$

for all $x \in X$.

Suppose first that f converges to l at c, and let $\epsilon > 0$ be a positive real number. Then there is some positive real number $\delta > 0$ so that

$$\rho_Y(f(x), l) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$. We observe that

$$d(g_l(x), 0) = \rho_Y(f(x), l) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$, and so g_l converges to 0 at c.

Conversely, now suppose that g_l converges to 0 at c, and let $\epsilon > 0$ be a positive real number. Then there is some positive real number $\delta > 0$ so that

$$d(g_l(x), 0) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$. We observe that

$$\rho_Y(f(x), l) = d(g_l(x), 0) < \epsilon$$

for all inputs $x \in X$ so that $0 < \rho_X(x, c) < \delta$, and so f converges to l at c. ■

So we can reduce the question of convergence for maps between metric spaces to that of Euclidean convergence of real-valued functions on metric spaces. We will study the important topic of convergence of such real-valued functions under the Euclidean metric on the real numbers in detail in Section **??** Limits in the Euclidean Metric.

The following theorem establishes another necessary and sufficient condition for map convergence; a map between metric spaces converges if and only if all convergent sequences in the domain can be "pushed through" to convergent sequences in the codomain.

Theorem 3.2.19 Sequential characterization of map convergence. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For all points $c \in X$ and $l \in Y$, f converges to l at c if and only if for all sequences $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$, if*

$$\lim_{n\to\infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = l.$$

Proof. First suppose that f converges to l at c, and consider a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$ which converges to c. We observe that $\rho_X(x_n, c) > 0$

for all indices $n \in \mathbb{N}$.

Let $\epsilon > 0$ be a positive real number. Then there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x\right), l\right) < \epsilon$$

for all points $x \in X$ so that $0 < \rho_X\left(x, c\right) < \delta$. Since $\delta > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$0 < \rho_X\left(x_n, c\right) < \delta$$

for all indices $n \geq n_\epsilon$. In particular, this implies that

$$\rho_Y\left(f\left(x_n\right), l\right) < \epsilon$$

for all indices $n \geq n_\epsilon$. We conclude that $\left(f\left(x_n\right)\right)_{n=0}^{\infty}$ converges to l.

Conversely, now suppose that f does not converge to l at c. Then there is a positive real number ϵ so that for all positive real numbers $\delta > 0$, there is a point $x \in X \setminus \{c\}$ so that

$$0 < \rho_X\left(x, c\right) < \delta$$

but

$$\rho_Y\left(f\left(x\right), l\right) \geq \epsilon.$$

In particular, let $\left(\delta_n\right)_{n=0}^{\infty}$ be a sequence of positive real numbers $\delta_n \geq 0$ which converges in the Euclidean metric to 0. For each index $n \in \mathbb{N}$, there is a point $x_n \in X \setminus \{c\}$ so that

$$0 < \rho_X\left(x_n, c\right) < \delta_n$$

but

$$\rho_Y\left(f\left(x_n\right), l\right) \geq \epsilon.$$

Since

$$d\left(\rho_X\left(x_n, c\right), 0\right) = \left|\rho_X\left(x_n, c\right) - 0\right| = \rho_X\left(x_n, c\right) < \delta_n$$

for all indices $n \in \mathbb{N}$, Proposition **??** Sequential convergence implies that $\left(x_n\right)_{n=0}^{\infty}$ converges to c but $\left(f\left(x_n\right)\right)_{n=0}^{\infty}$ does not converge to l. ■

The above result relates map convergence and sequential convergence. Note, however, that this theorem and the uniqueness of sequential limits in metric spaces do not necessarily imply the uniqueness of the limit of a convergent map between metric spaces. In fact, ***map limits need not be uniquely determined in some metric spaces***.

Example 3.2.20 Non-uniqueness of map limits. Consider the discrete metric δ_B on the set $B = \{0, 1\}$, and consider the function $f\colon B \to B$ defined by the formula $f\left(b\right) = b$. Then f converges to 0 at 0, and f converges to 1 at 0.

Indeed, let $c = 0$, and consider $l \in B$. For all positive real numbers $\epsilon > 0$, let $\delta = \frac{1}{2}$, and note that there are no points $b \in B$ so that $0 < \delta_B\left(b, c\right) < \frac{1}{2}$. Thus f converges to l at c vacuously. ▲

So the limit of a convergent map between metric spaces is not necessarily unique. However, this undesirable situation may be rectified by restricting our attention to certain points in the domain of such a map. Note that the uniqueness of the limit in Example **??** Non-uniqueness of map limits failed because the point $c = 0$ is far away from the rest of the set B; that is, there are no points close to but distinct from 0; equivalently, any sequence in B converging to 0 must eventually take only 0 as a value. Such points are called **isolated points**, and other points are called **limit points**.

Definition 3.2.21 Accumulation; isolation. Consider a subset $U \subseteq X$ of a metric space (X, ρ).

- *Limit point.*

  We say that U **accumulates** at a point $c \in X$ if for all positive real numbers $\epsilon > 0$, there is a point $u \in U$ so that $0 < \rho(u, c) < \epsilon$. In this case, c is called a **limit point** (or **accumulation point**) of U in X. The set of such limit points of U is called the **derived set** of U, and is denoted U'.

- *Isolated point.*

  We say that a point $c \in X$ is **isolated** from U if c is not a limit point of U; that is, c is isolated from U if there is a positive real number $\epsilon > 0$ so that $\rho(u, c) \geq \epsilon$ for all points $u \in U$ so that $u \neq c$. If $c \in U$ is isolated from U, then c is called an **isolated point** of U.

◆

Example 3.2.22 Limit points and isolated points. The following are examples of limit points and isolated points in metric spaces:

- *Euclidean space.*

  Let $n \geq 1$ be a positive integer, and consider a real number $1 \leq p < \infty$. If $\mathbf{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n$, then consider for each positive real number $\epsilon > 0$ the vector $\mathbf{x}_\epsilon = \left(x_1 + \frac{\epsilon}{2}, x_2, \ldots, x_n\right)$. We observe that

$$\begin{aligned} d_p(\mathbf{x}, \mathbf{x}_\epsilon) &= \left(\left|x_1 - \left(x_1 + \frac{\epsilon}{2}\right)\right|^p + |x_2 - x_2|^p + \cdots + |x_n - x_n|^p\right)^{\frac{1}{p}} \\ &= \left|-\frac{\epsilon}{2}\right| = \frac{\epsilon}{2}. \end{aligned}$$

  In particular,

$$0 < d_p(\mathbf{x}, \mathbf{x}_\epsilon) < \epsilon,$$

  and so $\mathbf{x}$ is a limit point of $\mathbb{R}^n$. Since $\mathbf{x}$ was chosen arbitrarily, we conclude that every point of $\mathbb{R}^n$ is a limit point.

- *Discrete space.*

  Let (X, δ_X) be a discrete metric space, and consider a point $x \in X$. Since the values of δ_X are 0 and 1, there is no point $y \in X$ so that $0 < \delta_X(x, y) < 1$. Thus x is an isolated point of X. Since x was chosen arbitrarily, we conclude that every point of (X, δ_X) is an isolated point.

- The only limit point of the set $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ in $\mathbb{R}$ is 0.

- The only limits point of the set $\left\{(-1)^n + \frac{1}{n}\right\}_{n=0}^{\infty}$ in $\mathbb{R}$ are 1 and -1.

▲

A point in a metric space is a limit point of a subset if and only if it is the limit of a sequence of points in that subset which are distinct from that point.

Proposition 3.2.23 Sequential characterization of limit points. *Let $U \subseteq X$ be a subset of a metric space (X, ρ). A point $c \in X$ is a limit point of U if and only if*

$$c = \lim_{n \to \infty} u_n$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U \setminus \{c\}$.

Proof. First suppose that c is a limit point of U. Then for each positive integer $n \geq 1$, $\frac{1}{n} > 0$, and so there is a point $u_n \in U \setminus \{c\}$ so that

$$0 < \rho(u_n, c) < \frac{1}{n}.$$

We claim that $(u_n)_{n=0}^{\infty}$ converges to c. Indeed, for any positive real number $\epsilon > 0$, the Archimedean principle implies that there is a natural number $n_\epsilon > \frac{1}{\epsilon}$. We observe that

$$\rho(u_n, c) < \frac{1}{n} \leq \frac{1}{n_\epsilon} < \epsilon$$

for all indices $n \geq n_\epsilon$. We conclude that

$$\lim_{n\to\infty} u_n = c.$$

Conversely, now suppose that

$$\lim_{n\to\infty} u_n = c$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U \setminus \{c\}$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon > 0$ so that $\rho(u_n, c) < \epsilon$ for all indices $n \geq n_\epsilon$. In particular,

$$0 < \rho(u_{n_\epsilon}, c) < \epsilon,$$

since $u_{n_\epsilon} \neq c$. We conclude that c is a limit point of U. ■

As we have seen, the limit of a convergent map between metric spaces at an isolated point of the domain need not be unique. However, the limit of such a map at a limit point of the domain is unique. This follows immediately from the above results.

Corollary 3.2.24 Uniqueness of map limits. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). If f converges at a limit point $c \in X$ of X, then it converges to a unique point $l \in Y$.*

Proof. Suppose that f converges to two points $l, m \in Y$ at a limit point $c \in X$ of X. Since c is a limit point of X,

$$c = \lim_{n\to\infty} x_n$$

for some sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X \setminus \{c\}$ by Proposition **??** Sequential characterization of limit points. Theorem **??** Sequential characterization of map convergence now implies that

$$l = \lim_{n\to\infty} f(x_n) = m.$$

■

Analogously to the case of unique sequential limits, the above corollary leads naturally to the following definition, which is that of the unique **limit** of a map between metric spaces at a limit point.

Definition 3.2.25 Map limit. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). If f converges at a limit point $c \in X$ of X, the unique point $l \in Y$ to which f converges at c is called the **limit** of f at c, and is denoted

$$\lim_{x\to c} f(x) = l.$$

◆

Again, note the ambiguity of the above notation; it is entirely unclear with respect to which metrics this convergence is satisfied. Indeed, as in the case of sequential limits, maps between sets may converge in one pair of metrics and diverge in another, or may converge to different limits in different metrics. We will try to avoid this ambiguity if at all possible. Analogously to the case of sequential limits, if the map inside the limit operator is a map between real coordinate spaces, it will usually be safe to assume that convergence is with respect to the relevant Euclidean metrics.

Example 3.2.26 Map limits. Consider the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = 2x + 1$. We claim that

$$\lim_{x \to c} f(x) = 2c + 1$$

for all points $c \in \mathbb{R}$.

Indeed, let $\epsilon > 0$ be a positive real number, and consider the positive real number $\delta = \frac{\epsilon}{2} > 0$. We observe that

$$\begin{aligned} d(f(x), 2c+1) &= |2x + 1 - (2c+1)| = |2(x-c)| \\ &= 2|x-c| = 2d(x,c) < 2\delta = \epsilon \end{aligned}$$

for all points $x \in \mathbb{R}$ so that $0 < d(x,c) < \delta$, and so f converges to $2c+1$ at c. Since c is a limit point of $\mathbb{R}$, we conclude that

$$\lim_{x \to c} f(x) = 2c + 1.$$

▲

In this section, we have seen several ways in which the convergence of objects in an arbitrary metric space can be reduced to convergence of related objects in the real line $\mathbb{R}$ taken with the Euclidean metric d. In this way, the topic of convergence in the Euclidean metric d on $\mathbb{R}$ plays a central role in the point-set topology of metric spaces. In the next section, we will investigate the limiting behavior of sequences of real numbers and of real-valued functions on metric spaces in greater depth.

3.2.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Convergence in your own words. Describe *in your own words* the given mode of convergence. ***Do not*** just copy the definition.

1. Describe *in your own words* what it means for a sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) to converge to a limit $x \in X$.
2. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge pointwise to a limit $f\colon X \to Y$.

3. Describe *in your own words* what it means for a sequence $(f_n)_{n=0}^{\infty}$ of maps $f\colon X \to Y$ from a set X to a metric space (Y, ρ) to converge uniformly to a limit $f\colon X \to Y$.

4. Describe *in your own words* what it means for a map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) to converge at a point $c \in X$ to a limit $l \in Y$.

Sequential convergence. Determine whether the following sequences of real numbers converge or diverge in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges, find the limit.

5. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real numbers $a_n \in \mathbb{R}$ defined by the formula

$$a_n = \frac{\cos(n\pi)}{n}.$$

Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

6. Consider the sequence $(b_n)_{n=0}^{\infty}$ of real numbers $b_n \in \mathbb{R}$ defined by the formula

$$b_n = n\cos(n\pi).$$

Determine whether the sequence $(b_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

7. Consider the sequence $(c_n)_{n=0}^{\infty}$ of real numbers $c_n \in \mathbb{R}$ defined by the formula

$$c_n = n\sin(n\pi).$$

Determine whether the sequence $(c_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

8. Consider the sequence $(d_n)_{n=0}^{\infty}$ of real numbers $d_n \in \mathbb{R}$ defined by the formula

$$d_n = \frac{2}{n+1}.$$

Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

9. Consider the sequence $(e_n)_{n=0}^{\infty}$ of real numbers $e_n \in \mathbb{R}$ defined by the formula

$$e_n = \frac{2n}{n+1}.$$

Determine whether the sequence $(e_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

10. Consider the sequence $(f_n)_{n=0}^{\infty}$ of real numbers $f_n \in \mathbb{R}$ defined by the formula

$$f_n = \sqrt{\frac{2n}{n+1}}.$$

Determine whether the sequence $(f_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

11. Consider the sequence $(g_n)_{n=0}^{\infty}$ of real numbers $g_n \in \mathbb{R}$ defined by the formula

$$g_n = \frac{3^n}{n!}.$$

Determine whether the sequence $(g_n)_{n=0}^{\infty}$ converges or diverges in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges, find the limit, and express your answer in limit notation.

Pointwise and uniform convergence. Determine whether the following sequences of real-valued functions converge or diverge both pointwise and uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If the given sequence converges pointwise, find the pointwise limit. Express your answer using limit notation. If the given sequence converges uniformly, find the uniform limit.

12. Consider the sequence $(a_n)_{n=1}^{\infty}$ of real-valued functions $a_n \colon \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$a_n(x) = \left(x - \frac{1}{n}\right)^2.$$

(a) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(a_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

13. Consider the sequence $(b_n)_{n=1}^{\infty}$ of real-valued functions $b_n \colon [-10, 10] \to \mathbb{R}$ defined by the formulae

$$b_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(b_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

14. Consider the sequence $(c_n)_{n=1}^{\infty}$ of real-valued functions $c_n \colon \mathbb{R} \to \mathbb{R}$ defined by the formulae

$$c_n(x) = \frac{x^2}{n}.$$

(a) Determine whether the sequence $(c_n)_{n=1}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(c_n)_{n=1}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

15. Consider the sequence $(d_n)_{n=0}^{\infty}$ of real-valued functions $d_n \colon [-10, 10] \to \mathbb{R}$ defined by the formulae

$$d_n(x) = \frac{1}{3^{x+n}}.$$

(a) Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges pointwise in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges pointwise, find the pointwise limit. Express your answer using limit notation.

(b) Determine whether the sequence $(d_n)_{n=0}^{\infty}$ converges or diverges uniformly in the Euclidean metric d on the real numbers $\mathbb{R}$. If it converges uniformly, find the uniform limit.

Limit points and isolated points. Find all limit points and isolated points of the given subset of the Euclidean plane $\mathbb{R}^2$.

16. Let U be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$U = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; 1 \leq \|\mathbf{x}\| < 2 \right\}.$$

Find all limit points and isolated points of U in $\mathbb{R}^2$.

17. Let V be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$V = \left\{ \mathbf{x} \in \mathbb{R}^2 \;\middle|\; \|\mathbf{x}\| \in \mathbb{N} \right\}.$$

Find all limit points and isolated points of V in $\mathbb{R}^2$.

18. Let W be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$W = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; x, y \in \mathbb{Z} \right\}.$$

Find all limit points and isolated points of W in $\mathbb{R}^2$.

19. Let X be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$X = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; x, y \in \mathbb{Q} \right\}.$$

Find all limit points and isolated points of X in $\mathbb{R}^2$.

20. Let Y be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Y = \left\{ (x, y) \in \mathbb{R}^2 \;\middle|\; x, y \notin \mathbb{Q} \right\}.$$

Find all limit points and isolated points of Y in $\mathbb{R}^2$.

21. Let Z be the subset of the Euclidean plane $\mathbb{R}^2$ defined by

$$Z = \left\{ \left(\frac{1}{m}, \frac{1}{n} \right) \in \mathbb{R}^2 \;\middle|\; m, n \in \mathbb{N} \setminus \{0\} \right\}.$$

Find all limit points and isolated points of Z in $\mathbb{R}^2$.

Map convergence. Determine whether or not the given real-valued function converges at the given input. If the function converges, find the limit and express your answer using limit notation.

22. Consider the real-valued function $f \colon \mathbb{R} \to \mathbb{R}$ defined piecewise by the formula

$$f(x) = \begin{cases} -1 & x < 0 \\ 0 & x = 0 \\ 1 & x > 0. \end{cases}$$

Determine whether or not f converges at 0. If f converges at 0, find the limit and express your answer using limit notation.

23. Consider the real-valued function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the formula

$$g(y) = y^2 - 2y + 1.$$

Determine whether or not g converges at 0. If g converges at 0, find the limit and express your answer using limit notation.

24. Consider the real-valued function $h\colon [0, \infty) \to \mathbb{R}$ defined by the formula

$$h(z) = \sqrt{z}.$$

Determine whether or not h converges at 4. If h converges at 4, find the limit and express your answer using limit notation.

Hint. Use the fact that $|\sqrt{x} - 2| \cdot |\sqrt{x} + 2| = |x - 4|$ for all non-negative real numbers $x \geq 0$.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.2.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Eventually constant sequences.** A sequence $(x_n)_{n=0}^{\infty}$ of elements $x_n \in X$ of a set X is called **eventually constant** if there is a point $x \in X$ and a natural number $n_0 \in \mathbb{N}$ so that $x_n = x$ for all indices $n \geq n_0$.

(a) Prove that any eventually constant sequence of points in a metric space is convergent.

(b) Prove that any sequence of points in a discrete metric space is convergent if and only if it is eventually constant.

(c) Give an example of a sequence of real numbers which converges in the Euclidean metric d on $\mathbb{R}$ but diverges in the discrete metric $\delta_{\mathbb{R}}$ on $\mathbb{R}$.

2. **Sequential convergence in the infinity metric.** Let $m \in \mathbb{N}$ be a natural number, and consider a sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ of points $\mathbf{x}_n = (x_{1,n}, \ldots, x_{m,n}) \in \mathbb{R}^m$. Prove that for any vector $\mathbf{x} = (x_1, \ldots, x_m) \in \mathbb{R}$, the following are equivalent:

(a) The sequence $(\mathbf{x}_n)_{n=0}^{\infty}$ converges to $\mathbf{x}$ in the ∞-metric d_∞ on $\mathbb{R}^m$.

(b) For all indices $1 \leq j \leq m$, the sequence $(x_{j,n})_{n=0}^{\infty}$ converges to x_j in the Euclidean metric d on $\mathbb{R}$.

Hint. You may want to revisit the proof of Theorem **??** Sequential convergence in the p-metric for inspiration.

3. **Alternative characterization of uniform convergence.** Let $(f_n)_{n=0}^{\infty}$ be a sequence of maps $f_n\colon X \to Y$ from a non-empty set $X \neq \emptyset$ to a metric space (Y, ρ). Prove that $(f_n)_{n=0}^{\infty}$ converges to a map $f\colon X \to Y$ from X to Y if and only if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$0 \leq \sup_{x \in X} \rho\left(f_n\left(x\right), f\left(x\right)\right) < \epsilon$$

for all indices $n \geq n_\epsilon$.

4. **Uniqueness of map limits.** Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) which has at least two distinct points. Prove the following partial converse to Corollary **??** Uniqueness of map limits: If f converges at a point $c \in X$ to a unique point $l \in Y$, then c is a limit point of X.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.3 Limits in the Euclidean Metric

In the previous section, we defined and explored the convergence and divergence of sequences and maps in the context of general metric spaces. In this section, however, we turn our attention to the limiting behavior in Euclidean spaces specifically. Euclidean spaces, being (at least in low dimensions) good models of the space in which we live, are the metric spaces in which our intuitions about distance are most accurate and useful.

3.3.1 Limits and Arithmetic

Much of our work in this text will rely on limits taken in Euclidean spaces, and the material introduced in this section is fundamental to the field of real analysis. Euclidean spaces are not just metric objects, however; as normed vector spaces, Euclidean spaces have a vector space structure as well as a norm and an inner product. We will see that Euclidean limits preserve this additional structure; for example, linear combinations of convergent sequences of real coordinate vectors converge in the Euclidean metric to the linear combination of the limits. In fact, this holds true in any normed vector space.

Proposition 3.3.1 Limits and vector arithmetic. *Let $\|\cdot\|$ be a norm on a real or complex vector space V, and consider the induced metric ρ on V.*

1. *Vector addition.*

   For all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ and $(\mathbf{w}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n, \mathbf{w}_n \in V$,

$$\lim_{n\to\infty} \left(\mathbf{v}_n + \mathbf{w}_n\right) = \lim_{n\to\infty} \mathbf{v}_n + \lim_{n\to\infty} \mathbf{w}_n.$$

2. *Scalar multiplication.*

For all scalars α and convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n \in V$,

$$\lim_{n\to\infty} \alpha\mathbf{v}_n = \alpha \lim_{n\to\infty} \mathbf{v}_n.$$

3. *Norm.*

   For all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n \in V$,

$$\lim_{n\to\infty} \|\mathbf{v}_n\| = \left\| \lim_{n\to\infty} \mathbf{v}_n \right\|.$$

4. *Inner product.*

   If the norm $\|\cdot\|$ is induced by an inner product $\langle \cdot\,,\,\cdot \rangle$ on V, then for all convergent sequences $(\mathbf{v}_n)_{n=0}^{\infty}$ and $(\mathbf{w}_n)_{n=0}^{\infty}$ of vectors $\mathbf{v}_n, \mathbf{w}_n \in V$,

$$\lim_{n\to\infty} \langle \mathbf{v}_n, \mathbf{w}_n \rangle = \left\langle \lim_{n\to\infty} \mathbf{v}_n, \lim_{n\to\infty} \mathbf{w}_n \right\rangle.$$

Proof.

1. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$ and $\mathbf{w} = \lim_{n\to\infty} \mathbf{w}_n$. Let $\epsilon > 0$ be a positive real number. Since $\frac{\epsilon}{2} > 0$ is a positive real number, there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{2}$$

   for all indices $n \geq m_\epsilon$ and

$$\rho(\mathbf{w}_n, \mathbf{w}) < \frac{\epsilon}{2}$$

   for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max(m_\epsilon, n_\epsilon) \in \mathbb{N}$, and note that

$$\begin{aligned}\rho(\mathbf{v}_n + \mathbf{w}_n, \mathbf{v} + \mathbf{w}) &= \|\mathbf{v}_n + \mathbf{w}_n - (\mathbf{v} + \mathbf{w})\| = \|\mathbf{v}_n - \mathbf{v} + \mathbf{w}_n - \mathbf{w}\| \\ &\leq \|\mathbf{v}_n - \mathbf{v}\| + \|\mathbf{w}_n - \mathbf{w}\| = \rho(\mathbf{v}_n, \mathbf{v}) + \rho(\mathbf{w}_n, \mathbf{w}) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon\end{aligned}$$

   for all indices $n \geq p_\epsilon$. We conclude that

$$\lim_{n\to\infty} (\mathbf{v}_n + \mathbf{w}_n) = \mathbf{v} + \mathbf{w}.$$

2. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$. Let $\epsilon > 0$ be a positive real number. Note that

$$\begin{aligned}\rho(\alpha\mathbf{v}_n, \alpha\mathbf{v}) &= \|\alpha\mathbf{v}_n - \alpha\mathbf{v}\| = |\alpha|\, \|\mathbf{v}_n - \mathbf{v}\| \\ &= |\alpha|\, \rho(\mathbf{v}_n, \mathbf{v})\end{aligned}$$

   for all indices $n \in \mathbb{N}$. If $\alpha = 0$, then

$$\rho(\alpha\mathbf{v}_n, \alpha\mathbf{v}) = |\alpha|\, \rho(\mathbf{v}_n, \mathbf{v}) = 0 \cdot \rho(\mathbf{v}_n, \mathbf{v}) = 0 < \epsilon$$

   for all indices $n \in \mathbb{N}$. We conclude that in this case,

$$\lim_{n\to\infty} \alpha\mathbf{v}_n = \alpha\mathbf{v}.$$

   Now suppose that $\alpha \neq 0$. Since $\frac{\epsilon}{|\alpha|} > 0$ is a positive real number, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{|\alpha|}$$

for all indices $n \geq m_\epsilon$. In particular,

$$\rho(\alpha \mathbf{v}_n, \alpha \mathbf{v}) = |\alpha| \rho(\mathbf{v}_n, \mathbf{v}) < |\alpha| \left(\frac{\epsilon}{|\alpha|}\right) = \epsilon.$$

We conclude that in this case,

$$\lim_{n\to\infty} \alpha \mathbf{v}_n = \alpha \mathbf{v}.$$

3. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number n_ϵ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \epsilon$$

for all indices $n \geq n_\epsilon$. We observe that Proposition **??** Reverse triangle inequality implies that

$$\begin{aligned} d(\|\mathbf{v}_n\|, \|\mathbf{v}\|) &= |\|\mathbf{v}_n\| - \|\mathbf{v}\|| = |\|\mathbf{v}_n - \mathbf{0}\| - \|\mathbf{v} - \mathbf{0}\|| \\ &= |\rho(\mathbf{v}_n, \mathbf{0}) - \rho(\mathbf{v}, \mathbf{0})| \leq \rho(\mathbf{v}_n, \mathbf{v}) < \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$, and so

$$\lim_{n\to\infty} \|\mathbf{v}_n\| = \|\mathbf{v}\|.$$

4. Let $\mathbf{v} = \lim_{n\to\infty} \mathbf{v}_n$ and $\mathbf{w} = \lim_{n\to\infty} \mathbf{w}_n$. Let $\epsilon > 0$ be a positive real number. We observe that

$$\begin{aligned} d(\langle \mathbf{v}_n, \mathbf{w}_n \rangle, \langle \mathbf{v}, \mathbf{w} \rangle) &= |\langle \mathbf{v}_n, \mathbf{w}_n \rangle - \langle \mathbf{v}, \mathbf{w} \rangle| \\ &= |\langle \mathbf{v}_n, \mathbf{w}_n \rangle - \langle \mathbf{v}, \mathbf{w}_n \rangle + \langle \mathbf{v}, \mathbf{w}_n \rangle - \langle \mathbf{v}, \mathbf{w} \rangle| \\ &= |\langle \mathbf{v}_n, \mathbf{w}_n \rangle - \langle \mathbf{v}, \mathbf{w}_n \rangle| + |\langle \mathbf{v}, \mathbf{w}_n \rangle - \langle \mathbf{v}, \mathbf{w} \rangle| \\ &= |\langle \mathbf{v}_n - \mathbf{v}, \mathbf{w}_n \rangle| + |\langle \mathbf{v}, \mathbf{w}_n - \mathbf{w} \rangle| \\ &\leq \|\mathbf{v}_n - \mathbf{v}\| \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \|\mathbf{w}_n - \mathbf{w}\| \\ &= \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) \end{aligned}$$

for all indices $n \in \mathbb{N}$ by the Cauchy--Schwarz inequality. There are four cases, depending on whether or not $\mathbf{v}$ and $\mathbf{w}$ are zero or non-zero vectors. If $\mathbf{v} = \mathbf{w} = \mathbf{0}$, then

$$d(\langle \mathbf{v}_n, \mathbf{w}_n \rangle, \langle \mathbf{v}, \mathbf{w} \rangle) \leq \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) = 0 + 0 = 0 < \epsilon.$$

If $\mathbf{v} = \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$, then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{v}_n, \mathbf{v}) < \frac{\epsilon}{\|\mathbf{w}\|}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} d(\langle \mathbf{v}_n, \mathbf{w}_n \rangle, \langle \mathbf{v}, \mathbf{w} \rangle) &\leq \rho(\mathbf{v}_n, \mathbf{v}) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho(\mathbf{w}_n, \mathbf{w}) \\ &< \left(\frac{\epsilon}{\|\mathbf{w}\|}\right) \|\mathbf{w}\| + 0 = \epsilon + 0 = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$.

If $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} = \mathbf{0}$, then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(\mathbf{w}_n, \mathbf{w}) < \frac{\epsilon}{\|\mathbf{v}\|}$$

for all indices $n \geq n_\epsilon$. We observe that

$$\begin{aligned} d\left(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle\right) &\leq \rho\left(\mathbf{v}_n, \mathbf{v}\right) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho\left(\mathbf{w}_n, \mathbf{w}\right) \\ &< 0 + \|\mathbf{v}\| \left(\frac{\epsilon}{\|\mathbf{w}\|}\right) \|\mathbf{w}\| = 0 + \epsilon = \epsilon \end{aligned}$$

for all indices $n \geq n_\epsilon$.

Finally, if $\mathbf{v} \neq \mathbf{0}$ and $\mathbf{w} \neq \mathbf{0}$, then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$\rho\left(\mathbf{v}_n, \mathbf{v}\right) < \frac{\epsilon}{2\|\mathbf{w}\|}$$

for all indices $n \geq m_\epsilon$ and

$$\rho\left(\mathbf{w}_n, \mathbf{w}\right) < \frac{\epsilon}{2\|\mathbf{v}\|}$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max\left(\{m_\epsilon, n_\epsilon\}\right)$. We observe that

$$\begin{aligned} d\left(\langle \mathbf{v}_n, \mathbf{w}_n\rangle, \langle \mathbf{v}, \mathbf{w}\rangle\right) &\leq \rho\left(\mathbf{v}_n, \mathbf{v}\right) \cdot \|\mathbf{w}\| + \|\mathbf{v}\| \cdot \rho\left(\mathbf{w}_n, \mathbf{w}\right) \\ &< \left(\frac{\epsilon}{2\|\mathbf{w}\|}\right) \|\mathbf{w}\| + \|\mathbf{v}\| \left(\frac{\epsilon}{2\|\mathbf{v}\|}\right) = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$.

In all cases, we conclude that

$$\lim_{n\to\infty} \langle \mathbf{v}_n, \mathbf{w}_n\rangle = \langle \mathbf{v}, \mathbf{w}\rangle.$$

■

Proposition **??** Limits and vector arithmetic details the ways in which the metric structure induced on a vector space by a norm or inner product interacts nicely with the additional structure of the vector space. In the real numbers $\mathbb{R}$, there are similar interactions between the metric structure and the standard ordering $\leq$.

Proposition 3.3.2 Comparison of limits. *Let $(x_n)_{n=0}^\infty$ and $(y_n)_{n=0}^\infty$ be convergent sequences of real numbers $x_n, y_n \in \mathbb{R}$. If $x_n \leq y_n$ for all indices $n \in \mathbb{N}$, then*

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n.$$

Proof. By contraposition. Let $x = \lim_{n\to\infty} x_n$ and $y = \lim_{n\to\infty} y_n$, and suppose that $x > y$. Consider the positive real number

$$\epsilon = \frac{x-y}{2}.$$

Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$d\left(x_n, x\right) < \epsilon$$

for all indices $n \geq m_\epsilon$ and

$$d\left(y_n, y\right) < \epsilon$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max\left(\{m_\epsilon, n_\epsilon\}\right) \in \mathbb{N}$. We observe that

$$x - x_{p_\epsilon} \leq \left|x_{p_\epsilon} - x\right| = d\left(x_{p_\epsilon}, x\right) < \epsilon = \frac{x-y}{2}$$

and

$$y_{p_\epsilon} - y \leq |y_{p_\epsilon} - y| = d\left(y_{p_\epsilon}, y\right) < \epsilon = \frac{x-y}{2},$$

so that

$$y_{p_\epsilon} < \frac{x+y}{2} < x_{p_\epsilon}.$$

In summary, we have shown that if $x > y$, then $x_n > y_n$ for some index $n \in \mathbb{N}$. Taking the contrapositive, we arrive at the desired result. ■

This is as good a place as any to introduce another useful result concerning the relationship between sequential convergence in the real line $\mathbb{R}$ and the standard ordering $\leq$. Called the Squeeze theorem, this result can be used cleverly to avoid lengthy convergence arguments.

Theorem 3.3.3 Squeeze theorem. *Let $(x_n)_{n=0}^{\infty}$, $(y_n)_{n=0}^{\infty}$, and $(z_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n, z_n \in \mathbb{R}$. If*

$$x_n \leq y_n \leq z_n$$

for all indices $n \in \mathbb{N}$ and $(x_n)_{n=0}^{\infty}$ and $(z_n)_{n=0}^{\infty}$ converge to the same real number $l \in \mathbb{R}$, then $(y_n)_{n=0}^{\infty}$ converges to l.

Proof. Let $\epsilon > 0$ be a positive real number. Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that

$$d\left(x_n, l\right) < \epsilon$$

for all indices $n \geq m_\epsilon$ and

$$d\left(z_n, l\right) < \epsilon$$

for all indices $n \geq n_\epsilon$. Let $p_\epsilon = \max\left(\{m_\epsilon, n_\epsilon\}\right) \in \mathbb{N}$, and note that

$$y_n - l \leq z_n - l \leq |z_n - l| = d\left(z_n, l\right) < \epsilon$$

and

$$l - y_n \leq l - x_n \leq |x_n - l| = d\left(x_n, l\right) < \epsilon$$

for all indices $n \geq p_\epsilon$. In particular, we conclude that

$$d\left(y_n, l\right) = |y_n - l| = \max\left(\{y_n - l, l - y_n\}\right) < \epsilon$$

for all indices $n \geq p_\epsilon$, and so $(y_n)_{n=0}^{\infty}$ converges to l. ■

Example 3.3.4 Squeeze theorem. Since $0 \leq \frac{1}{n^2} \leq \frac{1}{n}$ for all positive integers $n \geq 1$ and

$$\lim_{n\to\infty} 0 = \lim_{n\to\infty} \frac{1}{n} = 0,$$

we can conclude by the Squeeze theorem that

$$\lim_{n\to\infty} \frac{1}{n^2} = 0$$

without worrying about a lengthy convergence argument. ▲

Proposition **??** Limits and vector arithmetic and Proposition **??** Comparison of limits further justify the choice to emphasize the limiting behavior of objects on the real line $\mathbb{R}$ in the rest of this section; as we have already seen in Section **??** Convergence and Limits, these results further emphasize that knowledge of such limiting behavior can be used to make conclusions about the limiting behavior of a wide variety of metric objects.

3.3.2 Divergence to Infinity

In some ways, the binary distinction between convergence and divergence may be viewed as more coarse than an analyst might like. On the real line $\mathbb{R}$, we can make more qualitative statements about the limiting behavior of certain divergent sequences and maps to alleviate this issue somewhat. We now consider the special case of divergence of a sequence or map to infinity. Some sequences increase or decrease without bound. Such sequences can be said to have infinite limits, even though they don't converge to any real number.

Definition 3.3.5 Sequential divergence to infinity. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

- *Divergence to ∞.*

  The sequence $(x_n)_{n=0}^{\infty}$ is said to **diverge to** ∞ if for each real number $B \in \mathbb{R}$, there is a natural number $n_B \in \mathbb{N}$ so that $x_n > B$ for all indices $n \geq n_B$. In this case, we write

$$\lim_{n \to \infty} x_n = \infty.$$

- *Divergence to $-\infty$.*

  The sequence $(x_n)_{n=0}^{\infty}$ is said to **diverge to** $-\infty$ if for each real number $B \in \mathbb{R}$, there is a natural number $n_B \in \mathbb{N}$ so that $x_n < B$ for all indices $n \geq n_B$. In this case, we write

$$\lim_{n \to \infty} x_n = -\infty.$$

◆

Example 3.3.6 Sequential divergence to infinity. The following are examples and non-examples of sequences of real numbers which diverge to infinity.

- The sequence $(n)_{n=0}^{\infty} = (0, 1, 2, 3, \ldots)$ diverges to ∞; that is,

$$\lim_{n \to \infty} n = \infty.$$

- The sequence $((-1)^n \, n)_{n=0}^{\infty} = (0, -1, 2, -3, \ldots)$ diverges, but it does not diverge to ∞ or $-\infty$.

▲

There are also versions of divergence to infinity for real-valued functions on metric spaces.

Definition 3.3.7 Function divergence to infinity. Let $f \colon X \to \mathbb{R}$ be a real-valued function on a metric space (X, ρ).

- *Divergence to ∞.*

  The function f is said to **diverge to** ∞ at a point $c \in X$ if for each real number $B \in \mathbb{R}$, there is a positive real number $\delta > 0$ so that $f(x) > B$ for all points $x \in X$ so that $0 < \rho(x, c) < \delta$.

  In this case, if c is a limit point of X, then we write

$$\lim_{x \to c} f(x) = \infty.$$

- *Divergence to $-\infty$.*

The function f is said to **diverge to** $-\infty$ at a point $c \in X$ if for each real number $B \in \mathbb{R}$, there is a positive real number $\delta > 0$ so that $f(x) < B$ for all points $x \in X$ so that $0 < \rho(x, c) < \delta$.

In this case, if c is a limit point of X, then we write

$$\lim_{x \to c} f(x) = -\infty.$$

◆

Example 3.3.8 Function divergence to infinity. The following are examples and non-examples of real-valued functions on metric spaces which diverge to infinity

- Consider the real-valued function $f\colon \mathbb{R} \setminus \{1\} \to \mathbb{R}$ defined by the formula

$$f(x) = -\frac{1}{(x-1)^2}.$$

Then f diverges to $-\infty$ at 1; that is,

$$\lim_{x \to 1} -\frac{1}{x^2} = -\infty.$$

- Consider the real-valued function $g\colon \mathbb{R} \setminus \{0\} \to \mathbb{R}$ defined by the formula

$$g(y) = \frac{1}{y}.$$

Then g diverges at 0, but it does not diverge to ∞ or $-\infty$ at 0.

▲

Divergence of a sequence of real numbers or a real-valued function to infinity is a strong statement about the limiting behavior of such a sequence or function. However, since there is no metric on the extended real number system which agrees with the Euclidean metric on the real numbers, we cannot view divergence to infinity as convergence in a metric space of which the real line is a metric subspace. We will see in Chapter **??** Topological Spaces a way in which the situation of divergence to infinity can be said to be a form of convergence to an extended real number, namely the infinite quantities ∞ and $-\infty$.

In many ways, the symbols ∞ and $-\infty$ can be viewed as one-sided "reciprocals" of 0. While division by zero is left undefined for very important arithmetical reasons, it turns out that divergence of a sequence of nonzero real numbers to infinity implies the Euclidean convergence of the reciprocals to zero.

Proposition 3.3.9 Reciprocals of infinite limits. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of positive real numbers $x_n > 0$. Then*

$$\lim_{n \to \infty} x_n = \infty$$

if and only if

$$\lim_{n \to \infty} \frac{1}{x_n} = 0.$$

Proof. First suppose that $(x_n)_{n=0}^{\infty}$ diverges to ∞, and let $\epsilon > 0$. Then $\frac{1}{\epsilon} \in \mathbb{R}$

is a real number, and so there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$x_n > \frac{1}{\epsilon}$$

for all indices $n \geq n_\epsilon$. In particular,

$$d\left(\frac{1}{x_n}, 0\right) = \left|\frac{1}{x_n} - 0\right| = \left|\frac{1}{x_n}\right| = \frac{1}{x_n} < \epsilon$$

for all indices $n \geq n_\epsilon$, and so $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0.

Conversely, now suppose that $\left(\frac{1}{x_n}\right)_{n=0}^{\infty}$ converges to 0, and let $B \in \mathbb{R}$ be a real number. If $B \leq 0$ is non-positive, then

$$x_n > 0 \geq B$$

for all indices $n \in \mathbb{N}$. On the other hand, if $B > 0$ is positive, then $\frac{1}{B} > 0$ is a positive real number, and so there is a natural number n_B so that

$$d\left(\frac{1}{x_n}, 0\right) < \frac{1}{B}$$

for all indices $n \geq n_B$. In particular,

$$x_n = |x_n| = \frac{1}{\left|\frac{1}{x_n} - 0\right|} = \frac{1}{d\left(\frac{1}{x_n}, 0\right)} > B$$

for all indices $n \geq n_B$, and so $(x_n)_{n=0}^{\infty}$ diverges to ∞. ■

So, as far as limiting behavior of sequences of real numbers goes, the infinite quantities ∞ and $-\infty$ behave like one-sided reciprocals of zero. It's incredibly important to avoid assuming anything stronger than what is explicitly stated in the proposition above. Specifically, the hypothesis that the sequence in question have positive real number values is crucial, and the statement is not necessarily true without this requirement.

A sequence of real numbers diverges to ∞ or $-\infty$ if it increases or decreases, respectively, without bound. This notion of **boundedness** is directly related to the one introduced in Section **??** Introduction to Order Theory for subsets of partially ordered sets. Informally, a map with codomain a partially ordered set is considered to be bounded if its image is bounded.

Definition 3.3.10 Map boundedness. Let $f\colon X \to Y$ be a map from a set X to a partially ordered set $(Y, \preceq)$.

- *Upper bound.*

  An output $y \in Y$ is an **upper bound** for the map f if it is an upper bound for the image $\operatorname{im}(f) = f(X)$; that is, y is an upper bound for f if $f(x) \preceq y$ for all inputs $x \in X$.

  The map f is called **bounded from above** if there is an upper bound for f, and **unbounded from above** if there is no such upper bound.

- *Lower bound.*

  An output $y \in Y$ is a **lower bound** for the map f if it is a lower bound for the image $\operatorname{im}(f) = f(X)$; that is, y is a lower bound for f if $y \preceq f(x)$ for all inputs $x \in X$.

The map f is called **bounded from below** if there is a lower bound for f, and **unbounded from below** if there is no such lower bound.

- *Boundedness.*

  The map f is called **bounded** if it is both bounded from above and bounded from below, and f is called **unbounded** otherwise if it is either unbounded from above or unbounded from below (or both).

◆

Example 3.3.11 Sequential boundedness in the real line. A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded if and only if there are real numbers $a, b \in \mathbb{R}$ such that $a \leq x_n \leq b$ for all indices $n \in \mathbb{N}$. For example, the sequence $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is bounded, because

$$0 < \frac{1}{n} \leq 1$$

for all indices $n \geq 1$. ▲

Clearly, any sequence of real numbers or any real-valued function which diverges to infinity cannot be bounded. However, the converse is not true; unbounded sequences need not diverge to infinity. We have already seen one such example of this in Example **??** Function divergence to infinity.

Considered as a real-valued function on the set $\mathbb{N}$ of natural numbers, a sequence of real numbers is a map whose codomain is a totally ordered set. However, the domain of such a sequence is also totally ordered. Because the metric structure on the real line $\mathbb{R}$ can be described in terms of the standard ordering $\leq$, sequences which preserve these orders have special convergence properties. Such sequences are called **monotone**.

Definition 3.3.12 Monotonicity. Let $f\colon X \to Y$ be a map from a partially ordered set $(X, \preceq_X)$ to a partially ordered set $(Y, \preceq_Y)$.

- *Isotonicity.*

  The map f is called **non-decreasing**, **order-preserving**, or **isotone** if for all inputs $x_1, x_2 \in X$, if $x_1 \preceq_X x_2$, then $f(x_1) \preceq_Y f(x_2)$. The map f is called **increasing**, **strictly order-preserving**, or **strictly isotone** if it is non-decreasing and injective.

- *Antitonicity.*

  The map f is called **non-increasing**, **order-reversing**, or **antitone** if for all inputs $x_1, x_2 \in X$, if $x_1 \preceq_X x_2$, then $f(x_2) \preceq_Y f(x_1)$. The map f is called **decreasing**, **strictly order-reversing**, or **strictly antitone** if it is non-increasing and injective.

- *Monotonicity.*

  The map f is called **monotone** if it is either non-decreasing or non-increasing, and f is called **strictly monotone** if it is either increasing or decreasing.

◆

It turns out that all monotone sequences of real numbers have limits, in the sense that a monotone sequence of real numbers either converges to a real number or diverges to infinity. In fact, something slightly stronger is true; the limit of a non-decreasing sequence exists and is its supremum, and the limit of a non-increasing sequence exists and is its infimum. This useful fact is a property of the extended real number system $\mathbb{R}$. Other totally ordered number subsystems, such as the rational numbers $\mathbb{Q}$, do not enjoy this property, since suprema and infima of sets need not exist in these number systems, which are Dedekind incomplete in the sense introduced in Section **??** The Complete Ordered Field.

Theorem 3.3.13 Monotone convergence theorem. *Let* $(x_n)_{n=0}^{\infty}$ *be a sequence of real numbers* $x_n \in \mathbb{R}$*.*

1. *If* $(x_n)_{n=0}^{\infty}$ *is non-decreasing, then*

$$\lim_{n\to\infty} x_n = \sup_{n\in\mathbb{N}} x_n.$$

2. *If* $(x_n)_{n=0}^{\infty}$ *is non-increasing, then*

$$\lim_{n\to\infty} x_n = \inf_{n\in\mathbb{N}} x_n.$$

In particular, if $(x_n)_{n=0}^{\infty}$ *is bounded and monotone, then it is convergent.*

Proof.

1. We observe that $\{x_n\}_{n=0}^{\infty} \neq \emptyset$ is non-empty, and so $\sup_{n\in\mathbb{N}} x_n = \sup\left(\{x_n\}_{n=0}^{\infty}\right) \neq -\infty$. There are two cases; either $\sup_{n\in\mathbb{N}} x_n = \infty$ or $\sup_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number.

   First suppose that $\sup_{n\in\mathbb{N}} x_n = \infty$, and let $B \in \mathbb{R}$ be a real number. Since

$$B < \infty = \sup_{n\in\mathbb{N}} x_n,$$

   B is not an upper bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_B} > B$ for some natural number $n_B \in \mathbb{N}$. We observe that since $(x_n)_{n=0}^{\infty}$ is non-decreasing,

$$x_n \geq x_{n_B} > B$$

   for all indices $n \geq n_B$, and so

$$\lim_{n\to\infty} x_n = \infty = \sup_{n\in\mathbb{N}} x_n.$$

   On the other hand, now suppose that $S = \sup_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number, and let $\epsilon > 0$ be a positive real number. Since

$$S - \epsilon < S = \sup_{n\in\mathbb{N}} x_n,$$

   $S - \epsilon$ is not an upper bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_\epsilon} > S - \epsilon$ for some natural number $n_\epsilon \in \mathbb{N}$. Since $(x_n)_{n=0}^{\infty}$ is non-decreasing,

$$S - \epsilon < x_{n_\epsilon} \leq x_n \leq S$$

   for all indices $n \geq n_\epsilon$, and so

$$\begin{aligned} d\left(x_n, S\right) &= |x_n - S| = S - x_n \\ &< S - (S - \epsilon) = \epsilon \end{aligned}$$

   for all indices $n \geq n_\epsilon$. We conclude that

$$\lim_{n\to\infty} x_n = S = \sup_{n\in\mathbb{N}} x_n.$$

2. We observe that $\{x_n\}_{n=0}^{\infty} \neq \emptyset$ is non-empty, and so $\inf_{n\in\mathbb{N}} x_n = \inf(\{x_n\}_{n=0}^{\infty}) \neq \infty$. There are two cases; either $\inf_{n\in\mathbb{N}} x_n = -\infty$ or $\inf_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number.

   First suppose that $\inf_{n\in\mathbb{N}} x_n = -\infty$, and let $B \in \mathbb{R}$ be a real number. Since
$$B > -\infty = \inf_{n\in\mathbb{N}} x_n,$$
   B is not a lower bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_B} < B$ for some natural number $n_B \in \mathbb{N}$. We observe that since $(x_n)_{n=0}^{\infty}$ is non-increasing,
$$x_n \leq x_{n_B} < B$$
   for all indices $n \geq n_B$, and so
$$\lim_{n\to\infty} x_n = -\infty = \inf_{n\in\mathbb{N}} x_n.$$

   On the other hand, now suppose that $I = \inf_{n\in\mathbb{N}} x_n \in \mathbb{R}$ is a real number, and let $\epsilon > 0$ be a positive real number. Since
$$I + \epsilon > S = \inf_{n\in\mathbb{N}} x_n,$$
   $I + \epsilon$ is not a lower bound for $\{x_n\}_{n=0}^{\infty}$; that is, $x_{n_\epsilon} < I + \epsilon$ for some natural number $n_\epsilon \in \mathbb{N}$. Since $(x_n)_{n=0}^{\infty}$ is non-increasing,
$$I + \epsilon > x_{n_\epsilon} \geq x_n \geq I$$
   for all indices $n \geq n_\epsilon$, and so
$$\begin{aligned} d(x_n, I) &= |x_n - I| = x_n - I \\ &< I + \epsilon - I = \epsilon \end{aligned}$$
   for all indices $n \geq n_\epsilon$. We conclude that
$$\lim_{n\to\infty} x_n = I = \inf_{n\in\mathbb{N}} x_n.$$

■

What we have just shown will prove incredibly useful; in particular, we will use the fact that all bounded, monotone sequences of real numbers are convergent several times throughout this chapter. The converse is not true; while convergent sequences of real numbers are bounded, not all such sequences need be monotone.

3.3.3 Upper and Lower Limits

If a sequence of points in a metric space is convergent, then its image has exactly one limit point. More generally, studying the limit points of the image of a sequence of points is a fruitful exercise regardless of whether or not the series converges. We will see that the Dedekind completeness of the real numbers $\mathbb{R}$ allows us to characterize these limit points via the **upper** and **lower limits**.

Lemma 3.3.14 Monotonicity of suprema/infima. *Let $U, V \subseteq X$ be subsets of a partially ordered set $(X, \preceq)$, and suppose that $U \subseteq V$.*

1. *If U and V have suprema, then $\sup(U) \leq \sup(V)$.*

2. *If U and V have infima, then* $\inf(U) \geq \inf(V)$.

Proof.

1. If $U \subseteq V$, then every upper bound for V is an upper bound for U, and so

$$\begin{aligned}\sup(U) &= \min(\{x \in X \mid u \preceq x \text{ for all } u \in U\}) \\ &\leq \min(\{x \in X \mid v \preceq x \text{ for all } v \in V\}) = \sup(V).\end{aligned}$$

2. If $U \subseteq V$, then every lower bound for V is a lower bound for U, and so

$$\begin{aligned}\inf(U) &= \max(\{x \in X \mid x \preceq u \text{ for all } u \in U\}) \\ &\geq \min(\{x \in X \mid x \preceq v \text{ for all } v \in V\}) = \inf(V).\end{aligned}$$

■

Definition 3.3.15 Upper/lower limit. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

- *Upper limit.*

  The sequence $\left(\sup_{n\geq m} x_n\right)_{m=0}^{\infty}$ is non-increasing, and so by the Monotone convergence theorem has a well-defined limit in the extended real numbers $\mathbb{R}\cup\{\pm\infty\}$. This limit is called the **upper limit** or **limit supremum** of the sequence $(x_n)_{n=0}^{\infty}$, and is denoted

$$\limsup_{n\to\infty} x_n = \lim_{m\to\infty} \sup_{n\geq m} x_n = \inf_{m\in\mathbb{N}} \sup_{n\geq m} x_n.$$

- *Lower limit.*

  The sequence $(\inf_{n\geq m} x_n)_{m=0}^{\infty}$ is non-decreasing, and so by the Monotone convergence theorem has a well-defined limit in the extended real numbers $\mathbb{R}\cup\{\pm\infty\}$. This limit is called the **lower limit** or **limit infimum** of the sequence $(x_n)_{n=0}^{\infty}$, and is denoted

$$\liminf_{n\to\infty} x_n = \lim_{m\to\infty} \inf_{n\geq m} x_n = \sup_{m\in\mathbb{N}} \inf_{n\geq m} x_n.$$

◆

Example 3.3.16 Upper/lower limits. Consider the sequence

$$x_n = (-1)^n \left(1 + \frac{1}{n}\right) = (-1)^n \left(\frac{n+1}{n}\right).$$

Then $\limsup_{n\to\infty} x_n = 1$, and $\liminf_{n\to\infty} x_n = -1$. ▲

Proposition 3.3.17 Properties of upper limits. *Upper limits of sequences of real numbers have the following properties:*

1. *A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded from above if and only if*

$$\limsup_{n\to\infty} x_n < \infty.$$

2. *For any sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ of real numbers $x_n, y_n \in \mathbb{R}$,*

$$\limsup_{n\to\infty}(x_n + y_n) \leq \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n$$

whenever the right-hand side of the above inequality is well-defined.

3. *For any non-negative real number $\alpha \in [0, \infty)$ and sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$,*

$$\limsup_{n\to\infty} \alpha x_n = \alpha \limsup_{n\to\infty} x_n$$

whenever the right-hand side of the above equality is well-defined.

Proof.

1. Suppose that $(x_n)_{n=0}^{\infty}$ is bounded from above. Then there is a real number $B \in \mathbb{R}$ so that $x_n \leq B$ for all indices $n \in \mathbb{N}$. In particular, for all natural numbers $m \in \mathbb{N}$, $x_n \leq B$ for all indices $n \geq m$, and so $\sup_{n\geq m} x_n \leq B$. Proposition **??** Comparison of limits implies that

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup_{n\geq m} x_n \leq B < \infty.$$

Conversely, now suppose that $\limsup_{n\to\infty} x_n < \infty$. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n\geq m} x_n$, and let

$$S = \lim_{m\to\infty} S_m = \lim_{m\to\infty} \sup_{n\geq m} x_n = \limsup_{n\to\infty} x_n.$$

Since 1 is a positive real number, there is a natural number $m_1 \in \mathbb{N}$ so that $d(S_m, S) < 1$ for all indices $m \geq m_1$. In particular,

$$S_{m_1} - S \leq |S_{m_1} - S| = d(S_{m_1}, S) < 1,$$

and so $\sup_{n\geq m_1} x_n = S_{m_1} < S + 1$. Let

$$B = \max(\{x_0, x_1, \ldots, x_{m_1-1}, S+1\}),$$

and note that $x_n \leq B$ for all indices $n \in \mathbb{N}$. Since such an upper bound exists, we conclude that $(x_n)_{n=1}^{\infty}$ is bounded from above.

2. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n\geq m} x_n$ and $T_m = \sup_{n\geq m} y_n$. We observe that $x_n + y_n \leq S_m + T_m$ for all natural numbers $m \in \mathbb{N}$ and indices $n \geq m$. Thus

$$\sup_{n\geq m} (x_n + y_n) \leq S_m + T_m$$

for all natural numbers $m \in \mathbb{N}$, and so Proposition **??** Comparison of limits and (1) of Proposition **??** Limits and vector arithmetic together imply that

$$\begin{aligned}\limsup_{n\to\infty} (x_n + y_n) &= \lim_{n\to\infty} \sup_{n\geq m} (x_n + y_n) \leq \lim_{m\to\infty} (S_m + T_m) \\ &= \lim_{m\to\infty} S_m + \lim_{m\to\infty} T_m = \limsup_{n\to\infty} x_n + \limsup_{n\to\infty} y_n.\end{aligned}$$

3. If $\alpha = 0$, then our hypotheses imply that $\limsup_{n\to\infty} x_n \in \mathbb{R}$ is a real number, and so

$$\limsup_{n\to\infty} \alpha x_n = \limsup_{n\to\infty} 0 = 0 = 0 \limsup_{n\to\infty} x_n.$$

Thus we may suppose that $\alpha > 0$ is positive. In this case, multiplication by α is a strictly order-preserving bijection of $\mathbb{R} \cup \{\pm\infty\}$, and so

$$\sup_{n\geq m} \alpha x_n = \sup(\{\alpha x_n\}_{n=m}^{\infty}) = \alpha \sup(\{x_n\}_{n=m}^{\infty}) = \alpha \sup_{n\geq m} x_n$$

for all natural numbers $m \in \mathbb{N}$. We conclude that

$$\begin{aligned}\limsup_{n\to\infty} \alpha x_n &= \lim_{m\to\infty} \sup_{n\geq m} \alpha x_n = \lim_{m\to\infty} \alpha \sup_{n\geq m} x_n \\ &= \alpha \lim_{m\to\infty} \sup_{n\geq m} x_n = \alpha \limsup_{n\to\infty} x_n.\end{aligned}$$

■

One natural follow-up from Proposition **??** Properties of upper limits is the question of negative scalings of sequences: If $\alpha < 0$ is a negative real number and $(x_n)_{n=0}^{\infty}$ is a sequence of real numbers $x_n \in \mathbb{R}$, what is the upper limit $\limsup_{n\to\infty} \alpha x_n$?

Lemma 3.3.18 Upper and lower limits. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers. Then*

$$\limsup_{n\to\infty} (-x_n) = -\liminf_{n\to\infty} x_n$$

and

$$\liminf_{n\to\infty} (-x_n) = -\limsup_{n\to\infty} x_n.$$

Here we define $-(\infty) = -\infty$ and $-(-\infty) = \infty$.

Proof. We observe that

$$\begin{aligned}\sup_{n\geq m} (-x_n) &= \min\left(\{x \in [-\infty,\infty] \mid -x_n \leq x \text{ for all } n \geq m\}\right) \\ &= \min\left(\{x \in [-\infty,\infty] \mid -x \leq x_n \text{ for all } n \geq m\}\right) \\ &= \min\left(\{-x \in [-\infty,\infty] \mid x \leq x_n \text{ for all } n \geq m\}\right) \\ &= -\max\left(\{-x \in [-\infty,\infty] \mid x \leq x_n \text{ for all } n \geq m\}\right) = -\inf_{n\geq m} x_n\end{aligned}$$

for all natural numbers $m \in \mathbb{N}$, and so

$$\begin{aligned}\limsup_{n\to\infty} (-x_n) &= \lim_{n\to\infty} \sup_{n\geq m} (-x_n) \\ &= \lim_{n\to\infty} -\inf_{n\geq m} x_n \\ &= -\lim_{n\to\infty} \inf_{n\geq m} x_n = -\liminf_{n\to\infty} x_n.\end{aligned}$$

Applying this result to the sequence $(-x_n)_{n=0}^{\infty}$, we see that

$$\begin{aligned}-\limsup_{n\to\infty} x_n &= -\limsup_{n\to\infty} (-(-x_n)) \\ &= -\left(-\liminf_{n\to\infty} (-x_n)\right) = \liminf_{n\to\infty} (-x_n).\end{aligned}$$

■

In addition to answering our question about the behavior of upper limits under scaling by negative real numbers, we can use Lemma **??** Upper and lower limits to translate results about upper limits into results about lower limits. For example, each of the properties in Proposition **??** Properties of upper limits has an analogue for lower limits.

Corollary 3.3.19 Properties of lower limits. *Lower limits of sequences of real numbers have the following properties:*

1. *A sequence $(x_n)_{n=0}^{\infty}$ of real numbers $x_n \in \mathbb{R}$ is bounded from below if and*

only if

$$\liminf_{n\to\infty} x_n > -\infty.$$

2. *For any sequences* $(x_n)_{n=0}^{\infty}$ *and* $(y_n)_{n=0}^{\infty}$ *of real numbers* $x_n, y_n \in \mathbb{R}$,

$$\liminf_{n\to\infty} (x_n + y_n) \geq \liminf_{n\to\infty} x_n + \liminf_{n\to\infty} y_n$$

whenever the right-hand side of the above inequality is well-defined.

3. *For any non-negative real number* $\alpha \in [0, \infty)$ *and sequence* $(x_n)_{n=0}^{\infty}$ *of real numbers* $x_n \in \mathbb{R}$,

$$\liminf_{n\to\infty} \alpha x_n = \alpha \liminf_{n\to\infty} x_n$$

whenever the right-hand side of the above equality is well-defined.

Finally, we end with a characterization of convergence and divergence to infinity in terms of upper and lower limits.

Theorem 3.3.20 Convergence and upper and lower limits. *Let* $(x_n)_{n=0}^{\infty}$ *be a sequence of real numbers* $x_n \in \mathbb{R}$.

1. *Convergence.*

   For any real number $l \in \mathbb{R}$, *the sequence* $(x_n)_{n=0}^{\infty}$ *converges to* l *if and only if*

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

2. *Divergence to* ∞.

   The sequence $(x_n)_{n=0}^{\infty}$ *diverges to* ∞ *if and only if*

$$\liminf_{n\to\infty} x_n = \infty.$$

3. *Divergence to* $-\infty$.

   The sequence $(x_n)_{n=0}^{\infty}$ *diverges to* $-\infty$ *if and only if*

$$\limsup_{n\to\infty} x_n = -\infty.$$

Proof.

1. For each natural number $m \in \mathbb{N}$, let $S_m = \sup_{n\geq m} x_n$ and $I_m = \inf_{n\geq m} x_n$. First suppose that $(x_n)_{n=0}^{\infty}$ converges to l. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$l - \epsilon < x_n < l + \epsilon$$

   for all indices $n \geq n_\epsilon$. In particular,

$$l - \epsilon < I_m \leq S_m < l + \epsilon$$

   for any natural number $m \geq n_\epsilon$, and so

$$l - \epsilon \leq \lim_{m\to\infty} I_m \leq \lim_{m\to\infty} S_m \leq l + \epsilon$$

   by Proposition **??** Comparison of limits. Since this holds for all positive real numbers $\epsilon > 0$,

$$\limsup_{m\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

Conversely, now suppose that

$$\limsup_{m\to\infty} x_n = \liminf_{n\to\infty} x_n = l.$$

Let $\epsilon > 0$ be a positive real number. Then there are natural numbers $m_\epsilon, n_\epsilon \in \mathbb{N}$ so that $d(S_m, l) < \epsilon$ for all indices $m \geq m_\epsilon$ and $d(I_m, l) < \epsilon$ for all indices $m \geq n_\epsilon$. Let $p_\epsilon = \max(\{m_\epsilon, n_\epsilon\}) \in \mathbb{N}$, and note that

$$\begin{aligned} x_n - l &\leq S_{p_\epsilon} - l \leq |S_{p_\epsilon} - l| \\ &= d(S_{p_\epsilon}, l) < \epsilon \end{aligned}$$

and

$$\begin{aligned} l - x_n &\leq l - I_{p_\epsilon} \leq |I_{p_\epsilon} - l| \\ &= d(I_{p_\epsilon}, l) < \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$. We conclude that

$$\begin{aligned} d(x_n, l) &= |x_n - l| \\ &= \max(\{x_n - l, l - x_n\}) < \epsilon \end{aligned}$$

for all indices $n \geq p_\epsilon$, and so $(x_n)_{n=0}^{\infty}$ converges to l.

■

You will be given the opportunity to prove (2) and (3) of Theorem **??** Convergence and upper and lower limits in the problems for this section.

In Chapter **??** Topological Spaces, we will reexamine limits, convergence, and divergence to infinity in the extended real number system as an example of the more general context of topological spaces. For now, however, we will return to our study of the point-set topology of metric spaces.

In this section, we have seen that much of the metric structure of the real line $\mathbb{R}$ can be phrased in terms of the standard ordering $\leq$. In particular, we have seen that convergence in the Euclidean metric is deeply related to the properties of small intervals of real numbers. In the next section, we will examine the analogs of these intervals in general metric spaces.

3.3.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Limits and arithmetic. Compute the limits of the given sequences of real numbers.

1. Compute $\lim_{n\to\infty} \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3}$.

2. Compute $\lim_{n\to\infty} \frac{2n+1}{3n}$.

3. Compute $\lim_{n\to\infty} \frac{1-n^2}{n}$.

4. **Limits and strict comparison.** Give examples of convergent sequences $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ of real numbers $x_n, y_n \in \mathbb{R}$ so that $x_n < y_n$ for all indices $n \in \mathbb{N}$ but

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} y_n.$$

Computing upper/lower limits. Compute the upper and lower limits of the given sequences of real numbers.

5. Compute $\limsup_{n\to\infty} 2^n$ and $\liminf_{n\to\infty} 2^n$.

6. Compute $\limsup_{n\to\infty} 2^{-n}$ and $\liminf_{n\to\infty} 2^{-n}$.

7. Compute $\limsup_{n\to\infty} (-2)^n$ and $\liminf_{n\to\infty} (-2)^n$.

8. Compute $\limsup_{n\to\infty} (-2)^{-n}$ and $\liminf_{n\to\infty} (-2)^{-n}$.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.3.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Limits and scalar multiplication.** Let $\|\cdot\|$ be a norm on a real or complex vector space V. Prove that

$$\lim_{n\to\infty} \alpha_n \mathbf{v}_n = \lim_{n\to\infty} \alpha_n \lim_{n\to\infty} \mathbf{v}_n$$

for any convergent sequences $(\alpha_n)_{n=0}^{\infty}$ and $(\mathbf{v}_n)_{n=0}^{\infty}$ of scalars α_n and vectors $\mathbf{v}_n \in V$.

2. **Limits and division.** Prove that

$$\lim_{n\to\infty} \frac{1}{x_n} = \frac{1}{\lim_{n\to\infty} x_n}$$

for every sequence $(x_n)_{n=0}^{\infty}$ of non-zero real numbers $x_n \in \mathbb{R} \setminus \{0\}$ which converges to a non-zero real number limit

$$\lim_{n\to\infty} x_n \neq 0.$$

3. **Squeezing to infinity.** Let $(x_n)_{n=0}^{\infty}$ and $(y_n)_{n=0}^{\infty}$ be sequences of real numbers $x_n, y_n \in \mathbb{R}$.

(a) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(x_n)_{n=0}^{\infty}$ diverges to ∞, then $(y_n)_{n=0}^{\infty}$ diverges to ∞.

(b) Prove that if $x_n \leq y_n$ for all indices $n \in \mathbb{N}$ and $(y_n)_{n=0}^{\infty}$ diverges to $-\infty$, then $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$.

We can interpret the above results as versions of the Squeeze theorem for divergence to infinity.

4. Reciprocals of infinite limits. Prove that a sequence $(x_n)_{n=0}^{\infty}$ of negative real numbers $x_n < 0$ diverges to $-\infty$ if and only if

$$\lim_{n\to\infty} \frac{1}{x_n} = 0.$$

5. Divergence and upper/lower limits. In this problem, you will complete the proof of Theorem **??** Convergence and upper and lower limits. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$.

(a) Prove that $(x_n)_{n=0}^{\infty}$ diverges to ∞ if and only if

$$\liminf_{n\to\infty} x_n = \infty.$$

(b) Prove that $(x_n)_{n=0}^{\infty}$ diverges to $-\infty$ if and only if

$$\limsup_{n\to\infty} x_n = -\infty.$$

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.4 Open and Closed Sets

In this section, we generalize the notion of open and closed intervals to subsets of arbitrary metric spaces. Recall that the open interval (a, b) between two real numbers $a, b \in \mathbb{R}$ is the subset of the real numbers $\mathbb{R}$ defined as follows.

$$(a, b) = \{x \in \mathbb{R} : a < x < b\} \subsetneq \mathbb{R}.$$

Similarly, the closed interval $[a, b]$ between two real numbers $a, b \in \mathbb{R}$ is the subset of the real numbers $\mathbb{R}$ defined as follows.

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\} \subsetneq \mathbb{R}.$$

Soon, we will develop the more general notion of the open and closed subsets of a metric space, which will play a crucial role in the remainder of this text.

3.4.1 Neighborhoods in Metric Spaces

Informally, a subset of a metric space is **open** if it contains all points which are close to its elements and **closed** if it contains all of its limit points. These are one of many provably equivalent definitions of openness and closedness. In order to formalize this, we make the following preliminary definitions.

Definition 3.4.1 Open ball; closed disk; sphere. Fix a positive real number $r > 0$ and a point $c \in X$ in a metric space (X, ρ).

- *Open metric ball.*

  The **open (metric) ball** $\mathrm{B}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

  $$\mathrm{B}_r(c) = \{x \in X \mid \rho(x, c) < r\}.$$

- *Closed metric disk.*

The **closed (metric) disk** $\mathrm{D}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

$$\mathrm{D}_r(c) = \{x \in X \mid \rho(x, c) \leq r\}.$$

- *Open metric ball.*

  The **(metric) sphere** $\mathrm{S}_r(c)$ in X of **radius** r and **center** c is the subset of X defined by

  $$\mathrm{S}_r(c) = \{x \in X \mid \rho(x, c) = r\}.$$

◆

So an open ball contains precisely the points of distance strictly less than its radius from its center; a closed disk contains precisely the points of distance at most its radius from its center; and a sphere contains precisely the points of distance precisely its radius from its center. Just as the difference between an open and a closed interval is the question of the inclusion of its endpoints, the distinction between open balls and closed disks is that a closed disk contains all points of distance precisely its radius from its center, while an open ball does not.

We will see that in many ways, this section is an investigation of the properties of subsets of metric spaces which contain or are contained in certain open balls. The following definition, that of a **neighborhood**, will be a useful notion for both the rest of this chapter and the next chapter as well. Informally, a subset of a metric space is a neighborhood of a point if it contains all points which are very close to that point.

Definition 3.4.2 Neighborhood. A subset $U \subseteq X$ is a **neighborhood** of a point $x \in X$ in a metric space (X, ρ) if U contains an open ball centered at x; that is, U is a neighborhood of x if $\mathrm{B}_r(x) \subseteq U$ for some positive real number $r > 0$. ◆

Example 3.4.3 Neighborhoods. Every open ball and closed disk in a metric space is a neighborhood of its center. ▲

Remark 3.4.4 Locality. The notion of a neighborhood is our latest exposure to the conceptual metaphor of **locality**. We conceive of metric spaces by spatial reasoning, saying for example that for points $x, y \in X$ in a metric space (X, ρ), if $\rho(x, y)$ is a small positive number, then x and y are "close".

This idea is pervasive throughout point-set topology, and it implicitly informs much of the relevant terminology. Just as your neighborhood contains all of the people closest to where you live, a neighborhood in a metric space contains all nearby points.

We now have the terminology to make the crucial and titular definitions of this section; that of **open** and **closed subsets** of a metric space. A subset of a metric space is open if it is a neighborhood of all of its points, and is closed if its complement is open.

Definition 3.4.5 Open/closed subset. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- *Open subset.*

  U is **open** in X if it is a neighborhood of each of its points; that is, U is open if for each point $u \in U$, there is a positive real number $r > 0$ so that $\mathrm{B}_r(u) \subseteq U$.

- *Closed subset.*

  U is **closed** in X if its complement $X \setminus U$ is open in X; that is, U is closed if each point $x \in X \setminus U$ has a neighborhood in X which is disjoint from U.

◆

Lemma 3.4.6 Open balls and closed disks. *Open balls in metric spaces are open, and closed disks are closed.*

Proof. Fix a positive real number $r > 0$ and a point $c \in X$ in a metric space (X, ρ). Let $U = \mathrm{B}_r(c)$ and $V = \mathrm{D}_r(c)$ be the open ball and closed disk of radius r and center c, respectively. We will show that U and V are open and closed in X, respectively.

To that end, first let $u \in U$, so that $\rho(u, c) < r$. Let

$$s = r - \rho(u, c).$$

We note that for a point $x \in X$, if $\rho(x, u) < s$, then

$$\begin{aligned} \rho(x, c) &\leq \rho(x, u) + \rho(u, c) < s + r \\ &= r - \rho(u, c) + \rho(u, c) = r, \end{aligned}$$

and so $x \in U$. Thus $\mathrm{B}_s(u) \subseteq \mathrm{B}_r(c) = U$. Since U contains an open ball centered at each of its points, it is a neighborhood of each of its points, and so U is open in X.

On the other hand, now suppose that $x \in X \setminus V$, so that $\rho(x, c) > r$. Let

$$s = \rho(x, c) - r.$$

We note that for a point $y \in X$, if $\rho(y, x) < s$, then

$$\begin{aligned} r + s = \rho(x, c) &\leq \rho(x, y) + \rho(y, c) \\ &= \rho(y, x) + \rho(y, c) < s + \rho(y, c). \end{aligned}$$

We conclude that $\rho(y, c) > r$, and so $y \notin \mathrm{D}_r(c) = V$. Since $X \setminus V$ contains an open ball centered at each of its points, it is a neighborhood of each of its points, and so $X \setminus V$ is open in X; that is, V is closed in X. ■

Lemma **??** Open balls and closed disks illustrates that the names "open" and "closed" for balls and disks are well-deserved; open balls and closed disks in a metric space are open and closed, respectively. This immediately implies that open intervals and closed intervals of real numbers (with finite endpoints) are open and closed, respectively, in the Euclidean metric, since any such interval is either a open ball or a closed disk (depending on whether or not it includes its endpoints) around its center, which is the arithmetic mean of its endpoints.

We next develop a series of point-set topological definitions based on the notion of neighborhoods which will play an important role in both this and the next chapter. Given a subset of a metric space, we may consider open or closed sets which approximate the set as well as possible from the inside or outside, respectively.

Definition 3.4.7 Interior; closure; boundary. Fix a subset $U \subseteq X$ of a metric space (X, ρ).

- *Interior.*

  A point $x \in X$ is an **interior point** of U if U is a neighborhood of x. The set of interior points of U is called the **interior** of U and is denoted $\mathrm{Int}(U)$ or U°.

- *Closure.*

A point $x \in X$ is called a **point of closure** of U in X if $N \cap U \neq \emptyset$ for all neighborhoods N of x in X. The set of points of closure of U in X is called the **closure** of U in X and is denoted $\mathrm{Cl}\,(U)$ or $\overline{U}$.

- *Boundary.*

  A point $x \in X$ is called a **boundary point** of U in X if $N \cap U \neq \emptyset$ and $N \setminus U \neq \emptyset$ for all neighborhoods N of x in X. The set of all boundary points of U in X is called the **boundary** of U in X and is denoted $\mathrm{Bd}\,(U)$ or ∂U.

◆

The above definitions are equivalent if we quantify over open balls centered at the point in question rather than over neighborhoods of said point. However, the notion of neighborhood will extend into our discussion in Chapter **??** Topological Spaces, so we will continue in this spirit of generality, which we hope will ease our generalization of the above notions.

Example 3.4.8 Interior, closure, and boundary. Consider the half-closed, half-open interval

$$[0,1) = \{x \in \mathbb{R} \mid 0 \leq x < 1\}$$

of real numbers between 0 and 1.

- *Interior.*

  $[0,1)$ has interior the open interval

  $$\mathrm{Int}\,([0,1)) = (0,1) = \{x \in \mathbb{R} \mid 0 < x < 1\}\,.$$

- *Closure.*

  $[0,1)$ has closure the closed interval

  $$\mathrm{Cl}\,([0,1)) = [0,1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}\,.$$

- *Boundary.*

  $[0,1)$ has boundary the set $\mathrm{Bd}\,([0,1)) = \{0,1\}$.

▲

Clearly, a subset of a metric space is open if and only if it is its own interior. Perhaps less obvious, however, is the second half of the following result; a subset of a metric space is closed if and only if it is its own closure.

Proposition 3.4.9 Interior/closure. *Let $U \subseteq X$ be a subset of a metric space (X, ρ).*

1. *Interior.*

   U is open in X if and only if $U = \mathrm{Int}\,(U)$.

2. *Closure.*

   U is closed in X if and only if $U = \mathrm{Cl}\,(U)$.

Proof.

1. We first note that for any point $x \in \mathrm{Int}\,(U)$, there is some neighborhood N of x so that $N \subseteq U$. Since N is a neighborhood of x, it contains some open ball $\mathrm{B}_r\,(x) \subseteq N$. We conclude that

   $$x \in \mathrm{B}_r\,(x) \subseteq N \subseteq U,$$

and so $x \in U$. We conclude that $\mathrm{Int}(U) \subseteq U$. Therefore, it suffices to show that U is open in X if and only if $U \subseteq \mathrm{Int}(U)$.

To that end, we first suppose that U is open, and let $u \in U$. Then U is a neighborhood of u in X, and of course $U \subseteq U$. Thus $u \in \mathrm{Int}(U)$ is an interior point of U. We conclude that $U \subseteq \mathrm{Int}(U)$.

Conversely, now suppose that $U \subseteq \mathrm{Int}(U)$, and let $u \in U$. Then $u \in \mathrm{Int}(U)$, and so $N \subseteq U$ for some neighborhood N of U in X. Since N is a neighborhood of u, it contains some open ball $\mathrm{B}_r(u) \subseteq N$. We conclude that

$$\mathrm{B}_r(u) \subseteq N \subseteq U,$$

and so U is a neighborhood of u in X. Since this holds for all $u \in U$, U is open in X.

2. Let N be a neighborhood of a point $u \in U$ in X. Then N contains some open ball $\mathrm{B}_r(u)$, and so

$$u \in \mathrm{B}_r(u) \subseteq N.$$

In particular, $u \in N \cap U$, and so $N \cap U \neq \emptyset$ is non-empty. Thus $u \in \mathrm{Cl}(U)$ is a point of closure of U in X. We conclude that $U \subseteq \mathrm{Cl}(U)$. Therefore, it suffices to show that U is closed in X if and only if $\mathrm{Cl}(U) \subseteq U$.

To that end, we first suppose that U is closed, and let $x \in X$. If $x \in X \setminus U$, then our hypothesis that U is closed implies that x has a neighborhood N in X so that $N \cap U = \emptyset$. This implies that $x \in X \setminus \mathrm{Cl}(U)$. In summary, we have shown that $X \setminus U \subseteq X \setminus \mathrm{Cl}(U)$, and so $\mathrm{Cl}(U) \subseteq U$.

Conversely, now suppose that $\mathrm{Cl}(U) \subseteq U$, and let $x \in X \setminus U$. Then $x \notin \mathrm{Cl}(U)$ is not a point of closure of U in X, and so $N \cap U = \emptyset$ for some neighborhood N of x in X. This implies that $N \subseteq X \setminus U$, and so $X \setminus U$ is a neighborhood of x in X. Since this holds for all points $x \in X \setminus U$, we conclude that $X \setminus U$ is open in X; that is, U is closed in X.

■

So a subset of a metric space is open if and only if it is its own interior, and it is closed if and only if it is its own closure. One consequence of this is that the interior of a subset is open, and the closure of a set is closed. In order to show this, we make the following observation about the behavior of the openness and closedness conditions under the set operations.

Proposition 3.4.10 Topology of a metric space. *Let (X, ρ) be a metric space.*

1. *The empty set $\emptyset$ and the ground set X are both open and closed in X.*
2. *The union of any collection of open subsets of X is itself open in X.*
3. *The intersection of any collection of closed subsets of X is itself closed in X.*
4. *The intersection of finitely many open subsets of X is itself open in X.*
5. *The union of finitely many closed subsets of X is itself closed in X.*

Proof.

1. The empty set $\emptyset$ is vacuously a neighborhood of each of its points, and

so $\emptyset$ is open in X. On the other hand, let $x \in X$. Then

$$\mathrm{B}_1(x) = \{y \in X \mid \rho(y,x) < 1\} \subseteq X,$$

and so X is a neighborhood of x in X. We conclude that X is open in X.

Since the empty set $\emptyset$ and the ground set X are both open in X, their complements $X \setminus \emptyset = X$ and $X \setminus X = \emptyset$ are closed in X.

2. Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of open sets $U_j \subseteq X$ in X, and consider the union
$$U = \bigcup_{j \in \mathcal{J}} U_j.$$
Let $u \in U$. Then $u \in U_k$ for some index $k \in \mathcal{J}$. Since U_k is open in X, it is a neighborhood of x in X, and so $\mathrm{B}_r(x) \subseteq U_k$ for some positive real number $r > 0$.

   We observe that
$$\mathrm{B}_r(x) \subseteq U_k \subseteq \bigcup_{j \in \mathcal{J}} U_j = U,$$
and so U is a neighborhood of u in X. Since this holds for all points $u \in U$, U is open in X.

3. Let $(U_j)_{j \in \mathcal{J}}$ be an indexed family of closed sets $U_j \subseteq X$ in X, and consider the intersection
$$U = \bigcap_{j \in \mathcal{J}} U_j.$$
Since for each index $j \in \mathcal{J}$, $X \setminus U_j$ is open in X, (b) above implies that
$$X \setminus U = X \setminus \bigcap_{j \in \mathcal{J}} U_j = \bigcup_{j \in \mathcal{J}} (X \setminus U_j)$$
is open in X. We conclude that $U = X \setminus (X \setminus U)$ is closed in X.

4. Let $(U_j)_{j=1}^{m}$ be a finite sequence of open sets $U_j \subseteq X$ in X, and consider the intersection
$$U = \bigcap_{j=1}^{m} U_j.$$
Let $u \in U$, so that $u \in U_j$ for all indices $1 \leq j \leq m$. For each index $1 \leq j \leq m$, U_j is open in X, and so U_j is a neighborhood of u in X; that is, $\mathrm{B}_{r_j}(u) \subseteq U_j$.

   Let $r = \min_{1 \leq j \leq m} r_j > 0$, and note that
$$\mathrm{B}_r(u) \subseteq \mathrm{B}_{r_j}(u) \subseteq U_j$$
for all indices $1 \leq j \leq m$. We conclude that
$$\mathrm{B}_r(u) \subseteq \bigcap_{j=1}^{m} U_j = U,$$
so that U is a neighborhood of u in X. Since this holds for all points $u \in U$, U is open in X.

5. Let $(U_j)_{j=1}^m$ be a finite sequence of closed sets $U_j \subseteq X$ in X, and consider the union
$$U = \bigcup_{j=1}^m U_j.$$
Since for each index $1 \leq j \leq m$, $X \setminus U_j$ is open in X, (d) above implies that
$$X \setminus U = X \setminus \bigcup_{j=1}^m U_j = \bigcap_{j=1}^m (X \setminus U_j)$$
is open in X. We conclude that $U = X \setminus (X \setminus U)$ is closed in X.

■

Along with Proposition **??** Topology of a metric space, the following result implies that the interior of a subset of a metric space is the largest open set which is contained in it, the closure of such a subset is the smallest closed set in which it is contained.

Theorem 3.4.11 Interior/closure. *Let $U \subseteq X$ be a subset of a metric space (X, ρ).*

1. Interior.

U has interior
$$\mathrm{Int}\,(U) = \bigcup_{V \in \mathcal{O}_U} V,$$
where $\mathcal{O}_U$ is the collection of open subsets of X contained in U.

2. Closure.

U has closure
$$\mathrm{Cl}\,(U) = \bigcap_{V \in \mathcal{C}_U} V,$$
where $\mathcal{C}_U$ is the collection of closed subsets of X which contain U.

Proof.

1. Let $x \in X$, and first suppose that $x \in \mathrm{Int}\,(U)$ is an interior point to U in X. Then U is a neighborhood of x in X, and so $\mathrm{B}_r(x) \subseteq U$ for some positive real number $r > 0$. Lemma **??** Open balls and closed disks implies that $\mathrm{B}_r(x)$ is open in X, and so $\mathrm{B}_r(x) \in \mathcal{O}_U$. We conclude that
$$x \in \mathrm{B}_r(x) \subseteq \bigcup_{V \in \mathcal{O}_U} V.$$
Conversely, now suppose that $x \in \bigcup_{V \in \mathcal{O}_U} V$. Then $u \in V$ for some $V \in \mathcal{O}_U$. Since V is open in X, it is a neighborhood of x in X, and so $\mathrm{B}_r(x) \subseteq V$ for some positive real number $r > 0$. Since
$$\mathrm{B}_r(x) \subseteq V \subseteq U,$$
U is a neighborhood of x in X; that is, $x \in \mathrm{Int}\,(U)$ is an interior point of U in X.

In summary, we have shown that $x \in \mathrm{Int}\,(U)$ if and only if $x \in \bigcup_{V \in \mathcal{O}_U} V$, and so
$$\mathrm{Int}\,(U) = \bigcup_{V \in \mathcal{O}_U} V.$$

2. Consider an element $V \in \mathcal{C}_U$, and let $x \in X \setminus V$. Since V is closed in X, $X \setminus V$ is open in X, and so $X \setminus V$ is a neighborhood of each of its points. Moreover, since $U \subseteq V$, $(X \setminus V) \cap U = \emptyset$, and so none of the points of $X \setminus V$ are points of closure of U in X; that is, $X \setminus V \subseteq X \setminus \mathrm{Cl}\,(U)$. We conclude that $\mathrm{Cl}\,(U) \subseteq V$. Since this holds for all $V \in C$,

$$\mathrm{Cl}\,(U) \subseteq \bigcap_{V \in \mathcal{C}_U} V.$$

Conversely, now let $x \in X \setminus \mathrm{Cl}\,(U)$. Then $N \cap U = \emptyset$ for some neighborhood N of x in X. In particular, $\mathrm{B}_r\,(x) \cap U = \emptyset$ for some positive real number $r > 0$. Note that $X \setminus \mathrm{B}_r\,(x) \in \mathcal{C}_U$. However, $x \notin X \setminus \mathrm{B}_r\,(x)$, and so $x \notin \bigcap_{V \in \mathcal{C}_U} V$. We conclude that

$$\bigcap_{V \in \mathcal{C}_U} V \subseteq \mathrm{Cl}\,(U),$$

and so

$$\mathrm{Cl}\,(U) = \bigcap_{V \in \mathcal{C}_U} V..$$

■

Corollary 3.4.12 Idempotence of interior/closure. *Let (X, ρ) be a metric space.*

1. *Idempotence of interior.*

   $\mathrm{Int}\,(\mathrm{Int}\,(U)) = \mathrm{Int}\,(U)$ *for all subsets* $U \subseteq X$.

2. *Idempotence of closure.*

   $\mathrm{Cl}\,(\mathrm{Cl}\,(U)) = \mathrm{Cl}\,(U)$ *for all subsets* $U \subseteq X$.

Proof.

1. (1) of Theorem **??** Interior/closure implies that $\mathrm{Int}\,(U)$ is a union of open sets in X, and so $\mathrm{Int}\,(U)$ is itself open in X by (2) of Proposition **??** Topology of a metric space. Thus

$$\mathrm{Int}\,(\mathrm{Int}\,(U)) = \mathrm{Int}\,(U)$$

   by (1) of Proposition **??** Interior/closure.

2. (2) of Theorem **??** Interior/closure implies that $\mathrm{Cl}\,(U)$ is an intersection of closed sets in X, and so $\mathrm{Cl}\,(U)$ is itself closed in X by (3) of Proposition **??** Topology of a metric space. Thus

$$\mathrm{Cl}\,(\mathrm{Cl}\,(U)) = \mathrm{Cl}\,(U)$$

   by (2) of Proposition **??** Interior/closure.

■

For a subset $U \subseteq X$ of a metric space (X, ρ), the property of being a neighborhood of a point $x \in X$ is the existence of a certain kind lower bound on U in the power set $\mathcal{P}\,(X)$ of X. We now examine the dual condition, that of **metric boundedness**.

3.4.2 Metrics and Boundedness

In this section, we have introduced open balls and closed disks of positive radius. While the notion of radius does not easily generalize to arbitrary subsets of metric spaces (even for just open balls and closed disks), the related concept of **diameter** is easily generalized from its meaning for Euclidean circles and spheres of higher dimension. Informally, the diameter of a subset of a metric space is the largest possible value taken by the metric when considering only points in that subset.

Definition 3.4.13 Diameter. The diameter $\operatorname{diam}(U)$ of a subset $U \subseteq X$ of a metric space (X, ρ) is the extended real number defined by

$$\operatorname{diam}(U) = \begin{cases} 0 & U = \emptyset \\ \sup_{u,v \in U} \rho(u, v) & U \neq \emptyset. \end{cases}$$

◆

Lemma 3.4.14 Diameter and radius. *Let $r > 0$ be a positive real number and $c \in X$ be a point in a metric space (X, ρ). The open ball $\mathrm{B}_r(c)$, closed disk $\mathrm{D}_r(c)$, and sphere $\mathrm{S}_r(c)$ of radius r and center c have diameter at most $2r$.*

Proof. Since $c \in \mathrm{B}_r(c)$ the open ball $\mathrm{B}_r(c) \neq \emptyset$ is non-empty. Since

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) < r + r = 2r$$

for all points $u, v \in \mathrm{B}_r(c)$,

$$\operatorname{diam}(\mathrm{B}_r(c)) = \sup_{u,v \in \mathrm{B}_r(c)} \rho(u, v) \leq 2r.$$

Similarly, since $c \in \mathrm{D}_r(c)$ the closed disk $\mathrm{D}_r(c) \neq \emptyset$ is non-empty. Since

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) \leq r + r = 2r$$

for all points $u, v \in \mathrm{D}_r(c)$,

$$\operatorname{diam}(\mathrm{D}_r(c)) = \sup_{u,v \in \mathrm{D}_r(c)} \rho(u, v) \leq 2r.$$

If the sphere $\mathrm{S}_r(c) = \emptyset$ is empty, then $\operatorname{diam}(\mathrm{S}_r(c)) = \operatorname{diam}(\emptyset) = 0 < 2r$. On the other hand, if the sphere $\mathrm{S}_r(c) \neq \emptyset$ is non-empty, then we observe that

$$\rho(u, v) \leq \rho(u, c) + \rho(c, v) = \rho(u, c) + \rho(v, c) \leq r + r = 2r$$

for all points $u, v \in \mathrm{S}_r(c)$,

$$\operatorname{diam}(\mathrm{S}_r(c)) = \sup_{u,v \in \mathrm{D}_r(c)} \rho(u, v) \leq 2r.$$

■

So the diameter of an open ball, closed disk, or sphere in a metric space is bounded above by twice its radius. Note that, unlike in Euclidean spaces, the diameter of an open ball, closed disk, or sphere in arbitrary metric space need not actually reach this upper bound. This is because, unlike in Euclidean space, in which a given open ball or closed disk has a unique center and radius, balls and disks of different centers and radii may be equal. In particular, the geometry of open balls, closed disks, and spheres may be quite different in a given metric space than our intuition from Euclidean space might suggest.

Lemma 3.4.15 Monotonicity of the diameter. *Let $S, T \subseteq X$ be subsets of a metric space (X, ρ). If $S \subseteq T$, then* $\operatorname{diam}(S) \leq \operatorname{diam}(T)$.

Proof. If $T = \emptyset$, then $S = \emptyset$ as well, and so

$$\operatorname{diam}(S) = \operatorname{diam}(\emptyset) = \operatorname{diam}(T).$$

So we may suppose that $T \neq \emptyset$ is non-empty. We observe that $\rho(x, y) \geq 0$ for all points $x, y \in T$, and so

$$\operatorname{diam}(S) = 0 \leq \sup_{x,y \in T} \rho(x, y) = \operatorname{diam}(T).$$

Thus we may additionally suppose that $S \neq \emptyset$ is non-empty. In this case,

$$\begin{aligned} \operatorname{diam}(S) &= \sup(\{\rho(x, y) \in \mathbb{R} \mid x, y \in S\}) \\ &\leq \sup(\{\rho(x, y) \in \mathbb{R} \mid x, y \in T\}) = \operatorname{diam}(T) \end{aligned}$$

■

Much of this section thus far has been almost entirely concerned with the properties of subsets of metric spaces which contain certain open balls. We now consider the opposite case; what can we say about subsets which are themselves contained in open balls of finite radius? Such subsets of metric spaces are called **bounded**.

Definition 3.4.16 Boundedness. A subset $U \subseteq X$ of a metric space (X, ρ) is called **bounded** (resp. **unbounded**) if $\operatorname{diam}(U) < \infty$ (resp. $\operatorname{diam}(U) = \infty$). A metric space (X, ρ) is called **bounded** (resp. **unbounded**) if its ground set X is bounded (resp. unbounded). ◆

A priori, the above definition of boundedness does not appear to be related to the definition introduced in Section **??** Introduction to Order Theory. Namely, that kind of boundedness requires a partial order, and this one requires a metric. However, there is a way to reconcile these perspectives, and the two notions coincide in the real numbers, taken with their standard ordering $\leq$ and the Euclidean metric d.

Example 3.4.17 Boundedness. The following are examples of bounded and unbounded subsets of metric spaces:

- The empty set $\emptyset$ is a bounded subset of every metric space.
- Any discrete metric space is bounded, since its diameter is either 0 or 1.
- The n-dimensional Euclidean space $\mathbb{R}^n$ is bounded when $n = 0$ and unbounded when $n \geq 1$.

▲

Any subset of a bounded subset of a metric space is itself bounded, and any set which contains an unbounded set is itself unbounded. Moreover, any subset of a nonempty metric space is bounded if and only if it is contained in an open ball or closed disk of finite radius.

Proposition 3.4.18 Metric characterization of boundedness. *A subset of a nonempty metric space is bounded if and only if it is contained in an open ball of finite radius.*

Proof. Let $U \subseteq X$ be a subset of a non-empty metric space (X, ρ). First suppose that U is bounded, so that $\operatorname{diam}(U) < \infty$. Since $X \neq \emptyset$ is non-empty, it contains a point $x \in X$. If $U = \emptyset$ is empty, then $U = \emptyset \subseteq \mathrm{B}_r(x)$ for any positive real number $r > 0$.

On the other hand, if $U \neq \emptyset$ is non-empty, then it contains a point $u \in U$. Let $r = \operatorname{diam}(U) + 1$, and note that

$$\rho(u, v) \leq \sup_{s,t \in U} \rho(s, t) = \operatorname{diam}(U) < r$$

for all points $v \in U$, and so $U \subseteq \mathrm{B}_r(u)$.

Conversely, now suppose that $U \subseteq \mathrm{B}_r(x)$ for some point $x \in X$. Then Lemma **??** Monotonicity of the diameter and Lemma **??** Diameter and radius imply that

$$\operatorname{diam}(U) \leq \operatorname{diam}(\mathrm{B}_r(x)) \leq 2r < \infty,$$

so that U is bounded. ■

In fact, something stronger is true for normed vector spaces; a subset of a normed vector space is bounded if and only if it is contained in some open ball of finite radius centered at the origin. This kind of result is typical of analysis done on normed vector spaces; the interplay between the norm and the vector space structure implies that most point-set topological concerns can be reduced to questions which are answerable in a neighborhood of the origin.

Corollary 3.4.19 Norm characterization of boundedness. *A subset of a normed vector space is bounded if and only if it is contained in an open ball of finite radius centered at the origin.*

Proof. Let $(V, \|\cdot\|)$ be a normed vector space. By Proposition **??** Metric characterization of boundedness, it suffices to show that any bounded subset of V is contained in an open ball centered at $\mathbf{0}$. To that end, let $U \subseteq V$ be a bounded subset of V. Proposition **??** Metric characterization of boundedness implies that $U \subseteq \mathrm{B}_r(\mathbf{v})$ for some positive real number $r > 0$ and some vector $\mathbf{v}$.

Let $s = r + \|\mathbf{v}\|$, and observe that

$$\begin{aligned}\|\mathbf{u} - \mathbf{0}\| &= \|\mathbf{u} - \mathbf{v} + \mathbf{v}\| \leq \|\mathbf{u} - \mathbf{v}\| + \|\mathbf{v}\| \\ &< r + \|\mathbf{v}\| = s\end{aligned}$$

for all $\mathbf{u} \in U$. Thus $U \subseteq \mathrm{B}_s(\mathbf{0})$. ■

Any finite subset of a metric space is bounded. In fact, any finite union of bounded subsets of a metric space is itself bounded. In order to see this, it suffices to show that the binary union of two bounded subsets of a metric space is itself bounded.

Proposition 3.4.20 Unions of bounded sets. *The union of two bounded sets in a metric space is itself bounded.*

Proof. Let $U, V \subseteq X$ be bounded subsets of a metric space (X, ρ), and consider the union $U \cup V$. If $X = \emptyset$ is empty, then $U = V = \emptyset$, and so

$$\operatorname{diam}(U \cup V) = \operatorname{diam}(\emptyset \cup \emptyset) \operatorname{diam}(\emptyset) = 0 < \infty,$$

so that U and V are bounded. On the other hand, if $X \neq \emptyset$ is non-empty, then Proposition **??** Metric characterization of boundedness implies that $U \subseteq \mathrm{B}_r(x)$ and $V \subseteq \mathrm{B}_s(y)$ for some positive real numbers $r, s > 0$ and points $x, y \in X$. Let

$$t = \max(\{r, s + \rho(y, x)\}) > 0,$$

and consider a point $z \in U \cup V$. If $z \in U$, then

$$\rho(z, x) < r \leq t,$$

and if $z \in V$, then

$$\rho(z,x) \leq \rho(z,y) + \rho(y,x) < s + \rho(y,x) \leq t.$$

In either case, we conclude that $z \in \mathrm{B}_t(x)$, and so $U \cup V \subseteq \mathrm{B}_t(x)$. Proposition **??** Metric characterization of boundedness now implies that $U \cup V$ is bounded. ■

The following corollary follows immediately from the above result by induction. Essentially, it asserts that the property of boundedness is stable under finite unions.

Corollary 3.4.21 Unions of bounded sets. *The union of any finite collection of bounded subsets of a metric space is itself bounded.*

Before we move on to discuss the boundedness of maps between metric spaces, we will briefly mention one stronger kind of boundedness for subsets of metric spaces. A subset of a metric space is called **totally bounded** if it can be covered by finitely many open balls of arbitrarily small radius.

Definition 3.4.22 Total boundedness. A subset $U \subseteq X$ of a metric space (X,ρ) is called **totally bounded** if for all positive real numbers $r > 0$, there is a finite list of points $x_1, \ldots, x_n \in X$ so that

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_r(x_k).$$

A metric space (X,ρ) is called **totally bounded** if its ground set X is totally bounded. ◆

Example 3.4.23 Total boundedness. The following are examples of totally bounded and not totally bounded subsets of metric spaces:

- The empty set $\emptyset$ is a totally bounded subset of any metric space. More generally, any finite set of a metric space is totally bounded.
- Any infinite subset of a discrete metric space is bounded but not totally bounded.
- A subset of n-dimensional Euclidean space $\mathbb{R}^n$ is totally bounded if and only if it is bounded.

▲

The above examples illustrate that while boundedness and total boundedness are not in general equivalent for subsets of metric spaces, the two notions are also not necessarily entirely unrelated. In fact, total boundedness is a stronger condition than boundedness for subsets of a metric space.

Proposition 3.4.24 Total boundedness implies boundedness. *If a subset of a metric space is totally bounded, then it is bounded.*

Proof. Let $U \subseteq X$ be a totally bounded subset of a metric space (X,ρ). Since $1 > 0$ is a positive real number,

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_1(x_k)$$

for some finite list of points $x_1, \ldots, x_n \in X$. Lemma **??** Diameter and radius and Corollary **??** Unions of bounded sets together imply that U is bounded. ■

We have seen that infinite discrete metric spaces provide a counterexample to the converse of Proposition **??** Total boundedness implies boundedness. However, in Euclidean space, total boundedness and boundedness are equivalent. We offer this result without proof.

Proposition 3.4.25 Total boundedness in Euclidean space. *A subset of Euclidean space is totally bounded if and only if it is bounded.*

One way to see why Proposition **??** Total boundedness in Euclidean space is true is that any open ball or closed disk of finite radius in Euclidean space is a subset of a cube, which may be subdivided into finitely many appropriately small cubes. While boundedness and total boundedness are equivalent in Euclidean spaces, they are not the same for metric spaces in general. In some of the generalizations of theorems about Euclidean space to results about metric spaces in general, total boundedness is a more natural notion to discuss. We will further discuss totally bounded sets in the Section **??** Completeness and Compactness, in which we investigate **completeness** and **compactness** for metric spaces. For now, however, we move on in favor of exploring boundedness for maps between metric spaces.

3.4.3 Bounded maps between metric spaces

In Section **??** Introduction to Order Theory, we defined boundedness for subsets of partially ordered sets, and then we defined a map from a set to a partially ordered set to be bounded if and only if its image was a bounded subset of the codomain. In this section, we have introduced a different notion of boundedness for subsets of a metric space, but an there is an analogous way to extend this notion to maps to metric spaces. A map to a metric space is **bounded** if its image is bounded in the codomain, and unbounded otherwise.

We note that this is a different notion than a bounded linear transformation between normed vector spaces, which is not bounded in this sense. The homogeneity of a norm on a vector space implies that no nonzero linear transformation to a normed vector space can be bounded in the above sense.}

Definition 3.4.26 Map boundedness. Let $f\colon X \to Y$ be a map from a set X to a metric space (Y, ρ).

- *Boundedness.*

  The map f is called **bounded** if its image $\operatorname{im}(f) = f(X)$ is a bounded subset of Y.

- *Unboundedness.*

  The map f is called **unbounded** if its image $\operatorname{im}(f) = f(X)$ is an unbounded subset of Y.

The set of bounded maps from a set X to a metric space (Y, ρ) is denoted $\mathrm{B}(X, Y)$. In the special case $(Y, \rho) = (\mathbb{R}, d)$, the set of bounded real-valued functions on X is denoted $\mathrm{B}(X) = \mathrm{B}(X, \mathbb{R})$. ◆

Remark 3.4.27 Boundedness for linear transformations. We remark that the above definition of boundedness is different than the usual definition of boundedness for linear maps between normed vector spaces. Indeed, in the setting of normed vector spaces, the absolute homogeneity of the norm implies that no non-zero linear transformation to a normed vector space can be bounded in the above sense. Instead, a linear transformation is called **bounded** if its restriction to the unit sphere around the origin is bounded in the above sense.

Example 3.4.28 Bounded functions. Let $f\colon \mathbb{R} \to \mathbb{R}$ be the real-valued function defined by the formula

$$f(x) = e^{-x^2}.$$

Then f is a bounded function on the real line $\mathbb{R}$. ▲

Just as in the cases of sequential and map convergence, in attempting to understand boundedness in arbitrary metric spaces, it suffices to investigate boundedness in the real line $\mathbb{R}$. These links between arbitrary metric spaces and the metric space of the real numbers taken with the Euclidean metric stem from the definition of a metric; the structure of a real-valued function such as a metric can be seen to pull back some of the properties of its codomain $\mathbb{R}$ to its domain. In order to check that a metric space is bounded, it is both necessary and sufficient for the metric itself to be a bounded function.

Lemma 3.4.29 Map boundedness. *A metric space (X, ρ) is bounded if and only if the metric $\rho\colon X \times X \to \mathbb{R}$ is a bounded function.*

Proof. First suppose that (X, ρ) is bounded, so that $\operatorname{diam}(X) < \infty$. If $X = \emptyset$, then

$$\begin{aligned}\operatorname{diam}(\operatorname{im}(\rho)) &= \operatorname{diam}(\rho(\emptyset \times \emptyset)) = \operatorname{diam}(\rho(\emptyset))\\ &= \operatorname{diam}(\emptyset) = 0 < \infty,\end{aligned}$$

and so ρ is bounded. Thus we may suppose that $X \neq \emptyset$ is non-empty. In particular,

$$0 \leq \rho(x, y) \leq \operatorname{diam}(X)$$

for all points $x, y \in X$, which implies that

$$d(\rho(x, y), 0) = |\rho(x, y) - 0| = |\rho(x, y)| = \rho(x, y) \leq \operatorname{diam}(X)$$

for all points $x, y \in X$. In particular, this implies that $\operatorname{im}(\rho) \subseteq \mathrm{D}_{\operatorname{diam}(X)}(0)$. Now Lemma **??** Monotonicity of the diameter and Lemma **??** Diameter and radius imply that

$$\operatorname{diam}(\operatorname{im}(\rho)) \leq \operatorname{diam}\left(\mathrm{D}_{\operatorname{diam}(X)}(0)\right) \leq 2\operatorname{diam}(X) < \infty,$$

so that $\operatorname{im}(\rho)$ is bounded.

Conversely, now suppose that ρ is bounded. Then Corollary **??** Norm characterization of boundedness implies that $\operatorname{im}(\rho) \subseteq \mathrm{B}_r(0)$ for some positive real number $r > 0$. This implies that

$$\rho(x, y) = |\rho(x, y) - 0| = d(\rho(x, y), 0) < r$$

for all $x, y \in X$, so that $\operatorname{diam}(X) \leq r < \infty$; that is, (X, ρ) is bounded. ∎

The set of bounded maps to a metric space may inherit some of the structure of the codomain. We will see some examples of this shortly. First, however, a useful observation about the relationship between the boundedness of maps and the boundedness of their codomains.

Remark 3.4.30 Maps to bounded metric spaces. Let (Y, ρ) be a metric space. If Y is bounded, then Lemma **??** Monotonicity of the diameter implies that for any set X, any map $f\colon X \to Y$ is bounded; that is, $\mathrm{B}(X, Y) = Y^X$.

There is a canonical way of constructing a metric on the set of bounded maps to a metric space.

Definition 3.4.31 Uniform metric. Let X be a set and (Y, ρ) be a metric space. The **uniform metric** ρ_∞ on the set $\mathrm{B}(X, Y)$ of bounded maps from X to Y is the map $\rho_\infty \colon \mathrm{B}(X, Y) \times \mathrm{B}(X, Y) \to \mathbb{R}$ defined by the formula

$$\rho_\infty(f, g) = \begin{cases} 0 & X = \emptyset \\ \sup_{x \in X} \rho(f(x), g(x)) & X \neq \emptyset. \end{cases}$$

◆

Proposition 3.4.32 Uniform metric. *The uniform metric is a metric.*

Proof. Let X be a set and (Y, ρ) be a metric space. If $X = \emptyset$, then there is exactly one map from X to Y, and this map is bounded. The uniform metric ρ_∞ is identically zero in this case, and so is a metric on $\mathrm{B}(X, Y)$.

Thus we may suppose that $X \neq \emptyset$ is non-empty. Let $f, g \in \mathrm{B}(X, Y)$ be a bounded map from X to Y, and note that if $f = g$, then

$$\begin{aligned} \rho_\infty(f, g) &= \sup_{x \in X} \rho(f(x), g(x)) = \sup_{x \in X} \rho(f(x), f(x)) \\ &= \sup_{x \in X} 0 = 0. \end{aligned}$$

Conversely, if $\rho_\infty(f, g) = 0$, then

$$0 \leq \rho(f(x), g(x)) \leq \rho_\infty(f, g) = 0$$

for all inputs $x \in X$, so that $\rho(f(x), g(x)) = 0$. The non-degeneracy of the metric ρ implies that $f(x) = g(x)$ for all inputs $x \in X$, and so $f = g$. We conclude that ρ_∞ is non-degenerate.

We observe that the symmetry of the metric ρ implies that

$$\rho_\infty(f, g) = \sup_{x \in X} \rho(f(x), g(x)) = \sup_{x \in X} \rho(g(x), f(x)) = \rho_\infty(g, f)$$

for all bounded maps $f, g \in \mathrm{B}(X, Y)$, and so ρ_∞ is symmetric.

Finally, we observe that the triangle inequality for the metric ρ implies that

$$\begin{aligned} \rho_\infty(f, h) &= \sup_{x \in X} \rho(f(x), h(x)) \leq \sup_{x \in X} (\rho(f(x), g(x)) + \rho(g(x), h(x))) \\ &\leq \sup_{x \in X} \rho(f(x), g(x)) + \sup_{x \in X} \rho(g(x), h(x)) = \rho_\infty(f, g) + \rho_\infty(g, h) \end{aligned}$$

for all bounded maps $f, g, h \in \mathrm{B}(X, Y)$, and so ρ_∞ is satisfies the triangle inequality.

Since ρ_∞ is reflexive, symmetric, and satisfies the triangle inequality, it is a metric on $\mathrm{B}(X, Y)$. ■

You will have an opportunity to explore the uniform metric in the setting of normed vector spaces in the problems for this section. In particular, you will see that in this setting, the uniform metric is induced by a norm.

The similarity in the names of the uniform metric and uniformly convergent sequences of functions is no coincidence. In fact, it turns out that for bounded maps to a metric space, these two notions coincide.

Theorem 3.4.33 Convergence in the uniform metric. *Let $(f_n)_{n=0}^\infty$ be a sequence of bounded maps $f_n \in \mathrm{B}(X, Y)$ from a set X to a metric space (Y, ρ). For a bounded map $f \in \mathrm{B}(X, Y)$, $(f_n)_{n=0}^\infty$ converges uniformly to f if and only if it converges to f in the uniform metric.*

In the next section, we reintroduce one of the fundamental ideas in analysis and topology: that of **continuity**.

3.4.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Neighborhoods. Determine whether or not the given set is a neighborhood of the given point.

1. Let
$$A = (0, 2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$
be the open interval between 0 and 2. Determine whether or not A is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

2. Let
$$B = [0, 2] = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$
be the closed interval between 0 and 2. Determine whether or not B is a neighborhood of the point 1 in the metric space $(\mathbb{R}, d)$.

3. Let
$$C = [0, 2) = \{x \in \mathbb{R} \mid 0 \leq x < 2\}$$
be the half-closed, half-open interval between 0 and 2. Determine whether or not C is a neighborhood of the point 0 in the metric space $(\mathbb{R}, d)$.

4. Let
$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$
be the set of even integers. Determine whether or not D is a neighborhood of the point 0 in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

5. Let
$$E = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$
be the closed interval between 0 and 1. Determine whether or not E is a neighborhood of the point 0 in the metric subspace $([0, 2], d)$ of the real line $(\mathbb{R}, d)$.

Open and closed sets. Determine whether the given subset of a metric space is open, closed, both open and closed, or neither open nor closed.

6. Let
$$A = (0, 2) = \{x \in \mathbb{R} \mid 0 < x < 2\}$$
be the open interval between 0 and 2. Determine whether or not A is open or closed in the metric space $(\mathbb{R}, d)$.

7. Let
$$B = [0, 2] = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$
be the closed interval between 0 and 2. Determine whether or not B is open or closed in the metric space $(\mathbb{R}, d)$.

8. Let
$$C = [0, 2) = \{x \in \mathbb{R} \mid 0 \leq x < 2\}$$
be the half-closed, half-open interval between 0 and 2. Determine whether or not C is open or closed in the metric space $(\mathbb{R}, d)$.

9. Let
$$D = 2\mathbb{Z} = \{2k \in \mathbb{Z} \mid k \in \mathbb{Z}\} = \{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$$
be the set of even integers. Determine whether or not D is open or closed in the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

10. Let
$$E = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$
be the closed interval between 0 and 1. Determine whether or not E is open or closed in the metric subspace $([0, 2], d)$ of the real line $(\mathbb{R}, d)$.

11. Let
$$F = [0, 1] = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$
be the closed interval between 0 and 1. Determine whether or not F is open or closed in the metric subspace $([0, 1] \cup [2, 3], d)$ of the real line $(\mathbb{R}, d)$.

Diameter. Compute the diameter of the given subsets of metric spaces.

12. Let $A = \{0, 1, 2\}$. Compute the diameter of A as a subset of the metric space $(\mathbb{R}, d)$.

13. Let $B = \{0, 1, 2, 5\}$. Compute the diameter of B as a subset of the discrete metric space $(\mathbb{R}, \delta_{\mathbb{R}})$.

14. Let
$$C = \{0, 1, 2\} \cup \left\{3 - \frac{1}{n}\right\}_{n=1}^{\infty}.$$
Compute the diameter of C as a subset of the metric space $(\mathbb{R}, d)$.

15. Let
$$D = 3\mathbb{Z} = \{3k \in \mathbb{Z}\} \, k \in \mathbb{Z}$$
be the set of integers which are divisible by 3. Compute the diameter of D as a subset of the metric space $(\mathbb{R}, d)$.

16. Let $E = \{\pi\}$. Compute the diameter of E as a subset of the metric space $(\mathbb{R}, d)$.

17. Let $F = \{\pi\}$. Compute the diameter of F as a subset of the discrete metric space $(\mathbb{R}, \delta_X)$.

Bounded and unbounded subsets of metric spaces. Let
$$U = (0, 1) = \{x \in \mathbb{R} \mid 0 < x < 1\}$$
be the open interval between 0 and 1. Determine whether U is a bounded or unbounded subset of the given metric space.

18. Determine whether U is bounded or unbounded in the metric space $(\mathbb{R}, d)$.

19. Determine whether U is bounded or unbounded in the subspace $((0, \infty), d)$ of the metric space $(\mathbb{R}, d)$.

20. Determine whether U is bounded or unbounded in the metric space $((0,\infty),\rho)$, where ρ is the metric on $(0,\infty)$ defined by the formula

$$\rho(x,y) = \left|\ln\left(\frac{x}{y}\right)\right|.$$

21. Determine whether U is bounded or unbounded in the discrete metric space $\left((0,\infty),\delta_{(0,\infty)}\right)$.

The uniform metric. Compute the distance between the given bounded maps in the uniform metric.

22. Let $f,g\colon \mathbb{R}\to\mathbb{R}$ be the constant real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = \pi$. Compute the uniform distance $d_\infty(f,g)$.

23. Let $f,g\colon [-1,1]\to\mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2$ and $g(x) = x^2$. Compute the uniform distance $d_\infty(f,g)$.

24. Let $f,g\colon (0,2)\to\mathbb{R}$ be the real-valued functions defined by the formulae $f(x) = 2-x$ and $g(x) = 1+3x$. Compute the uniform distance $d_\infty(f,g)$.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.4.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. **Open and closed sets in metric subspaces.** Let $U\subseteq X$ be a subset of a metric space (X,ρ).

(a) Prove that a subset $V\subseteq U$ is open in the metric subspace (U,ρ) if and only if $V = U\cap W$ for some open set $W\subseteq X$ in X.

(b) Prove that if a subset $V\subseteq U$ is open in X, then it is open in U.

(c) Construct an explicit counterexample to the converse of (b) above.

2. **Interior, closure, and boundary of complements.** Let $U\subseteq X$ be a subset of a metric space (X,ρ).

(a) Prove that $\operatorname{Int}(X\setminus U) = X\setminus \operatorname{Cl}(U)$.

(b) Prove that $\operatorname{Cl}(X\setminus U) = X\setminus \operatorname{Int}(U)$.

(c) Prove that $\operatorname{Bd}(X\setminus U) = \operatorname{Bd}(U) = \operatorname{Cl}(U)\cap \operatorname{Cl}(X\setminus U)$.

(d) Prove that $\operatorname{Bd}(U) = \operatorname{Cl}(U)\setminus \operatorname{Int}(U)$.

3. Closure and limit points. Let $U \subseteq X$ be a subset of a metric space (X, ρ). Prove that $\mathrm{Cl}(U) = U \cup U'$.

Hint. Recall that the **derived set** U' of a subset $U \subseteq X$ of a metric space (X, ρ) is the set of all limit points of U in X.

4. Open balls in ultrametric spaces. Prove that any point in an open ball in an ultrametric space is its center.

Hint. You may find (c) of Problem 3.1.5.3 Ultrametrics to be of use.

5. Diameter in the real line. Let $\emptyset \subsetneq U \subseteq \mathbb{R}$ be a non-empty subset of the real line $(\mathbb{R}, d)$. Prove that U has diameter

$$\mathrm{diam}(U) = \sup(U) - \inf(U)$$

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.5 Notions of Metric Continuity

In this section, we present three distinct but related notions of continuity on a metric space. Informally, these forms of continuity are conditions related to the nature of the convergence of a function to its value at a point. We will in this section make this precise, and investigate the structure of continuous maps between metric spaces.

3.5.1 Continuous Maps and Functions

In calculus, you may have learned that a function of the real numbers is continuous if the function commutes with limits, if limits of the function agree with the values of the function. We will see that this is also the case for maps of metric spaces. More precisely, map of metric spaces is **continuous** at a point in its domain if its values can be made arbitrarily close to the image of that point just by choosing points close to that point.

Proposition 3.5.1 Characterizations of metric continuity. *Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). For a point $c \in X$, the following are equivalent:*

1. *Convergence characterization of continuity.*

   The map f converges at c to $f(c)$.

2. *ϵ-δ characterization of continuity.*

   For all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$.

3. *Neighborhood characterization of continuity.*

   The pre-image $f^{-1}(N)$ of any neighborhood N of $f(c)$ in Y is a neighborhood of c in X.

4. *Sequential characterization of continuity.*

   For every sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$, if

$$\lim_{n\to\infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Proof. First suppose that f converges at c to $f(c)$, and let $\epsilon > 0$. Then there exists a positive real number $\delta > 0$ so that for all points $x \in X$, if $0 < \rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$. We note that if $\rho_X(x,c) = 0$, then the non-degeneracy of ρ_X implies that $x = c$. In this case, $f(x) = f(c)$, and so $\rho_Y(f(x), f(c)) = 0$.

Now suppose that for all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$, and let $N \subseteq Y$ be a neighborhood of $f(c)$ in Y. Then $\mathrm{B}_\epsilon(f(c)) \subseteq N$ for some positive real number $\epsilon > 0$.

Since $\epsilon > 0$ is positive, there is a positive real number $\delta > 0$ so that for all points $x \in X$, if $\rho_X(x,c) < \delta$, then $\rho_Y(f(x), f(c)) < \epsilon$. In particular,

$$f(\mathrm{B}_\delta(c)) \subseteq \mathrm{B}_\epsilon(f(c)) \subseteq N,$$

and so $\mathrm{B}_\delta(c) \subseteq f^{-1}(N)$; that is, $f^{-1}(N)$ is a neighborhood of c in X.

Now suppose that the pre-image $f^{-1}(N)$ of any neighborhood N of $f(c)$ in Y is a neighborhood of c in X, and let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ so that

$$\lim_{n\to\infty} x_n = c.$$

Let $\epsilon > 0$ be a positive real number. Then $\mathrm{B}_\epsilon(f(c))$ is a neighborhood of $f(c)$ in Y, and so $f^{-1}(\mathrm{B}_\epsilon(f(c)))$ is a neighborhood of c in X. Thus

$$\mathrm{B}_\delta(c) \subseteq f^{-1}(\mathrm{B}_\epsilon(f(c)))$$

for some positive real number $\delta > 0$. Since $(x_n)_{n=0}^{\infty}$ converges to c, there is some natural number $n_\epsilon \in \mathbb{N}$ so that $\rho_X(x_n, c) < \delta$ for all indices $n \geq n_\epsilon$. This implies that

$$f(x_n) \in f(\mathrm{B}_\delta(c)) \subseteq \mathrm{B}_\epsilon(f(c))$$

for all indices $n \geq n_\epsilon$, and so $\rho_Y(f(x_n), f(c)) < \epsilon$ for all such indices. We conclude that

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Finally, suppose that for every sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$, if

$$\lim_{n\to\infty} x_n = c,$$

then

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Then Theorem **??** Sequential characterization of map convergence implies that f converges at c to $f(c)$. ■

Definition 3.5.2 Continuity. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y).

- *Continuity.*

  f is **continuous** at a point $c \in X$ if it satisfies any (and therefore all) of the conditions in Proposition **??** Characterizations of metric continuity. f is **continuous** if it is continuous at every point $c \in X$.

- *Discontinuity.*

  f is **discontinuous** at a point $c \in X$ if it does not satisfies any of the conditions in Proposition **??** Characterizations of metric continuity. f is **discontinuous** if it is discontinuous at some point $c \in X$.

The set of continuous maps from X to Y is denoted $\mathrm{C}(X, Y)$. In the special case $(Y, \rho_Y) = (\mathbb{R}, d)$, the set of continuous, real-valued functions on a metric space (X, ρ_X) is denoted $\mathrm{C}(X) = \mathrm{C}(X, \mathbb{R})$. ◆

Example 3.5.3 Continuity. The following are examples of continuous and discontinuous maps.

- Any constant map between metric spaces is continuous.
- The identity map $\mathrm{id}_X : X \to X$ on a metric space (X, ρ) is continuous. More generally, the inclusion map $\iota : U \to X$ of a metric subspace $U \subseteq X$ is continuous.
- Any map $f : X \to Y$ from a discrete metric space (X, δ_X) to a metric space (Y, ρ_Y) is continuous.
- Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be the real-valued function defined by the formula

$$f(x) = \frac{1}{x}.$$

  Then f is continuous. In particular, we note that f is neither continuous nor discontinuous at 0, since 0 is not in the domain of f.

  You may have been taught that f has a discontinuity at the point 0, but we will require more specific language. We can describe the behavior of f near 0 by saying that f does not have a continuous extension to $\mathbb{R}$.

▲

A map of metric spaces is continuous at a point if and only if the outputs of the map can be made close to the image of that point solely by choosing inputs close to the point. Perhaps unsurprising in light of Theorem **??** Sequential characterization of map convergence is the fact that continuity is totally determined by the behavior of the map on convergent sequences.In fact, unlike the more general spaces we will investigate in Chapter **??** Topological Spaces, the answers to all point-set topological questions about metric spaces are captured by the behavior of sequences. Perhaps more surprising is the following characterization of continuity in terms of open sets.

Corollary 3.5.4 Open set characterization of metric continuity. *A map $f : X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is continuous if and only if the pre-image of any open subset of Y is an open subset of X.*

Proof. Suppose first that f is continuous, and let $V \subseteq Y$ be an open subset of Y. Let $c \in f^{-1}(V)$. Since V is open, it is a neighborhood of $f(c)$, and so the neighborhood characterization of continuity in Proposition **??** Characterizations of metric continuity implies that $f^{-1}(V)$ is a neighborhood of c. Since this is true of every point $c \in f^{-1}(V)$, $f^{-1}(V)$ is open in X.

Conversely, now suppose that the pre-image of every open subset of Y is an open subset of X, and consider a point $c \in X$ and a neighborhood $N \subseteq Y$ of $f(c)$ in Y. Then $\mathrm{B}_r(f(c)) \subseteq N$ for some positive real number $r > 0$. Since $\mathrm{B}_r(c)$ is open in Y, $f^{-1}(\mathrm{B}_r(f(c)))$ is open in X and is therefore a neighborhood of each of its points.

In particular, $f(c) \in \mathrm{B}_r(f(c))$, and so $c \in f^{-1}(\mathrm{B}_r(f(c)))$. We conclude that $f^{-1}(\mathrm{B}_r(f(c)))$ is a neighborhood of c. The neighborhood characterization of continuity in Proposition ?? Characterizations of metric continuity now implies that f is continuous at c. Since this holds for all points $c \in X$, f is continuous. ■

In general, proving that a given map between metric spaces is continuous can be annoyingly difficult. For example, if you want to use the ϵ-δ characterization of continuity, then finding a correct positive real number $\delta > 0$ for each positive real number $\epsilon > 0$ is not necessarily an easy task. The following results allow us to decompose such a problem into smaller parts, which may be much easier to solve than taking on the entire problem at once.

Proposition 3.5.5 Composition of continuous maps. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). If f is continuous at a point $c \in X$ and g is continuous at $f(c)$, then the composition $g \circ f\colon X \to Z$ is continuous at c.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f is continuous at c,

$$\lim_{n\to\infty} f(x_n) = f(c).$$

Since g is continuous at $f(c)$,

$$\lim_{n\to\infty} (g \circ f)(x_n) = \lim_{n\to\infty} g(f(x_n)) = g(f(c)) = (g \circ f)(c).$$

Thus $g \circ f$ is continuous at c. ■

Proposition 3.5.6 Vector arithmetic of continuous maps. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   Let $f, g\colon X \to V$ be maps from X to V. If f and g are continuous at a point $c \in X$, then their pointwise sum $f + g$ is continuous at c.

2. *Scalar multiplication.*

   Let α be a scalar and $f\colon X \to V$ be a map from X to V. If f is continuous at a point $c \in X$, then the scalar multiplication αf is continuous at c.

3. *Norm.*

   Let $f\colon X \to V$ be a map from X to V. If f is continuous at a point $c \in X$, then the norm $\|f\| = \|\cdot\| \circ f$ is continuous at c.

4. *Inner product.*

   Let $f, g\colon X \to V$ be maps from X to V. If the norm $\|\cdot\|$ on V is induced by an inner product $\langle\cdot,\cdot\rangle$ and f and g are continuous at a point c, then the pointwise inner product $\langle f, g\rangle$ is continuous at c.

Proof.

1. Let $\epsilon > 0$ be a positive real number. Since f is continuous at c, there is a positive real number $\delta_f > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta_f$, then $\|f(x) - f(c)\| < \frac{\epsilon}{2}$. Similarly, since g is continuous at c, there is a positive real number $\delta_g > 0$ so that for all points $x \in X$, if $\rho_X(x, c) < \delta_g$, then $\|g(x) - g(c)\| < \frac{\epsilon}{2}$.

   Let $\delta = \min(\{\delta_f, \delta_g\}) > 0$, and note that for all points $x \in X$, if

$\rho_X(x, c) < \delta$, then

$$\begin{aligned}\|(f+g)(x) - (f+g)(c)\| &= \|f(x) + g(x) - f(c) - g(c)\| \\ &\leq \|f(x) - f(c)\| + \|g(x) - g(c)\| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.\end{aligned}$$

We conclude that $f + g$ is continuous at c.

2. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f is continuous at c, we observe that

$$\begin{aligned}\lim_{n\to\infty} (\alpha f)(x_n) = \lim_{n\to\infty} \alpha f(x_n) &= \alpha \lim_{n\to\infty} f(x_n) \\ &= \alpha f(c) = (\alpha f)(c),\end{aligned}$$

and so αf is continuous at c.

3. By Proposition **??** Composition of continuous maps, it suffices to show that $\|\cdot\| : V \to \mathbb{R}$ is a continuous real-valued function on V. This follows directly from (3) of Proposition **??** Limits and vector arithmetic.

4. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$, and suppose that

$$\lim_{n\to\infty} x_n = c.$$

Since f and g are continuous at c, we observe that

$$\begin{aligned}\lim_{n\to\infty} \langle f, g\rangle (x_n) = \lim_{n\to\infty} \langle f(x_n), g(x_n)\rangle &= \left\langle \lim_{n\to\infty} f(x_n), \lim_{n\to\infty} g(x_n)\right\rangle \\ &= \langle f(c), g(c)\rangle = \langle f, g\rangle (c),\end{aligned}$$

and so $\langle f, g\rangle$ is continuous at c.

■

Corollary 3.5.7 Arithmetic of continuous functions. *Let $f, g \colon X \to \mathbb{R}$ be real-valued functions on a metric space (X, ρ). If f and g are continuous at a point $c \in X$, then $f + g$, $f - g$, and fg are continuous at c.*

Moreover, if f and g are continuous at at point $c \in X$ and $g(x) \neq 0$ for all points $x \in X$, then $\frac{f}{g}$ is continuous at c.

Having introduced the notion of continuity for maps between metric spaces, one natural question is to describe the relationship between continuity and the limiting behavior of sequences of maps. At first glance, it may seem plausible that the limit of a sequence of continuous functions which converges pointwise must also be continuous. This, however, is not the case.

Example 3.5.8 Discontinuous limits of sequences of continuous functions. Consider the sequence $(f_n)_{n=0}^{\infty}$ of continuous real-valued functions $f_n \colon [0, 1] \to \mathbb{R}$ defined by the formulae

$$f_n(x) = x^n.$$

We recall that $(f_n)_{n=0}^{\infty}$ converges pointwise to the function $f \colon [0, 1] \to \mathbb{R}$ defined piecewise by the formula

$$f(x) = 1 \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1. \end{cases}$$

Even though each function f_n is continuous, the pointwise limit f is not continuous at 1. ▲

So the pointwise limit of a sequence of continuous maps between metric spaces need not be continuous. However, if such a sequence of continuous maps converges uniformly, then the limit is a continuous map.

Theorem 3.5.9 Uniform limits of continuous maps. *The limit of a uniformly convergent sequence of continuous maps between metric spaces is continuous.*

Proof. Let $(f_n)_{n=0}^{\infty}$ be a sequence of continuous maps $f_n \colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and suppose that $(f_n)_{n=0}^{\infty}$ converges uniformly to a map $f \colon X \to Y$.

Fix a point $c \in X$, and let $\epsilon > 0$. Since $(f_n)_{n=0}^{\infty}$ converges uniformly to f, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho_Y(f_n(x), f(x)) < \frac{\epsilon}{3}$$

for all indices $n \geq n_\epsilon$ and points $x \in X$. Fix one such index $n \geq n_\epsilon$. Since f_n is continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y(f_n(x), f_n(c)) < \frac{\epsilon}{3}$$

for all points $x \in X$ so that $\rho_X(x, c) < \delta$. We observe that

$$\begin{aligned} \rho_Y(f(x), f(c)) &\leq \rho_Y(f(x), f_n(x)) + \rho_Y(f_n(c), f_n(c)) + \rho_Y(f_n(c), f(c)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon \end{aligned}$$

for all points $x \in X$ so that $\rho_X(x, c) < \delta$, and so f is continuous at c. Since this holds for all points $c \in X$, we conclude that f is continuous. ■

So, while the pointwise limit of a sequence of continuous maps between metric spaces need not in general be continuous, any such sequence which converges uniformly must converge to a continuous limit. Uniformity, or the relative independence of the "size" of neighborhoods of points in a space from position, is a useful property which will allow us to reconcile complex limiting behavior where several limits interact. We now turn to a strengthening of the notion of continuity based on this idea of uniformity.

3.5.2 Stronger Forms of Metric Continuity

Continuity is a fundamental and useful tool for mathematical analysis of maps between metric spaces. However, there are many pathological examples of maps which are continuous and yet do not behave as well as we might like. Therefore, we now turn our attention to a stronger notion of continuity, that of uniform continuity. At the beginning of this section, we saw a definition for metric continuity which posited the existence of a positive real number δ for each point in the domain and each positive real number ϵ. Informally, a map between metric spaces is **uniformly continuous** if the δ associated to a given ϵ may be chosen independent of position in the domain.

Definition 3.5.10 Uniform continuity. A map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is called **uniformly continuous** if for all positive real numbers $\epsilon > 0$, there is a positive real number $\delta > 0$ so that

$$\rho_Y (f (x_1) , f (x_2)) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X (x_1, x_2) < \delta$.

The set of uniformly continuous maps from X to Y is denoted $\mathrm{UC}(X, Y)$. In the special case $(Y, \rho_Y) = (\mathbb{R}, d)$, the set of uniformly continuous real-valued functions on X is denoted $\mathrm{UC}(X) = \mathrm{UC}(X, \mathbb{R})$. ◆

Note that uniform continuity does not apply at a specific point, but rather over the entire domain of a map. ***It does not make sense to say that a map between metric spaces is uniformly continuous at a point.***

Example 3.5.11 Uniform continuity. The following are examples of uniformly continuous and not uniformly continuous maps between metric spaces:

- Any constant map between metric spaces is uniformly continuous. Indeed, fix metric spaces (X, ρ_X) and (Y, ρ_Y), and consider a point $y \in Y$. The constant map $f\colon X \to Y$ from X to Y defined by the formula $f(x) = y$ satisfies the condition

$$\rho_Y (f (x_1) , f (x_2)) = \rho_Y (y, y) = 0$$

  for all points $x_1, x_2 \in X$. Thus for a given positive real number $\epsilon > 0$, we may choose any positive real number $\delta > 0$ and check that this choice verifies the uniform continuity condition.

- The inclusion map of a metric subspace is uniformly continuous. Indeed, fix a subset $U \subseteq X$ of a metric space (X, ρ), and note that the inclusion map $\iota\colon U \to X$ satisfies the condition

$$\rho (\iota (u_1) , \iota (u_2)) = \rho (u_1, u_2)$$

  for all points $u_1, u_2 \in U$. Thus for a given positive real number $\epsilon > 0$, we may choose $\delta = \epsilon > 0$ and check that this choice verifies the uniform continuity condition.

- The real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x^2$ is continuous but not uniformly continuous. Indeed, for any positive real number $\delta > 0$, let

$$x_1 = \frac{1}{2\delta} \qquad\qquad x_2 = \frac{1+\delta^2}{2\delta}.$$

  We observe that

$$|x_1 - x_2| = \left|\frac{1}{2\delta} - \frac{1+\delta^2}{2\delta}\right| = \left|\frac{\delta^2}{2\delta}\right| = \frac{\delta}{2} < \delta,$$

  but

$$|x_1 + x_2| = \left|\frac{1}{2\delta} + \frac{1+\delta^2}{2\delta}\right| = \left|\frac{2+\delta^2}{2\delta}\right| = \frac{2+\delta^2}{2\delta} = \frac{1}{\delta} + \frac{\delta}{2},$$

  which implies that

$$\begin{aligned} |f (x_1) - f (x_2)| &= \left|x_1^2 - x_2^2\right| = |(x_1 - x_2) (x_1 + x_2)| = |x_1 - x_2| \cdot |x_1 + x_2| \\ &= \left(\frac{\delta}{2}\right)\left(\frac{1}{\delta} + \frac{\delta}{2}\right) = \frac{1}{2} + \frac{\delta^2}{4} > \frac{1}{2}. \end{aligned}$$

  In particular, for the positive real number $\epsilon = \frac{1}{2}$, there is no positive real number $\delta > 0$ for which the uniform continuity criterion is satisfied.

▲

The definitions of continuity and uniform continuity are very similar at first glance. In fact, despite the above example of a continuous but not uniformly continuous function, it's not difficult to see that uniform continuity implies continuity.

Proposition 3.5.12 Uniform continuity implies continuity. *Any uniformly continuous map between metric spaces is continuous.*

Proof. Let $f\colon X \to Y$ be a uniformly continuous map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). Consider a point $x \in X$, and let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X\left(x_1, x_2\right) < \delta$. In particular, we observe that

$$\rho_Y\left(f\left(x\right), f\left(x\right)\right) < \epsilon$$

for all points $x \in X$ so that $\rho_X\left(x, c\right) < \delta$, and so f is continuous at c. Since c was chosen to be an arbitrary point in X, f is continuous. ∎

Continuous maps between metric spaces (or other more general spaces) are fundamental objects in point-set topology. This is because lots of topological properties of spaces, which we will examine in the following section and chapter, are preserved under continuous maps. However, some of the more geometric properties, which start to approach notions of size, are not preserved under all continuous maps in general, but only under uniformly continuous maps. In the previous section, we introduced a notion of boundedness for subsets. While the image of a bounded subset of a metric space under a continuous map is not necessarily bounded, the image of a totally bounded subset of a metric space under a uniformly continuous map is totally bounded.

Proposition 3.5.13 Uniform continuity and total boundedness. *The image of a totally bounded subset of a metric space under a uniformly continuous map between metric spaces is totally bounded.*

Proof. Let $f\colon X \to Y$ be a uniformly continuous map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and consider a totally bounded subset $U \subseteq X$. We wish to show that $f\left(U\right)$ is totally bounded in Y.

To that end, let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X\left(x_1, x_2\right) < \delta$. In particular, this implies that

$$f\left(\mathrm{B}_\delta\left(x\right)\right) \subseteq \mathrm{B}_\epsilon\left(f\left(x\right)\right)$$

for all points $x \in X$.

Since U is totally bounded in X, there is a finite list $x_1, \ldots, x_n \in X$ so that

$$U \subseteq \bigcup_{k=1}^{n} \mathrm{B}_\delta\left(x_k\right).$$

We observe that

$$f\left(U\right) \subseteq f\left(\bigcup_{k=1}^{n} \mathrm{B}_\delta\left(x_k\right)\right) = \bigcup_{k=1}^{n} f\left(\mathrm{B}_\delta\left(x_k\right)\right)$$

$$\subseteq \bigcup_{k=1}^{n} \mathrm{B}_\epsilon\left(f\left(x_k\right)\right),$$

and so $f(U)$ is totally bounded in Y. ■

So the uniformly continuous image of a totally bounded subset of a metric space is totally bounded. Just as in the case of plain continuity, the set of uniformly continuous maps from a metric space to a normed vector space itself forms a vector space under pointwise addition and scalar multiplication.

Proposition 3.5.14 Vector arithmetic and uniform continuity. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   If two maps $f, g \colon X \to V$ from X to Y are uniformly continuous, then their pointwise sum $f + g \colon X \to V$ is also uniformly continuous.

2. *Scalar multiplication.*

   If a map $f \colon X \to V$ is uniformly continuous, then any pointwise scalar multiple αf is also uniformly continuous.

3. *Norm.*

   If a map $f \colon X \to V$ is uniformly continuous, then its pointwise norm $\|f\| = \|\cdot\| \circ f$ is also uniformly continuous.

Proof.

1. Let $\epsilon > 0$ be a positive real number. Since f and g are continuous, there are positive real numbers $\delta_f, \delta_g > 0$ so that

   $$\|f(x_1) - f(x_2)\| < \frac{\epsilon}{2}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta_f$ and

   $$\|g(x_1) - g(x_2)\| < \frac{\epsilon}{2}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta_g$. Let $\delta = \min(\{\delta_f, \delta_g\}) > 0$. We observe that

   $$\begin{aligned} \|(f+g)(x_1) - (f+g)(x_2)\| &= \|f(x_1) + g(x_1) - f(x_2) - g(x_2)\| \\ &\leq \|f(x_1) - f(x_2)\| + \|g(x_1) - g(x_2)\| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, and so $f + g$ is uniformly continuous.

2. Let $\epsilon > 0$ be a positive real number. If $\alpha = 0$, then $\alpha f = 0f = \mathbf{0}$ is a constant map and therefore is uniformly continuous. We may therefore suppose that $\alpha \neq 0$ is non-zero.

   Let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

   $$\|f(x_1) - f(x_2)\| < \frac{\epsilon}{|\alpha|}$$

   for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$. We observe that

   $$\|(\alpha f)(x_1) - (\alpha f)(x_2)\| = \|\alpha f(x_1) - \alpha f(x_2)\| = \|\alpha(f(x_1) - f(x_2))\|$$

$$= |\alpha| \|f(x_1) - f(x_2)\| < |\alpha| \left(\frac{\epsilon}{|\alpha|}\right) = \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, and so αf is uniformly continuous.

3. Let $\epsilon > 0$ be a positive real number. Since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\|f(x_1) - f(x_2)\| < \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$. We observe that Proposition **??** Reverse triangle inequality implies that

$$\begin{aligned} |\|f\|(x_1) - \|f\|(x_2)| &= |\|f(x_1)\| - \|f(x_2)\|| = |\|f(x_1)\| - \|f(x_2)\|| \\ &= |\|f(x_1) - \mathbf{0}\| - \|f(x_2) - \mathbf{0}\|| \\ &\leq \|f(x_1) - f(x_2)\| < \epsilon \end{aligned}$$

for all points $x_1, x_2 \in X$ so that $\rho(x_1, x_2) < \delta$, so that $\|f\|$ is uniformly continuous.

■

A natural question to ask is whether the same holds for the pointwise product of uniformly continuous real-valued functions on a metric space. In fact, we have seen that this is not necessarily the case: the real-valued function $f\colon \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$ is uniformly continuous, but the pointwise product $(f \cdot f)(x) = f(x) \cdot f(x) = x^2$ is not uniformly continuous. However, since uniformly continuous maps are continuous and because the product of two continuous, real-valued functions is continuous, the pointwise product of two uniformly continuous, real-valued functions on a metric space is continuous.

Just as in the case of continuous maps, the composition of uniformly continuous maps is uniformly continuous. This allows us to simplify the proofs of uniform continuity for a vast class of maps.

Proposition 3.5.15 Composition of uniformly continuous maps. *The composition of uniformly continuous maps between metric spaces is uniformly continuous.*

Proof. Let $f\colon X \to Y$ and $g\colon Y \to Z$ be uniformly continuous maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). Let $\epsilon > 0$ be a positive real number. Since g is uniformly continuous, there is a positive real number $\eta > 0$ so that

$$\rho_Z(g(y_1), g(y_2)) < \epsilon$$

for all points $y_1, y_2 \in Y$ so that $\rho_Y(y_1, y_2) < \eta$. Moreover, since f is uniformly continuous, there is a positive real number $\delta > 0$ so that

$$\rho_Y(f(x_1), f(x_2)) < \eta$$

for all points $x_1, x_2 \in X$ so that $\rho_X(x_1, x_2) < \delta$.

We observe that for all points $x_1, x_2 \in X$ if $\rho_X(x_1, x_2) < \delta$, then $\rho_Y(f(x_1), f(x_2)) < \eta$, and so

$$\rho_Z((g \circ f)(x_1), (g \circ f)(x_2)) = \rho_Z(g(f(x_1)), g(f(x_2))) < \epsilon.$$

We conclude that $g \circ f$ is uniformly continuous. ■

Uniform continuity of a map has strong implications for its properties. However, there are even stronger notions of continuity.

3.5.3 Even Stronger Forms of Metric Continuity

We conclude this section with a look at a stronger form of continuity than uniform continuity. A map between metric spaces is called **Lipschitz continuous** if it changes the distance between points by at most some positive scalar multiple.

Definition 3.5.16 Lipschitz continuity. A map f from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is called **Lipschitz continuous** if there is a non-negative real number $L \geq 0$ so that

$$d_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L d_X\left(x_1, x_2\right)$$

for all points $x_1, x_2 \in X$.

The set of Lipschitz continuous maps from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is denoted $\mathrm{LC}\left(X, Y\right)$, and the set of Lipschitz continuous real-valued functions on (X, ρ_X) is denoted $\mathrm{LC}\left(X\right) = \mathrm{LC}\left(X, \mathbb{R}\right)$. ◆

As with uniform continuity, Lipschitz continuity does not apply at a specific point, but rather over the entire domain of a map. ***It does not make sense to say that a map between metric spaces is Lipschitz continuous at a point.***

Example 3.5.17 Norms are Lipschitz continuous. Let $(V, \|\cdot\|)$ be a normed vector space. The Reverse triangle inequality implies that

$$d\left(\left\|\mathbf{v}_1\right\|, \left\|\mathbf{v}_2\right\|\right) = \left|\left\|\mathbf{v}_1\right\| - \left\|\mathbf{v}_2\right\|\right| \leq \left\|\mathbf{v}_1 - \mathbf{v}_2\right\| = 1\left\|\mathbf{v}_1 - \mathbf{v}_2\right\|$$

for all vectors $\mathbf{v}_1, \mathbf{v}_2 \in V$, and so the norm $\|\cdot\| : V \to \mathbb{R}$ is Lipschitz continuous. ▲

For any Lipschitz continuous map between metric spaces, there is a unique smallest non-negative real number such that the above condition is satisfied. This unique smallest constant is called the **Lipschitz constant** for the map.

Definition 3.5.18 Lipschitz constant. The **Lipschitz constant** L_f of a map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is the infimum

$$L_f = \inf\left(\left\{L \in [0, \infty) \mid \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L\rho_X\left(x_1, x_2\right) \text{ for all } x_1, x_2 \in X\right\}\right).$$

◆

Proposition 3.5.19 Lipschitz constants and Lipschitz continuity. *A map $f\colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is Lipschitz continuous if and only if $L_f < \infty$, in which case*

$$\rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L_f \rho_X\left(x_1, x_2\right)$$

for all points $x_1, x_2 \in X$.

Proof. We observe that

$$C = \left\{L \in [0, \infty) \mid \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \leq L\rho_X\left(x_1, x_2\right) \text{ for all } x_1, x_2 \in X\right\}$$

is a subset of the real numbers, and so $L_f = \inf\left(C\right) < \infty$ if and only if $C \neq \emptyset$ is non-empty. That $C \neq \emptyset$ is non-empty is precisely the Lipschitz continuity condition for f, and so f is Lipschitz continuous if and only if $L_f < \infty$.

If $C \neq \emptyset$ is non-empty, let $(l_n)_{n=0}^{\infty}$ be a sequence of elements $l_n \in L$ which converge to $L_f = \inf(C)$. We observe that

$$\begin{aligned}\rho_Y(f(x_1), f(x_2)) &= \lim_{n\to\infty} \rho_Y(f(x_1), f(x_2)) \\ &\leq \lim_{n\to\infty} l_n \rho_X(x_1, x_2) = L_f \rho(x_1, x_2)\end{aligned}$$

for all points $x_1, x_2 \in X$. ■

A map is Lipschitz continuous if it increases the distance between points by at most some nonnegative constant multiplier. This implies that Lipschitz continuity is a stronger condition than uniform continuity, and therefore is also a stronger condition than continuity.

Proposition 3.5.20 Lipschitz continuity implies uniform continuity. *If a map between metric spaces is Lipschitz continuous, then it is uniformly continuous and continuous.*

Proof. Let $f\colon X \to Y$ be a map from a metric space (X, ρ_X) to a metric space (Y, ρ_Y), and consider a positive real number $\epsilon > 0$. There are two cases; either $L_f = 0$, or $0 < L_f < \infty$.

If $L_f = 0$, then Proposition **??** Lipschitz constants and Lipschitz continuity implies that

$$0 \leq \rho_Y(f(x_1), f(x_2)) \leq L_f \rho_X(x_1, x_2) = 0\rho_X(x_1, x_2) = 0 < \epsilon$$

for all points $x_1, x_2 \in X$, and so $f(x_1) = f(x_2)$ for all points $x_1, x_2 \in X$. In particular, this implies that f is constant and therefore uniformly continuous.

If on the other hand $0 < L_f < \infty$, let $\delta = \frac{\epsilon}{L_f}$, and note that Proposition **??** Lipschitz constants and Lipschitz continuity implies that

$$0 \leq \rho_Y(f(x_1), f(x_2)) \leq L_f \rho_X(x_1, x_2) < L_f\left(\frac{\epsilon}{L_f}\right) = \epsilon$$

for all points $x_1, x_2 \in X$ so that $\rho_X(x_1, x_2) < \delta$. Thus f is uniformly continuous. We conclude that f is continuous by Proposition **??** Uniform continuity implies continuity. ■

Example 3.5.21 Uniform continuity does not imply Lipschitz continuity. Consider the real-valued function $f\colon [0,\infty) \to \mathbb{R}$ defined by the formula $f(x) = \sqrt{x}$. For each positive real number $\epsilon > 0$, let $\delta = \epsilon^2 > 0$. We observe that

$$|\sqrt{x_1} + \sqrt{x_2}| \leq |\sqrt{x_1}| + |\sqrt{x_2}| = \sqrt{x_1} + \sqrt{x_2} = |\sqrt{x_1} + \sqrt{x_2}|$$

for all points $x_1, x_2 \in [0,\infty)$, and so

$$\begin{aligned}d(f(x_1), f(x_2))^2 &= |\sqrt{x_1} - \sqrt{x_2}|^2 \leq |\sqrt{x_1} + \sqrt{x_2}| \cdot |\sqrt{x_1} - \sqrt{x_2}| = |x_1 - x_2| \\ &= d(x_1, x_2) < \delta = \epsilon^2\end{aligned}$$

for all points $x_1, x_2 \in [0,\infty)$ so that $d(x_1, x_2) < \delta$. In particular, $d(f(x_1), f(x_2)) < \epsilon$ for all points $x_1, x_2 \in [0,\infty)$ so that $d(x_1, x_2) < \delta$, and so f is uniformly continuous.

On the other hand, we observe that

$$\frac{d\left(f\left(\frac{1}{(L+1)^2}\right), f(0)\right)}{d\left(\frac{1}{(L+1)^2}, 0\right)} = \frac{\left|\sqrt{\frac{1}{(L+1)^2}} - \sqrt{0}\right|}{\left|\frac{1}{(L+1)^2} - 0\right|} = \left|\frac{\left(\frac{1}{L+1}\right)}{\left(\frac{1}{(L+1)^2}\right)}\right|$$

$$= \left| \frac{(L+1)^2}{L+1} \right| = |L+1| = L+1 > L$$

for any non-negative real number $0 \leq L < \infty$, so that

$$d\left(f\left(\frac{1}{(L+1)^2}\right), f(0)\right) > Ld\left(\frac{1}{(L+1)^2}, 0\right).$$

Thus $L_f > L$. Since this holds for each non-negative real number $0 \leq L < \infty$, $L_f = \infty$ and so f is not Lipschitz continuous by Proposition **??** Lipschitz constants and Lipschitz continuity. ▲

Just as in the case of continuity and uniform continuity, Lipschitz continuity is preserved by composition, although possibly at the cost of increasing the Lipschitz constant.

Lemma 3.5.22 Lipschitz continuity and composition. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be maps between metric spaces (X, ρ_X), (Y, ρ_Y), and (Z, ρ_Z). If f and g are Lipschitz continuous, then so is their composition $g \circ f\colon X \to Z$, and $L_{g\circ f} \leq L_g L_f$.*

Proof. We observe that

$$\begin{aligned} \rho_Z\left((g \circ f)(x_1), (g \circ f)(x_2)\right) &= \rho_Z\left(g\left(f\left(x_1\right)\right), g\left(f\left(x_2\right)\right)\right) \\ &\leq L_g \rho_Y\left(f\left(x_1\right), f\left(x_2\right)\right) \\ &\leq L_g L_f \rho_X\left(x_1, x_2\right) \end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the composition $g \circ f$ is Lipschitz continuous and $L_{g\circ f} \leq L_g L_f$. ■

Just as in the case of continuity and uniform continuity, Lipschitz continuous maps from a metric space to a normed vector space form a vector space under the pointwise operations of addition and scalar multiplication, although possibly at the cost of increasing the Lipschitz constant.

Proposition 3.5.23 Vector arithmetic and Lipschitz continuity. *Let (X, ρ) be a metric space and $(V, \|\cdot\|)$ be a normed vector space.*

1. *Vector addition.*

   If two maps $f, g\colon X \to V$ from X to Y are Lipschitz continuous, then their pointwise sum $f + g\colon X \to V$ is also Lipschitz continuous, and $L_{f+g} \leq L_f + L_g$.

2. *Scalar multiplication.*

   If a map $f\colon X \to V$ is Lipschitz continuous, then any pointwise scalar multiple αf is also Lipschitz continuous, and $L_{\alpha f} \leq |\alpha| L_f$.

3. *Norm.*

   If a map $f\colon X \to V$ is uniformly continuous, then its pointwise norm $\|f\| = \|\cdot\| \circ f$ is also uniformly continuous, and $L_{\|f\|} \leq L_f$.

Proof.

1. We observe that

$$\begin{aligned} \left\|(f+g)(x_1) - (f+g)(x_2)\right\| &= \left\|f(x_1) + g(x_1) - f(x_2) - g(x_2)\right\| \\ &\leq \left\|f(x_1) - f(x_2)\right\| + \left\|g(x_1) - g(x_2)\right\| \end{aligned}$$

$$\begin{aligned} &\leq L_f \rho(x_1, x_2) + L_g \rho(x_1, x_2) \\ &= (L_f + L_g)\, \rho(x_1, x_2) \end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the composition $g \circ f$ is Lipschitz continuous and $L_{f+g} \leq L_f + L_g$.

2. We observe that

$$\begin{aligned} \|(\alpha f)(x_1) - (\alpha f)(x_2)\| &= \|\alpha f(x_1) - \alpha f(x_2)\| = \|\alpha (f(x_1) - f(x_2))\| \\ &\leq |\alpha| \, \|f(x_1) - f(x_2)\| \leq |\alpha| \, L_f \rho(x_1, x_2) \end{aligned}$$

for all points $x_1, x_2 \in X$. We conclude that the pointwise scalar multiple αf is Lipschitz continuous and $L_{\alpha f} \leq |\alpha| \, L_f$.

3. Recall from Example **??** Norms are Lipschitz continuous that the norm $\|\cdot\| : V \to \mathbb{R}$ is Lipschitz continuous with Lipschitz constant $L_{\|\cdot\|} \leq 1$. Thus Lemma **??** Lipschitz continuity and composition implies that the composition $\|f\| = \|\cdot\| \circ f$ is Lipschitz continuous, with Lipschitz constant
$$L_{\|f\|} = L_{\|\cdot\| \circ f} \leq L_{\|\cdot\|} L_f \leq 1 L_f = L_f.$$

■

You will have several opportunities to explore applications of Lipschitz continuity in the problems for both this section and Section **??** Completeness and Compactness.

In this section, we have examined several nonequivalent notions of continuity for maps between metric spaces. In the next and final section of this chapter, we will introduce the notions of **completeness** and **compactness** for metric spaces, which both have to do with the convergence of sequences which morally should have a limit, but might not.

3.5.4 Exercises

The following exercises are designed to help you practice basic techniques and verify/solidify your retention of the core ideas presented in each section. Answering these questions will typically involve applying known ideas in familiar ways, recalling information presented in this section, and building fluency with the basic tools of the course. You should approach these exercises as opportunities to make sure you can recognize, recall, and use the fundamental concepts accurately and efficiently before you are asked to answer questions which require complex problem-solving strategies.

Continuity at a point. Determine whether or not the given map between metric spaces is continuous at the given point.

1. Let $f \colon \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(x) = 3x - 1$. Determine whether or not f is continuous at 2.

2. Let $g \colon \mathbb{R} \to \mathbb{R}$ be the linear function defined by the formula $f(y) = 2y^2 - 1$. Determine whether or not g is continuous at 4.

3. Let $h \colon [0, \infty) \to \mathbb{R}$ be the linear function defined by the formula $h(z) = \sqrt{z}$. Determine whether or not h is continuous at 1.

4. **Uniform and Lipschitz continuity.** Let $-\infty < a < b < \infty$ be real numbers, and consider the real-valued function $f \colon [a, b] \to \mathbb{R}$ defined by the formula
$$f(x) = x^3.$$

(a) Verify that f is not uniformly continuous.

(b) Without appealing to Proposition **??** Lipschitz continuity implies uniform continuity, verify that f is not Lipschitz continuous.

(c) Verify that the restriction of f to a closed interval $[a, b] \subsetneq \mathbb{R}$ is uniformly continuous.

(d) Verify that the restriction of f to a closed interval $[a, b] \subsetneq \mathbb{R}$ is Lipschitz continuous.

Short answers and/or in-depth solutions to these exercises are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to answer the above questions, revisit the content of this section rather than spoiling the answers.

3.5.5 Problems

In contrast with the exercises, solving the following problems will require you to engage more deeply with the course material. These questions may require you to analyze a situation, connect multiple ideas, or develop a logical argument from first principles. Many problems will involve writing proofs, explaining your reasoning clearly, and justifying each step. You should expect these to be less routine and more exploratory: there may not be an obvious starting point, and *persistence is part of the task*.

1. Continuity of rational functions.

(a) Prove using one or more characterizations of continuity that the real-valued function $f: \mathbb{R} \to \mathbb{R}$ defined by the formula $f(x) = x$ is continuous.

(b) Argue that any polynomial function is continuous.

(c) Argue that any rational function is continuous on its implicit domain.

2. Uniform limits of uniformly continuous maps. Let $(f_n)_{n=0}^{\infty}$ be a sequence of uniformly continuous maps $f_n: X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y). Prove that if $(f_n)_{n=0}^{\infty}$ converges uniformly, the uniform limit is also uniformly continuous.

3. Continuous pointwise limits of discontinuous maps. We have seen in Theorem **??** Uniform limits of continuous maps and Problem 3.5.5.2 Uniform limits of uniformly continuous maps that uniform limits preserve continuity and uniform continuity. In this problem, we will construct counterexamples to the converses of these results.

(a) Give an example of a sequence $(f_n)_{n=0}^{\infty}$ of continuous maps $f_n: X \to Y$ from a metric space X to a metric space Y which converges pointwise but not uniformly to a continuous map $f: X \to Y$ from X to Y.

(b) Give an example of a sequence $(g_n)_{n=0}^{\infty}$ of discontinuous maps $g_n: X \to Y$ from a metric space X to a metric space Y which converges uniformly to a continuous map $g: X \to Y$ from X to Y.

4. Uniform limits of Lipschitz continuous maps. Consider the sequence $(f_n)_{n=1}^{\infty}$ of real-valued functions $f_n: [0, \infty) \to \mathbb{R}$ defined by the formula

$$f_n(x) = \sqrt{x + \frac{1}{n}}.$$

(a) Show that f_n is Lipschitz continuous for each index $n \geq 1$.

(b) Show that the sequence $(f_n)_{n=0}^{\infty}$ converges uniformly, but the uniform limit is not Lipschitz continuous.

(c) Prove that the limit of a uniformly convergent sequence $(g_n)_{n=0}^{\infty}$ of Lipschitz continuous maps $g_n \colon X \to Y$ from a metric space (X, ρ_X) to a metric space (Y, ρ_Y) is Lipschitz continuous if the sequence $(L_{g_n})_{n=0}^{\infty}$ of Lipschitz constants L_{g_n} is bounded from above.

In-depth solutions to these problems are available in Appendix **??** Selected Solutions. However, these are provided in order to check your work. If you are unable to solve the above problems, discuss your questions with classmates and/or instructor rather than spoiling the answers.

3.6 Completeness and Compactness

In this section, we will investigate two related notions which, like much of the material of this chapter, have to do with the convergence of sequences in metric spaces. First, we will reexamine the limiting behavior of sequences from a more intrinsic perspective, drawing a distinction between convergent sequences and **Cauchy sequences**. We will then examine the properties of **complete metric spaces**, in which this distinction does not exist. Next, we will discuss **compact metric spaces**, which are those in which, even if a sequence diverges, it has a convergent **subsequence**. Finally, to conclude this chapter, we will use both of these notions to prove several powerful results about metric spaces.

3.6.1 Cauchy Sequences and Metric Completeness

Thus far, we have examined the limiting behavior of sequences through an extrinsic perspective, focusing on the distance between the values of a given sequence and the points of the metric space. We now turn our attention to a more broad class of sequences, which are called **Cauchy sequences**, and which are viewed in a more intrinsic way. While a sequence in a metric space is convergent if its values eventually grow close to a specific point, it is Cauchy if its values eventually grow close to each other.

Definition 3.6.1 Cauchy sequence. A sequence $(x_n)_{n=0}^{\infty}$ of points $x_n \in X$ in a metric space (X, ρ) is called **Cauchy** if for all positive real numbers $\epsilon > 0$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_m, x_n) < \epsilon$ for all indices $m, n \geq n_\epsilon$. ◆

The above requirement is sometimes called the **Cauchy criterion**, so a sequence in a metric space satisfies the Cauchy criterion if the distances between its values eventually becomes negligible. Note the distinction between this notion and that of convergence: The entries of a convergent sequence must approach a specific point but the entries of a Cauchy sequence must approach each other (while not necessarily approaching a point of the metric space). At first glance, these notions might seem to be equivalent characterizations of sequences. However, we will see that in some metric spaces, this is not the case.

Example 3.6.2 Cauchy sequence. For each natural number $k \in \mathbb{N}$, let $d_k \in \{0, 1, 2, \ldots, 9\}$ denote the kth digit in the decimal expansion of $\sqrt{2} \approx 1.414213562$, so that

$$d_0 = 1 \qquad d_1 = 4 \qquad d_2 = 1 \qquad d_3 = 4 \qquad d_4 = 2 \qquad \ldots.$$

Consider the sequence $(x_n)_{n=0}^{\infty}$ of rational numbers $x_n \in \mathbb{Q}$ defined by the formula

$$x_n = \sum_{k=0}^{n} \frac{d_k}{10^k} = d_0 + \frac{d_1}{10} + \frac{d_2}{100} + \cdots + \frac{d_n}{10^n}.$$

Let $\epsilon > 0$ be a positive real number. The Archimedean principle implies that

$$n_\epsilon > \log\left(\frac{1}{\epsilon}\right)$$

for some natural number $n_\epsilon \in \mathbb{N}$. We observe that

$$\begin{aligned} d(x_m, x_n) = |x_m - x_n| &= x_{\max(\{m,n\})} - x_{\min(\{m,n\})} \\ &= \sum_{k=0}^{\max(\{m,n\})} \frac{d_k}{10^k} - \sum_{k=0}^{\min(\{m,n\})} \frac{d_k}{10^k} = \sum_{k=\min(\{m,n\})+1}^{\max(\{m,n\})} \frac{d_k}{10^k} \\ &\le \frac{1}{10^{\min(\{m,n\})}} \le \frac{1}{10^{n_\epsilon}} < \epsilon \end{aligned}$$

for all indices $m, n \geq n_\epsilon$. Thus $(x_n)_{n=0}^{\infty}$ is Cauchy. ▲

The above example of Cauchy sequences shows that Cauchy sequences in a metric space need not converge; the given Cauchy sequence of rational numbers converges to $\sqrt{2}$ and thus diverges in the metric space $(\mathbb{Q}, d)$. However, any convergent sequence is Cauchy.

Proposition 3.6.3 Convergent sequences are Cauchy. *Any convergent sequence of points in a metric space is Cauchy.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ), and suppose that $(x_n)_{n=0}^{\infty}$ converges to a limit $x \in X$.

Let $\epsilon > 0$ be a positive real number. Since $(x_n)_{n=0}^{\infty}$, there is a natural number $n_\epsilon \in \mathbb{N}$ so that

$$\rho(x_n, x) < \frac{\epsilon}{2}$$

for all indices $n \geq n_\epsilon$. In particular, we observe that

$$\rho(x_m, x_n) \le \rho(x_m, x) + \rho(x, x_n) = \rho(x_m, x) + \rho(x_n, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $m, n \geq n_\epsilon$. We conclude that the sequence $(x_n)_{n=0}^{\infty}$ is Cauchy. ■

So the convergence of a sequence implies that it satisfies the Cauchy criterion. A natural follow-up question to the above result is whether or not it is possible for the reverse implication holds. The answer to this question actually depends on the particulars of the metric space. Such a metric space in which Cauchy sequences converge is called **complete**.

Definition 3.6.4 Metric completeness. Let $U \subseteq X$ be a subset of a metric space (X, ρ).

- *Completeness.*

  The subset U is called **complete** in X if every Cauchy sequence of points in U converges in X to a limit in U.

- *Incompleteness.*

  The subset U is called **incomplete** in X if some Cauchy sequence of points in U does not converge in X to a limit in U.

A metric space (X, ρ) is called **complete** (resp. **incomplete**) if X is complete (resp. incomplete) in X. A complete normed vector space is called a **Banach space**. ◆

Complete metric spaces are ideal for performing mathematical analysis, as it is often much easier to verify that a given sequence is Cauchy (and therefore has a limit) than to actually compute its limiting value.

Example 3.6.5 Metric completeness. The following are examples of complete and incomplete metric spaces.

- Any discrete metric space is complete, since any Cauchy sequence in a discrete metric space is eventually constant. More generally, any metric space whose points are all isolated points (for example, any finite metric space) is complete.

- The rational numbers $\mathbb{Q}$ are not complete in the real line $(\mathbb{R}, d)$. Indeed, we have already seen an example of a Cauchy sequence of rational numbers which converges in $\mathbb{R}$ to an irrational number $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$.

▲

The above fact about the rational number system is one of the reasons why it doesn't serve as a good context in which to perform mathematical analysis. However, as we will shortly see, the real number system (and, more generally, Euclidean space of any dimension) is a complete metric space, and so there is a rich theory of real analysis. These facts are themselves intimately related to the fact that the rational number system is not Dedekind complete, while the real number system is. This relationship between Dedekind completeness and completeness as a metric space is true for subspaces of the real numbers because of the relationship between the standard ordering of the real numbers and the point-set topology of the Euclidean metric, which we will examine in more detail in Chapter **??** Topological Spaces. We will prove that the real numbers are complete in the Euclidean metric shortly.

First, however, let us examine some of the properties of complete subsets of metric spaces in general. The following two results will be useful in determining whether or not a subset of a given metric space is complete. First, notice that any complete subset of a metric space needs to contain all of its limit points, and so must be closed.

Proposition 3.6.6 Complete subsets are closed. *A complete subset of a metric space is closed.*

Proof. By contraposition. Let $U \subseteq X$ be a subset of a metric space (X, ρ), and suppose that U is not closed in X. Then $U \subsetneq \mathrm{Cl}\,(U) = U \cup U'$ by (2) of Proposition **??** Interior/closure and Problem 3.4.5.3 Closure and limit points, and so $U' \nsubseteq U$; that is, there is some limit point $x \in U'$ which is not an element of U.

Since $x \in U'$ is a limit point of U in X,

$$x = \lim_{n\to\infty} u_n$$

for some sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$. Since $(u_n)_{n=0}^{\infty}$ converges in X, it is Cauchy. However, the limit of this Cauchy sequence of points in U does not lie in U. Thus U is not complete in X. ■

So any complete subset of a metric space is closed. The converse to this statement is not, in general, true, since any metric space is closed in the metric, but not every metric space is complete. However, there is a partial converse to the above result; any closed subset of a complete metric space is also complete.

Proposition 3.6.7 Closed subsets of complete metric spaces. *A subset of a complete metric space is complete if and only if it is closed.*

Proof. Let $U \subseteq X$ be a subset of a complete metric space (X, ρ). We have already seen in Proposition **??** Complete subsets are closed that if U is complete in X, then it is closed in X. It remains to prove the converse implication. To that end, suppose that U is not complete in X, so that some Cauchy sequence $(u_n)_{n=0}^{\infty}$ of points $u_n \in U$ does not converge to a limit in U.

However, since $(u_n)_{n=0}^{\infty}$ is a Cauchy sequence in a complete metric space (X, ρ), it converges to a limit

$$x = \lim_{n\to\infty} u_n \in X.$$

Since $u_n \neq x$ for all indices $n \in \mathbb{N}$, we conclude that $x \in U'$ is a limit point of U in X which is not contained in U. Problem 3.4.5.3 Closure and limit points implies that $x \in \mathrm{Cl}\,(U)$ is a point of closure of U in X, and so U is a proper subset of its closure $\mathrm{Cl}\,(U)$. (2) of Proposition **??** Interior/closure now implies that U is not closed in X. ■

We now would like to prove the completeness of various metric spaces associated with the real number system $\mathbb{R}$. First, however, let us consider the notion of a **subsequence**. Informally, a subsequence of a sequence is a sequence which takes some, but not necessarily all, of the same entries, and these entries appear in the same relative order. We can form a subsequence of a sequence by selecting some infinite subset of indices to include in the subsequence. In fact, any subsequence can be formed in this manner. In order to formalize this notion, recall that a sequence is formally a map whose domain is the natural numbers or the positive integers, and that an increasing function is an injective and non-decreasing function.

Definition 3.6.8 Subsequence. A sequence $y\colon \mathbb{N} \to X$ of points in a metric space (X, ρ) is a **subsequence** of a sequence $x\colon \mathbb{N} \to X$ of points in X if $y = x \circ f$ for some increasing function $f\colon \mathbb{N} \to \mathbb{N}$. In this case, if $x = (x_n)_{n=0}^{\infty}$ and $f = (n_k)_{k=0}^{\infty}$, then we write $y = (x_{n_k})_{k=0}^{\infty}$. ◆

Example 3.6.9 Subsequences. The following are examples of subsequences:

- The sequences
$$(n)_{n=3}^{\infty} = (n+3)_{n=0}^{\infty} = (3, 4, 5, 6, \ldots)$$
and
$$(2n+1)_{n=0}^{\infty} = (1, 3, 5, 7, \ldots)$$
are subsequences of the sequence $(n)_{n=0}^{\infty} = (0, 1, 2, 3, 4, \ldots)$.

- The sequence
$$\left(\frac{1}{n^2}\right)_{n=1}^{\infty} = \left(1, \frac{1}{4}, \frac{1}{9}, \ldots\right)$$
is a subsequence of the sequence
$$\left(\frac{1}{n}\right)_{n=1}^{\infty} = \left(1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right).$$

- The convergent sequence

$$(1)_{n=0}^{\infty} = \left((-1)^{2n}\right)_{n=0}^{\infty} = (1, 1, 1, \ldots)$$

  is a subsequence of the divergent sequence $((-1)^n)_{n=0}^{\infty} = (1, -1, 1, -1, 1, \ldots)$.

▲

The above example shows that divergent sequences may have convergent subsequences. In fact, much of this section will be devoted to examining the properties of metric spaces in which this is the only possible situation. First, however, note that any subsequence of a convergent sequence in a metric space converges to the same limit. In order to show this, it helps to see the following fact about strictly increasing functions on the natural numbers.

Lemma 3.6.10 Increasing functions. *$f(n) \geq n$ for any increasing function $f\colon \mathbb{N} \to \mathbb{N}$.*

Proof. By induction on $n \in \mathbb{N}$. For the base case $n = 0$, we observe that $f(0) \in \mathbb{N}$ is a natural number, and so

$$f(n) = f(0) \geq 0 = n.$$

For the inductive step, suppose for the inductive hypothesis that $f(n) \geq n$ for some natural number $n \in \mathbb{N}$. We observe that $n \leq n+1$, and so $f(n) \leq f(n+1)$, since f is non-decreasing. Moreover, $n \neq n+1$, and so $f(n) \neq f(n+1)$, since f is injective. In particular,

$$f(n+1) > f(n) \geq n$$

by the inductive hypothesis, and so $f(n+1) \geq n+1$. ∎

We are now ready to show that a subsequence of a convergent sequence in a metric space is itself convergent, and converges to the same limit as the larger sequence. So while a subsequence of a divergent sequence might converge, the reverse situation cannot occur.

Proposition 3.6.11 Subsequences of convergent sequences. *Let $(x_n)_{n=0}^{\infty}$ be a sequence of points $x_n \in X$ in a metric space (X, ρ). If $(x_n)_{n=0}^{\infty}$ converges in X, then any subsequence $(x_{n_k})_{k=0}^{\infty}$ also converges in X, and*

$$\lim_{k\to\infty} x_{n_k} = \lim_{n\to\infty} x_n.$$

Proof. Let

$$x = \lim_{n\to\infty} x_n,$$

and consider a positive real number $\epsilon > 0$. There is some natural number $n_\epsilon \in \mathbb{N}$ so that $\rho(x_n, x) < \epsilon$ for all indices $n \geq n_\epsilon$.

We observe that Lemma **??** Increasing functions implies that $n_k \geq n_\epsilon$ for all indices $k \geq n_\epsilon$, and so

$$\rho(x_{n_k}, x) < \epsilon$$

for all such indices. Thus

$$\lim_{k\to\infty} x_{n_k} = x = \lim_{n\to\infty} x_n.$$

∎

An analogous argument can be used to show that any subsequence of a sequence of real numbers which diverges to infinity must itself diverge to infinity. Taken together with Proposition **??** Subsequences of convergent sequences, these results state that a subsequence of a sequence in a metric space with a limit shares that limit. We have seen that the opposite is not in general true; divergent sequences which do not approach a limit may have subsequences which do. However, this cannot happen if the original sequence was Cauchy; a Cauchy sequence in a metric space with a convergent subsequence converges to the limit of this subsequence.

Proposition 3.6.12 Cauchy sequences with convergent subsequences. *If a Cauchy sequence of points in a metric space has a convergent subsequence, then it converges.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of points $x_n \in X$ in a metric space X, and suppose that some subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to a point $x \in X$. We will show that the entire sequence $(x_n)_{n=0}^{\infty}$ also converges to x.

To that end, let $\epsilon > 0$ be a positive real number. Since $(x_n)_{n=0}^{\infty}$ is Cauchy, there is a natural number $m_\epsilon \in \mathbb{N}$ so that

$$\rho(x_m, x_n) < \frac{\epsilon}{2}$$

for all indices $m, n \geq m_\epsilon$. Since the subsequence $(x_{n_k})_{k=0}^{\infty}$ converges to x, there is a natural number $k_\epsilon \in \mathbb{N}$ so that

$$\rho(x_{n_k}, x) < \frac{\epsilon}{2}$$

for all indices $k \geq k_\epsilon$. Let $n_\epsilon = \max(\{m_\epsilon, k_\epsilon\})$. We observe that Lemma **??** Increasing functions implies that $n_k \geq n_\epsilon$ for all indices $k \geq n_\epsilon$. In particular,

$$\rho(x_n, x) \leq \rho(x_n, x_{n_n}) + \rho(x_{n_n}, x) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

for all indices $n \geq n_\epsilon$. Thus the sequence $(x_n)_{n=0}^{\infty}$ converges to x. ■

The above result will come in handy in proving the completeness of the real numbers in the Euclidean metric. So will the following foundational theorem, the content of which is that any bounded sequence of real numbers has a subsequence which is convergent in the Euclidean metric.

Theorem 3.6.13 Bolzano-Weierstrass theorem. (Bernard Bolzano; Karl Weierstrass) *Any bounded sequence of real numbers has a convergent subsequence.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a sequence of real numbers $x_n \in \mathbb{R}$ which is bounded in the Euclidean metric on $\mathbb{R}$. For each natural number $m \in \mathbb{N}$, consider the set

$$T_m = \{x_n \in \mathbb{R} \mid n \geq m\}.$$

There are two cases: either T_m has a maximum for every index $m \in \mathbb{N}$ or T_m does not have a maximum for some index $m \in \mathbb{N}$. We will show that in either case, $(x_n)_{n=0}^{\infty}$ has a monotone subsequence.

First suppose that T_m has a maximum for each index $m \in \mathbb{N}$. Consider the sequence $(n_k)_{k=0}^{\infty}$ of natural numbers $n_k \in \mathbb{N}$ defined recursively by

$$\begin{aligned} n_0 &= \min(\{n \in \mathbb{N} \mid x_n = \max(T_0)\}) \\ n_{k+1} &= \min\left(\left\{n \in \mathbb{N} \,\middle|\, n \geq n_k + 1 \text{ and } x_n = \max\left(T_{n_{k+1}}\right)\right\}\right). \end{aligned}$$

Since $n_{m+1} \geq n_m + 1 > n_m$, the sequence $(n_k)_{n=0}^{\infty}$ is increasing, and so $(x_{n_k})_{k=0}^{\infty}$

is a subsequence of $(x_n)_{n=0}^{\infty}$. Moreover, we observe that

$$x_{n_{k+1}} = \max\left(T_{n_{k+1}}\right) \leq \min\left(T_{n_k}\right) = x_{n_k}$$

for all indices $k \in \mathbb{N}$, and so $(x_{n_k})_{k=0}^{\infty}$ is non-increasing.

On the other hand, now suppose that T_m has no maximum for some index $m \in \mathbb{N}$. Consider the sequence $(n_k)_{k=0}^{\infty}$ of natural numbers $n_k \in \mathbb{N}$ defined recursively by

$$\begin{aligned} n_0 &= m \\ n_{k+1} &= \min\left(\{n \in \mathbb{N} \mid n \geq n_k + 1 \text{ and } x_n > x_{n_k}\}\right). \end{aligned}$$

Since $n_{m+1} \geq n_m + 1 > n_m$, the sequence $(n_k)_{n=0}^{\infty}$ is increasing, and so $(x_{n_k})_{k=0}^{\infty}$ is a subsequence of $(x_n)_{n=0}^{\infty}$. Moreover, we observe that

$$x_{n_{k+1}} > x_{n_k}$$

for all indices $k \in \mathbb{N}$, and so $(x_{n_k})_{k=0}^{\infty}$ is increasing.

In either case, $(x_n)_{n=0}^{\infty}$ has a monotone subsequence $(x_{n_k})_{k=0}^{\infty}$. Since $(x_n)_{n=0}^{\infty}$ is bounded, so too is $(x_{n_k})_{k=0}^{\infty}$, and so Theorem **??** Monotone convergence theorem implies that $(x_{n_k})_{k=0}^{\infty}$ converges. ■

So any bounded sequence of real numbers has a subsequence which converges in the Euclidean metric. We will see that this implies that the real line is complete, and so is a Banach space under the Euclidean norm.

Lemma 3.6.14 Cauchy sequences are bounded. *Any Cauchy sequence of points in a metric space is bounded.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of points in a metric space (X, ρ). Then there is a natural number $n_1 \in \mathbb{N}$ so that $\rho(x_m, x_n) < 1$ for all indices $n \geq n_1$. Let

$$r = \max\left(\{\rho(x_1, x_n), \ldots, \rho(x_{n_1-1}, x_0), \rho(x_{n_1}, x_0) + 1\}\right) > 0.$$

For all indices $n \in \mathbb{N}$, if $n < n_1$, then $\rho(x_n, x_0) \leq r$. On the other hand, if $n \geq n_1$, then

$$\rho(x_n, x_0) \leq \rho(x_n, x_{n_1}) + \rho(x_{n_1}, x_0) \leq \rho(x_n, x_{n_1}) + 1 \leq r.$$

Thus $\{x_n\}_{n=0}^{\infty} \subseteq \mathrm{D}_r(x_0)$, and so the sequence $(x_n)_{n=0}^{\infty}$ is bounded. ■

Corollary 3.6.15 Completeness of the real line. *The real line* $(\mathbb{R}, d)$ *is complete, and so* $(\mathbb{R}, \|\cdot\|)$ *is a Banach space.*

Proof. Let $(x_n)_{n=0}^{\infty}$ be a Cauchy sequence of real numbers $x_n \in \mathbb{R}$. Then $(x_n)_{n=0}^{\infty}$ is bounded by Lemma **??** Cauchy sequences are bounded, and so it has a convergent subsequence by the Bolzano-Weierstrass theorem. Since $(x_n)_{n=0}^{\infty}$ is a Cauchy sequence with a convergent subsequence, it is itself convergent by Proposition **??** Cauchy sequences with convergent subsequences. Since every Cauchy sequence in $(\mathbb{R}, d)$ converges, this metric space is complete. ■

So the real line is a complete metric space. As the Euclidean metric on $\mathbb{R}$ is induced by the Euclidean norm, this implies that $\mathbb{R}$ is a Banach space. That this completeness result is dependent on the Dedekind completeness of the real number system is not immediately obvious. However, in our proof of the Bolzano-Weierstrass theorem, we used the fact that a bounded monotone sequence of real numbers converges to a real number, which requires the existence of upper and lower limits and is not true in ordered number systems such at the rational numbers $\mathbb{Q}$ in which these upper and lower limits do not necessarily exist.

In fact, the completeness of the real line implies that the normed vector spaces associated to the p-norms are all also Banach spaces.

Corollary 3.6.16 Completeness of the p-norm. $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ *is a Banach space for all natural numbers* $n \in \mathbb{N}$ *and extended real numbers* $1 \leq p \leq \infty$.

Proof. Let $(\mathbf{x}_m)_{m=0}^{\infty}$ be a Cauchy sequence of points $\mathbf{x}_m = (x_{m,k})_{k=1}^{n} \in \mathbb{R}^n$ in the metric space $(\mathbb{R}^n, d_p)$. Let $\epsilon > 0$ be a positive real number. Then there is a natural number $m_\epsilon > 0$ so that

$$d\left(\mathbf{x}_m, \mathbf{x}_p\right) < \epsilon$$

for all indices $m, p \geq m_\epsilon$. We observe that for each index $1 \leq k \leq n$,

$$d\left(x_{m,k}, x_{p,k}\right) = |x_{m,k} - x_{p,k}| \leq \|\mathbf{x}_m - \mathbf{x}_p\|_p = d_p\left(\mathbf{x}_m, \mathbf{x}_p\right) < \epsilon$$

for all indices $m, p \geq m_\epsilon$, and so $(x_{m,k})_{m=0}^{\infty}$ is a Cauchy sequence of real numbers.

Corollary **??** Completeness of the real line now implies that this sequence converges to some real number

$$x_k = \lim_{m\to\infty} x_{m,k}.$$

Let $\mathbf{x} = (x_k)_{k=1}^{n} \in \mathbb{R}^n$. Theorem **??** Sequential convergence in the p-metric and Problem 3.2.5.2 Sequential convergence in the infinity metric now imply that $(\mathbf{x}_m)_{m=0}^{\infty}$ converges to $\mathbf{x}$.

Since every Cauchy sequence in $(\mathbb{R}^n, d_p)$ converges, this metric space is complete. We conclude that $\left(\mathbb{R}^n, \|\cdot\|_p\right)$ is a Banach space. ■

It may appear from the above result that almost all commonly appearing metric spaces are complete, but this is certainly not the case. The rational numbers, $\mathbb{Q}$, for example, form a metric subspace of the real line $\mathbb{R}$ which are not complete. We showed this by exhibiting a Cauchy sequence of rational numbers which converge to an irrational number, but this could also be observed by noting that the rational numbers are not a closed subset of the real numbers.

As the properties of complete metric spaces are incredibly useful for performing analysis, it is almost always preferable to work in a complete metric space. Shortly, we will see that this is usually possible, by passing to the **completion** of an incomplete metric space.

3.6.2 Completion of a Metric Space [SKIP]

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3.6.3 Metric Compactness

When is it the case that a sequence in a metric space has a convergent subsequence? This is a natural question to ask about metric spaces. We have seen so far that this occurs when the original sequence converges, and when the sequence is a bounded sequence of real numbers, but this phenomenon is not limited to the case of convergent sequences. In some metric spaces,